

Model Testing Report

Experiment: model_testing

Generated: 2026-04-08 18:05:53

Segmentation Column: value_date

Segments Defined: 4

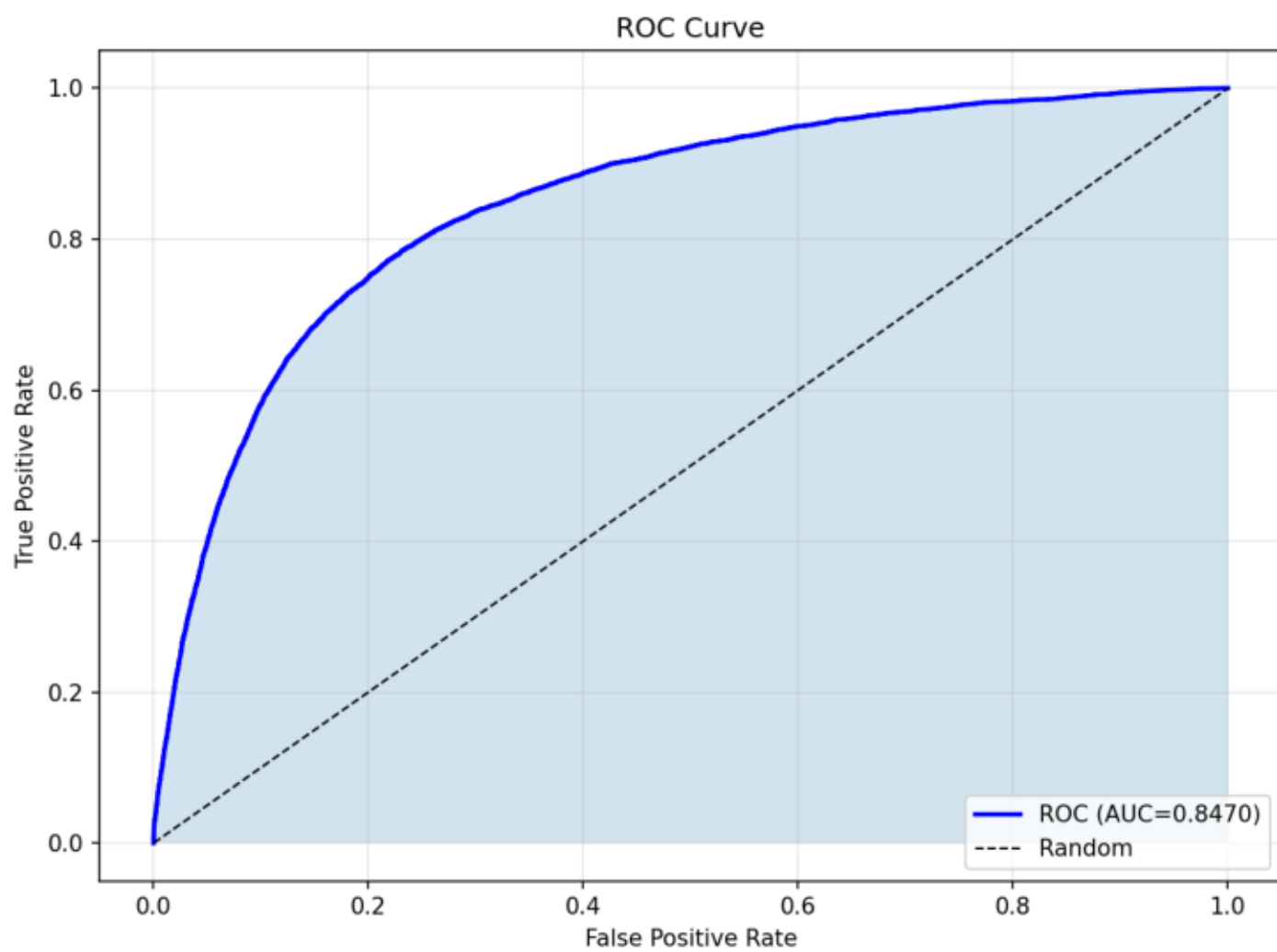
- value_date_quarter=2024Q1: 10801 rows (completed)
- value_date_quarter=2024Q2: 12206 rows (completed)
- value_date_quarter=2024Q3: 13077 rows (completed)
- value_date_quarter=2024Q4: 13916 rows (completed)

Overall - 1) Model Effectiveness

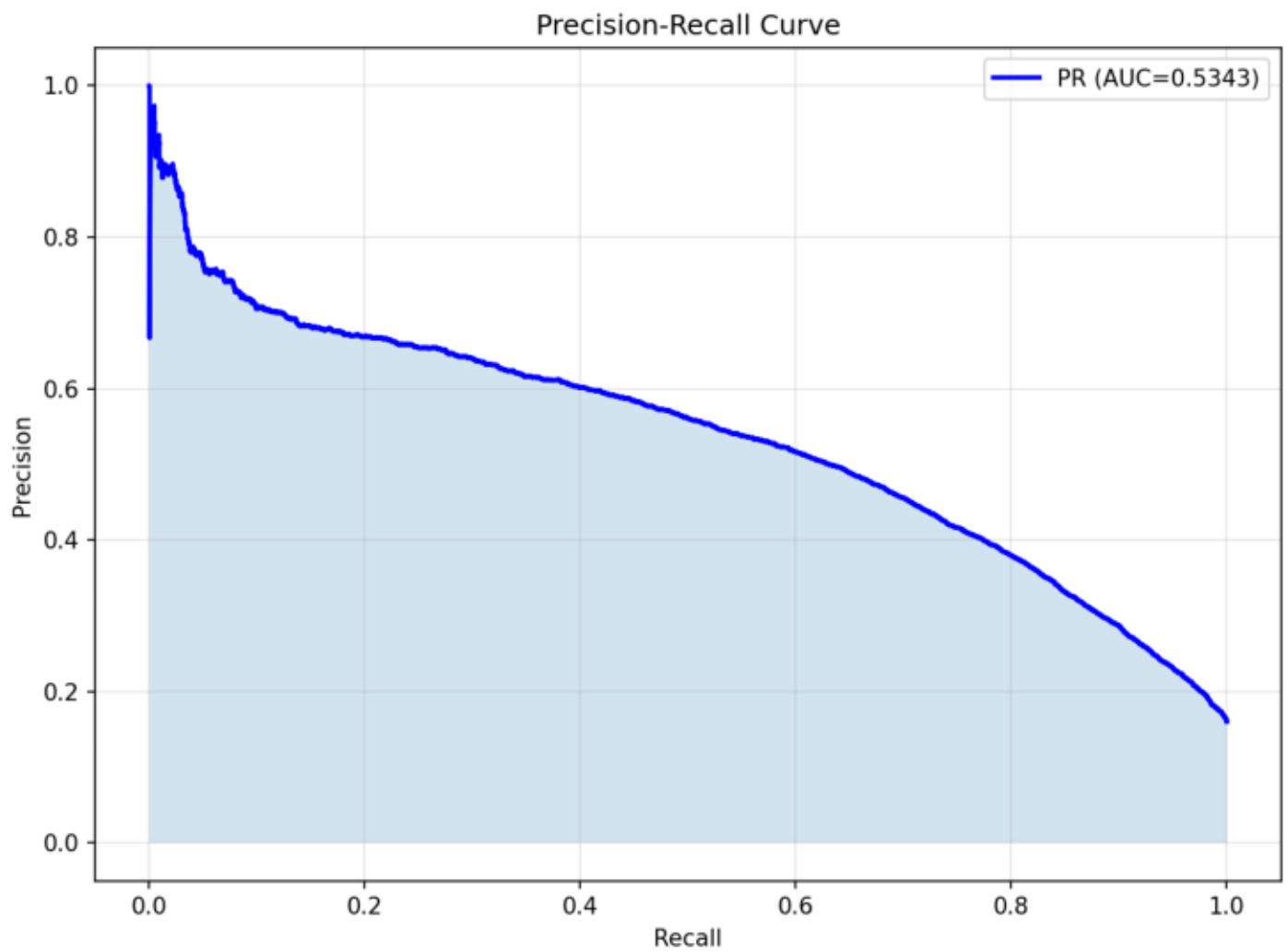
- auc_roc: 0.8470
- auc_pr: 0.5343
- precision: 0.6121
- recall: 0.3701
- f1: 0.4613
- ks_statistic: 0.5528
- ks_threshold: 0.1500
- confusion_matrix.TN: 40126
- confusion_matrix.FP: 1876
- confusion_matrix.FN: 5038
- confusion_matrix.TP: 2960
- precision_at_k.10: 0.9000
- precision_at_k.50: 0.9200
- precision_at_k.100: 0.9000
- precision_at_k.200: 0.8900
- precision_at_k.500: 0.7720
- recall_at_k.10: 0.0011
- recall_at_k.50: 0.0058
- recall_at_k.100: 0.0113
- recall_at_k.200: 0.0223
- recall_at_k.500: 0.0483

Overall - 1) Model Effectiveness - Explanations

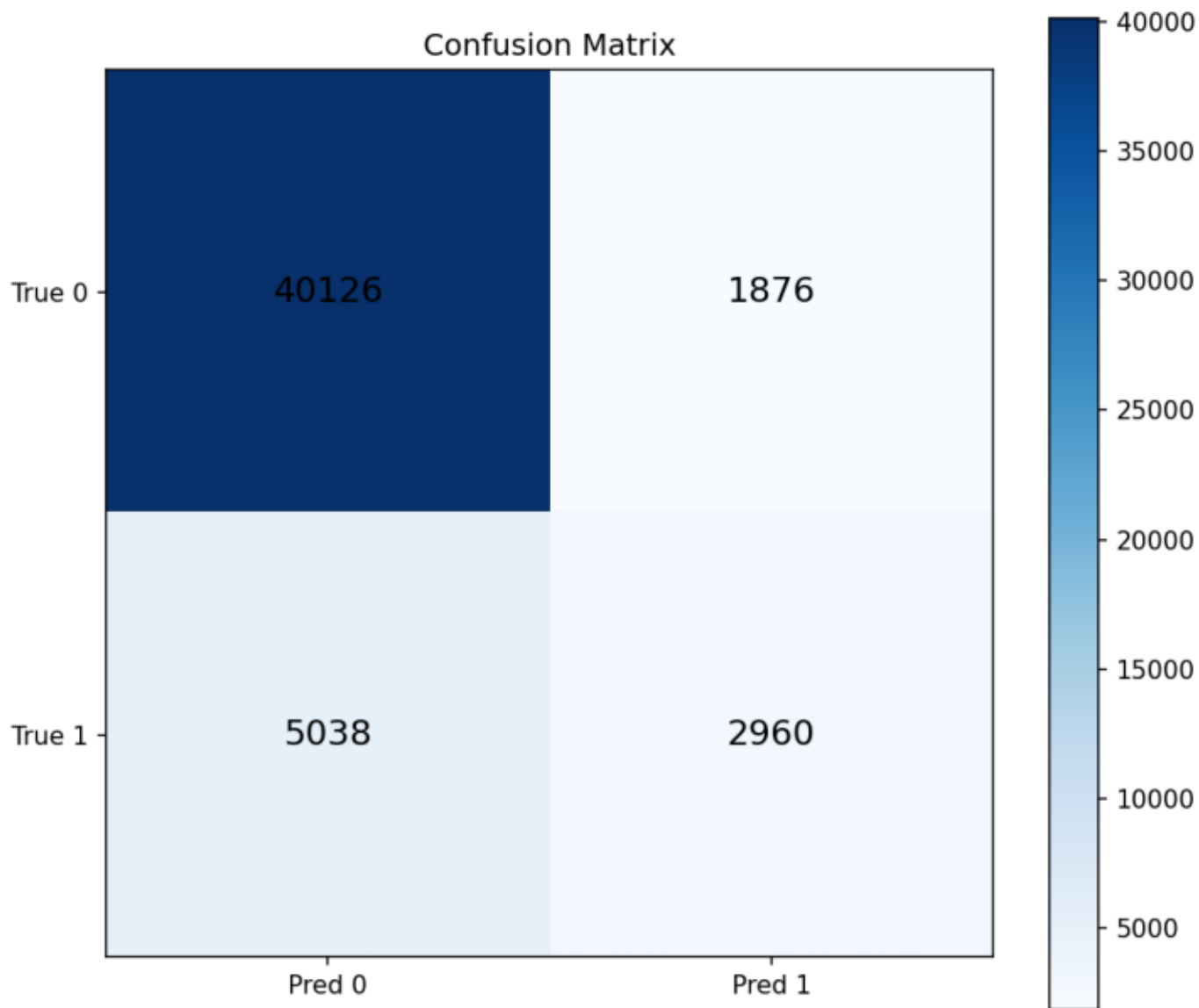
- Positive rate is 16.00%; AUC-ROC is 0.8470 (strong ranking).
- AUC-PR is 0.5343, above the base rate 16.00%, indicating meaningful lift.
- At threshold 0.50, precision=0.6121, recall=0.3701, F1=0.4613 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=2960, FP=1876, FN=5038, TN=40126 (FPR=4.47%, FNR=62.99%).
- KS is 0.5528 at threshold 0.15, indicating very strong separation.
- Top-10: precision=0.9000, recall=0.0011 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.7720, recall=0.0483 shows how much coverage you get with a larger queue.



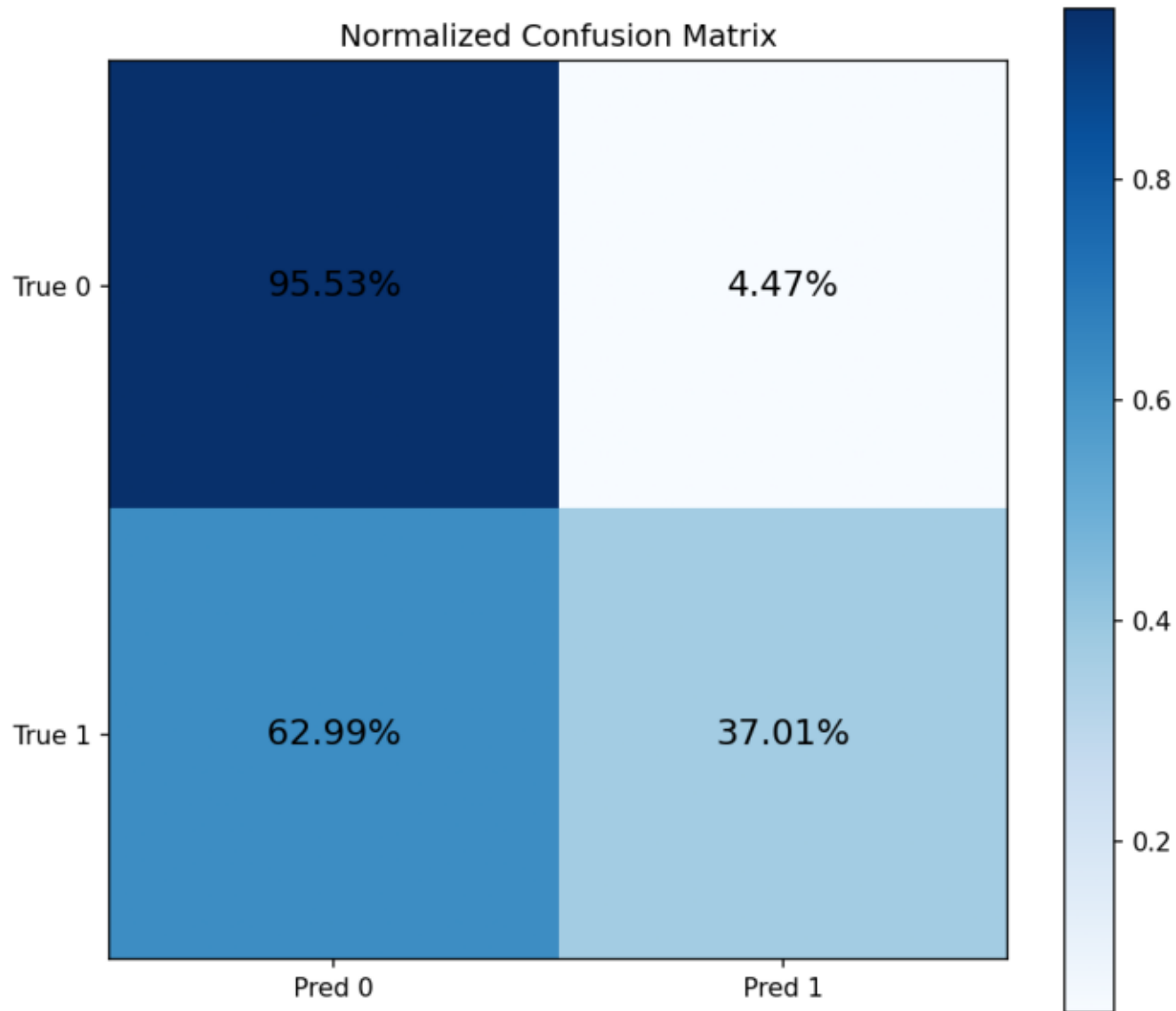
roc_curve: AUC-ROC=0.8470; curve above diagonal indicates strong separation.



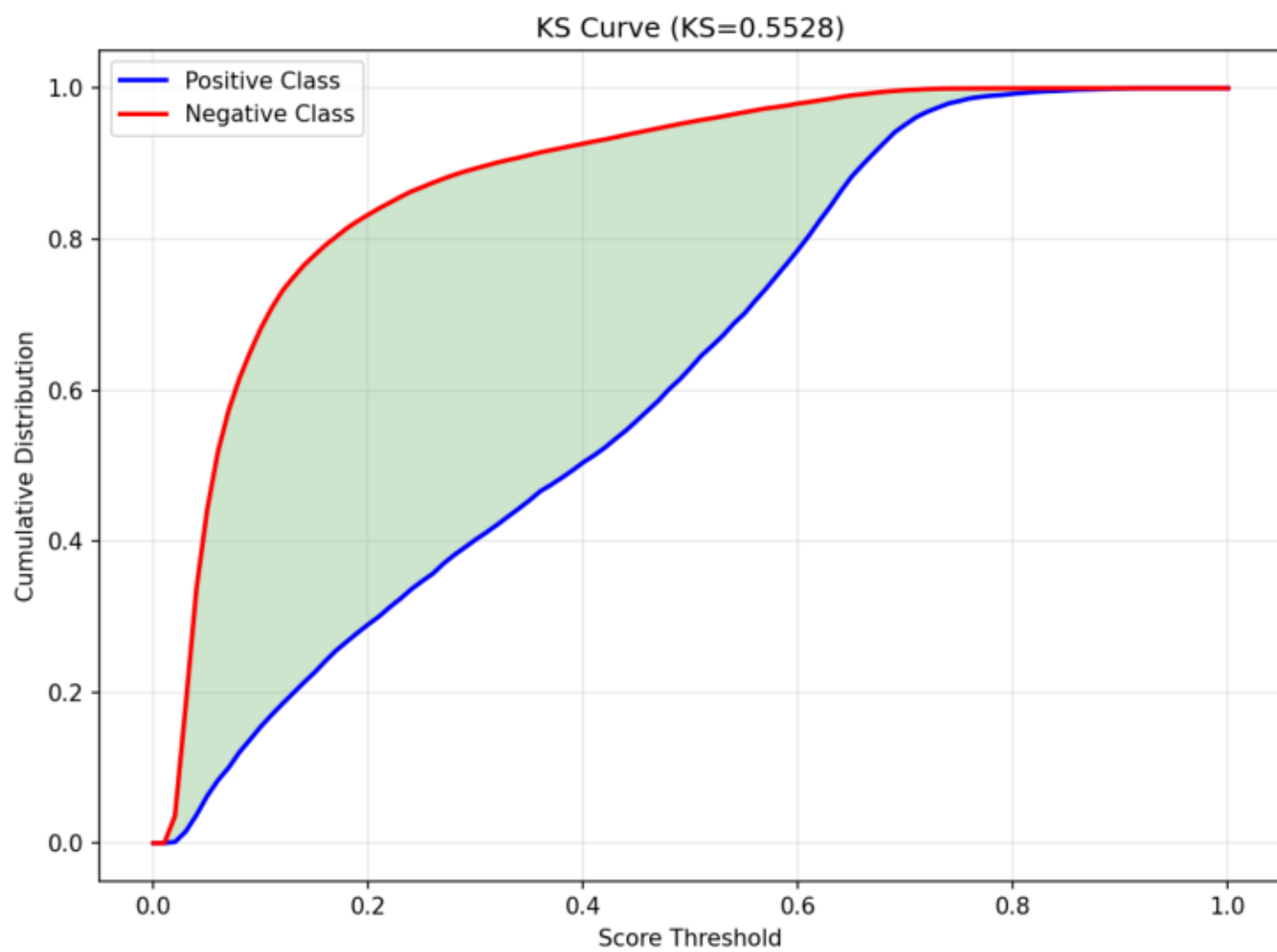
pr_curve: AUC-PR=0.5343 vs baseline 16.00%; higher curve implies better precision at recall.



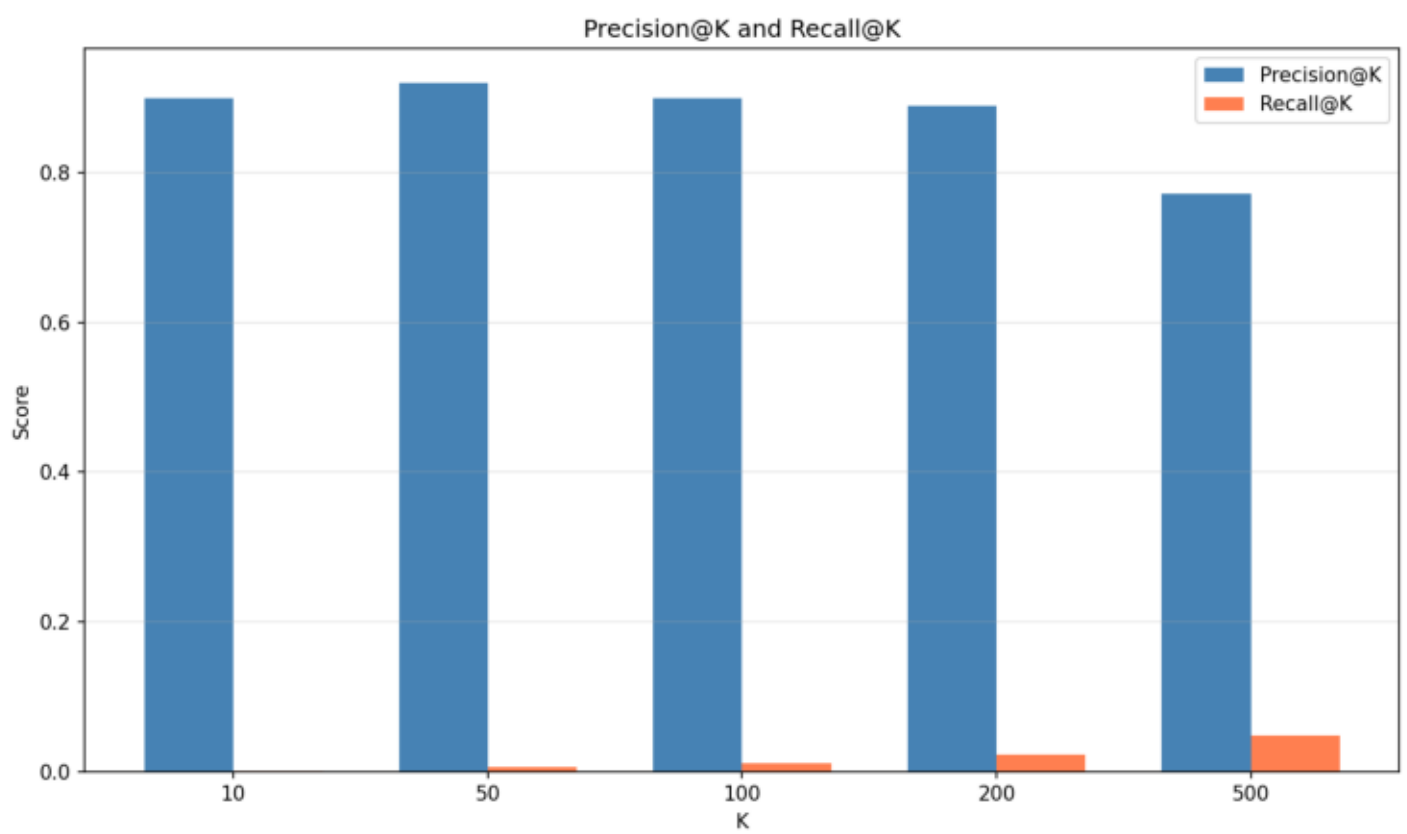
confusion_matrix: At threshold 0.50: TP=2960, FP=1876, FN=5038, TN=40126.



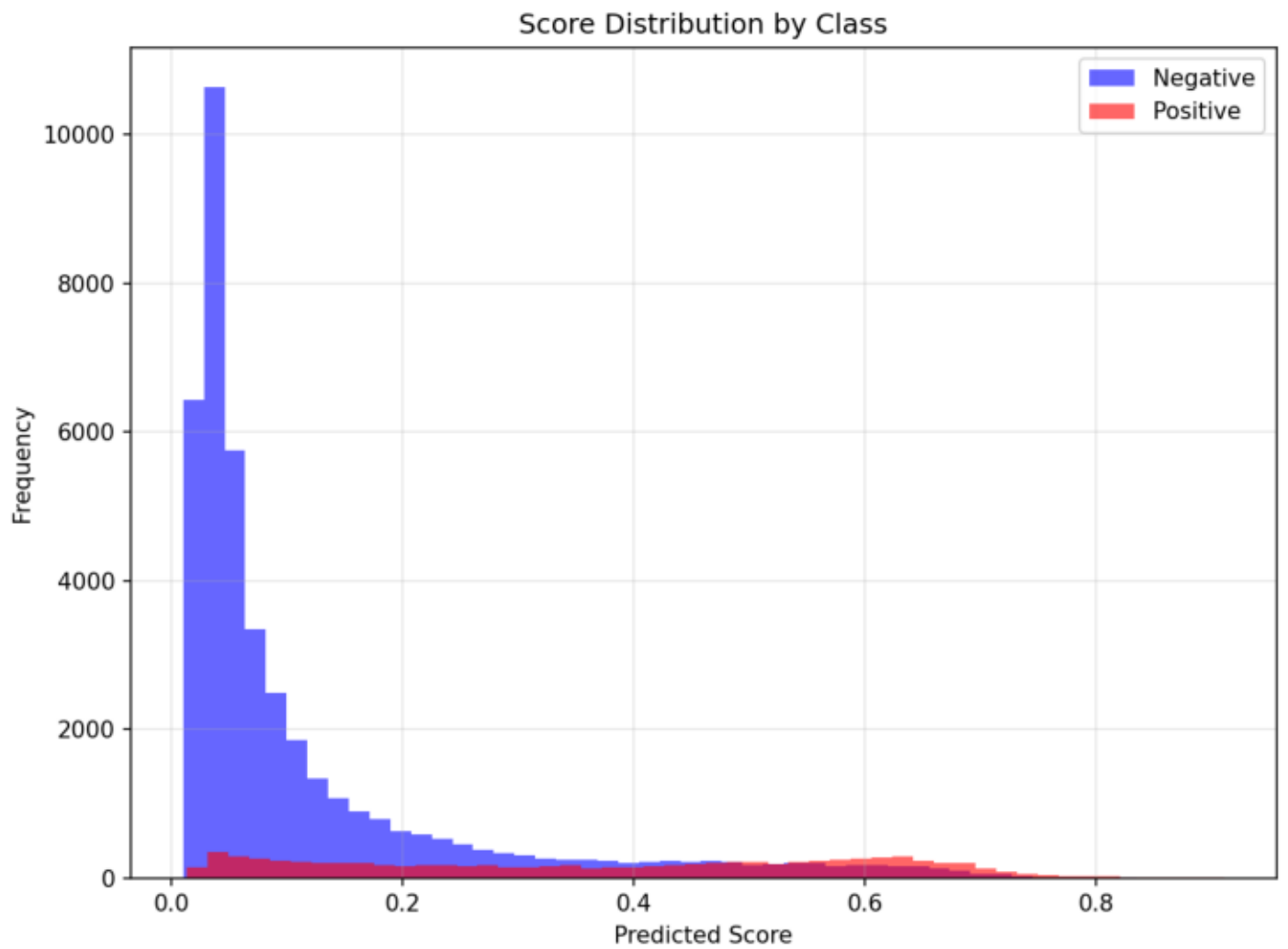
confusion_matrix_norm: Normalized rates: TPR=37.01%, TNR=95.53%.



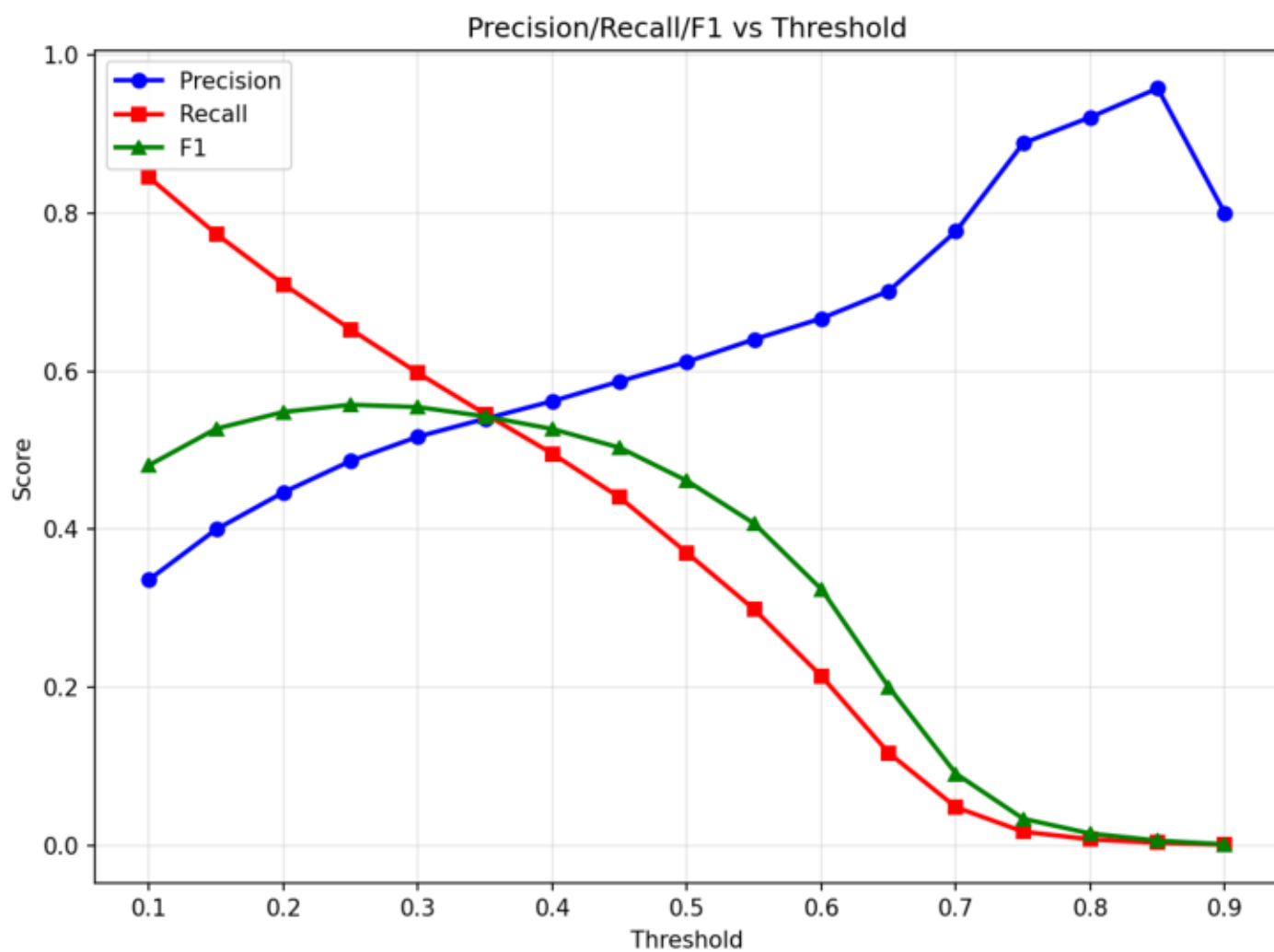
ks_curve: Maximum separation KS=0.5528 at threshold 0.15.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.379, negative=0.118 (larger gap is better).



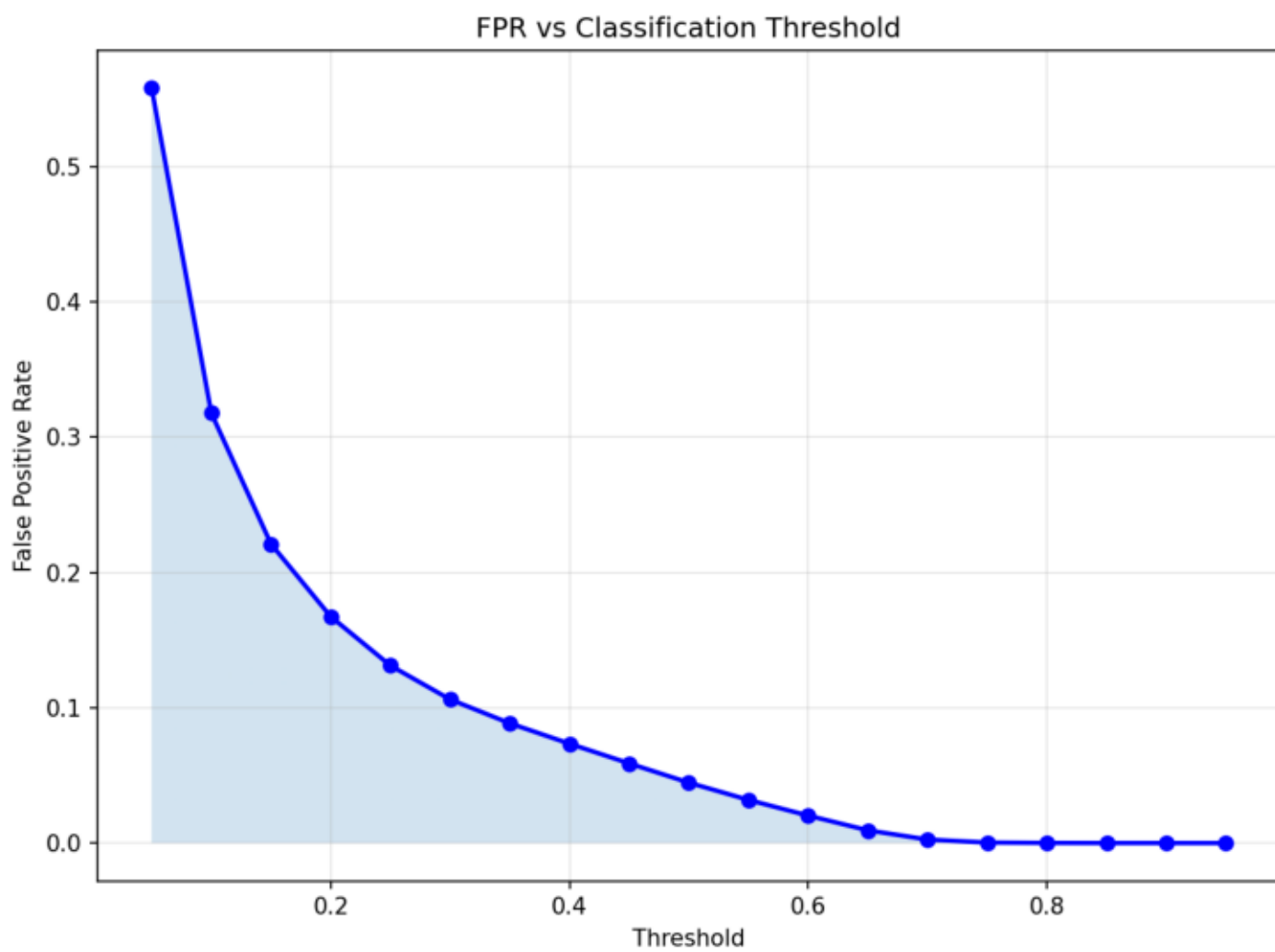
threshold_analysis: Best F1 in tested grid is 0.5577 at threshold 0.25.

Overall - 2) Model Efficiency

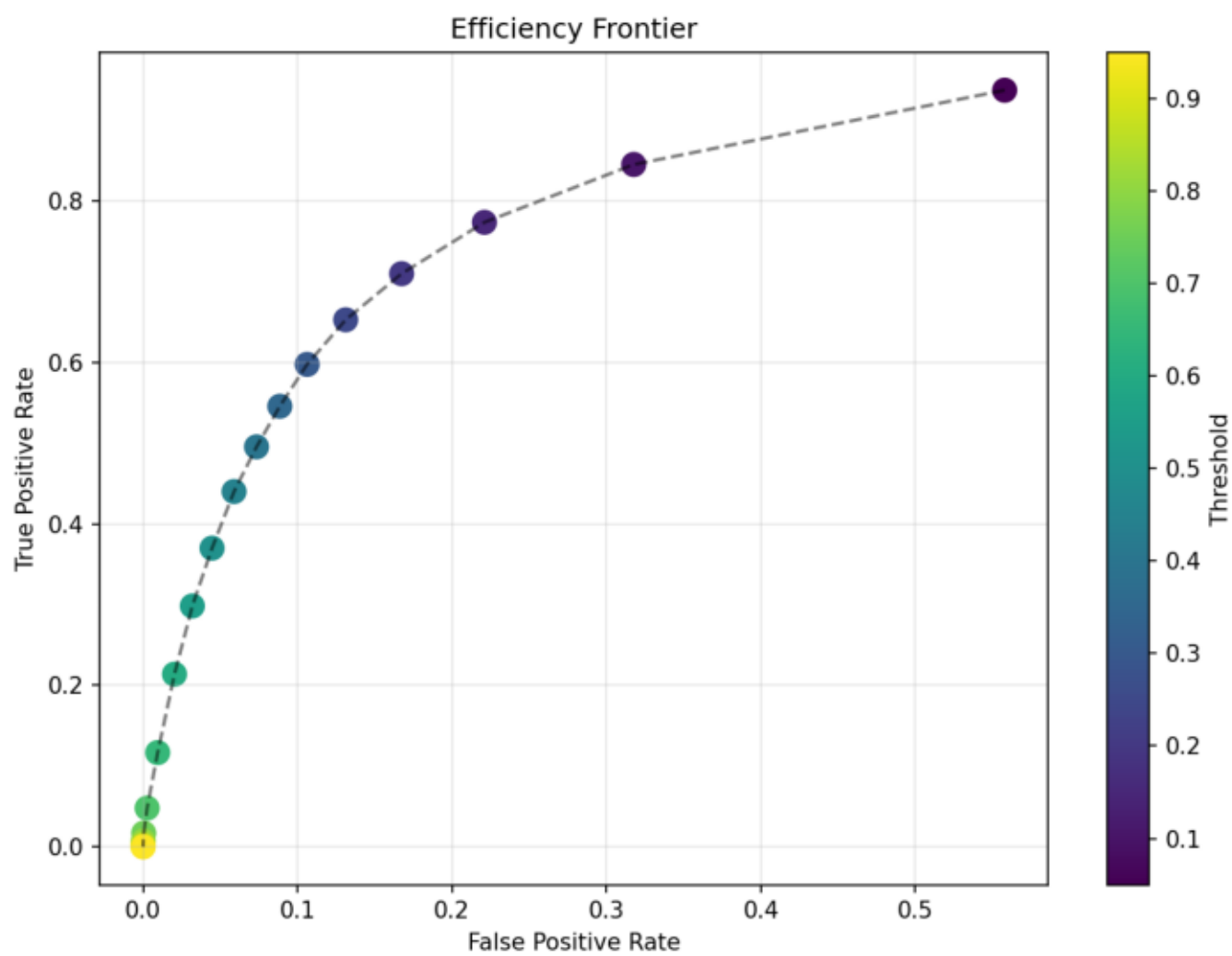
- fpr: 0.0447
- tn: 40126
- fp: 1876
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.5579
- fpr_at_thresholds.t_0.10: 0.3177
- fpr_at_thresholds.t_0.15: 0.2210
- fpr_at_thresholds.t_0.20: 0.1674
- fpr_at_thresholds.t_0.25: 0.1312
- fpr_at_thresholds.t_0.30: 0.1062
- fpr_at_thresholds.t_0.35: 0.0885
- fpr_at_thresholds.t_0.40: 0.0735
- fpr_at_thresholds.t_0.45: 0.0589
- fpr_at_thresholds.t_0.50: 0.0447
- fpr_at_thresholds.t_0.55: 0.0319
- fpr_at_thresholds.t_0.60: 0.0203
- fpr_at_thresholds.t_0.65: 0.0095
- fpr_at_thresholds.t_0.70: 0.0026
- fpr_at_thresholds.t_0.75: 0.0004
- fpr_at_thresholds.t_0.80: 0.0001
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

Overall - 2) Model Efficiency - Explanations

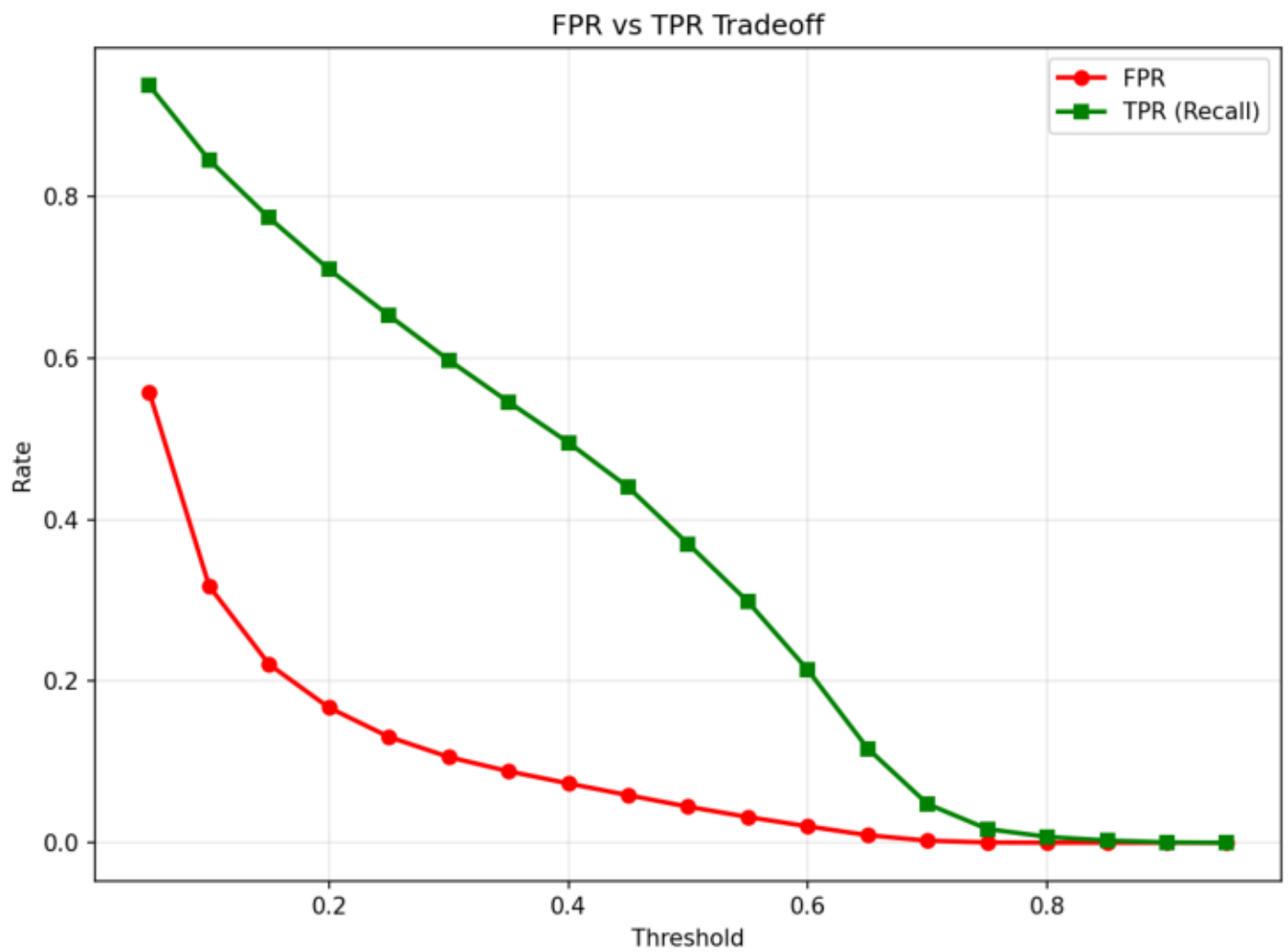
- At threshold 0.50, FPR=0.0447 (low); FP=1876 out of 42002 negatives.
- At the same threshold, TPR=0.3701 with TP=2960 and FN=5038, showing the capture rate of positives.
- A threshold near 0.50 yields FPR $\approx$ 0.0447 with TPR $\approx$ 0.3701 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.



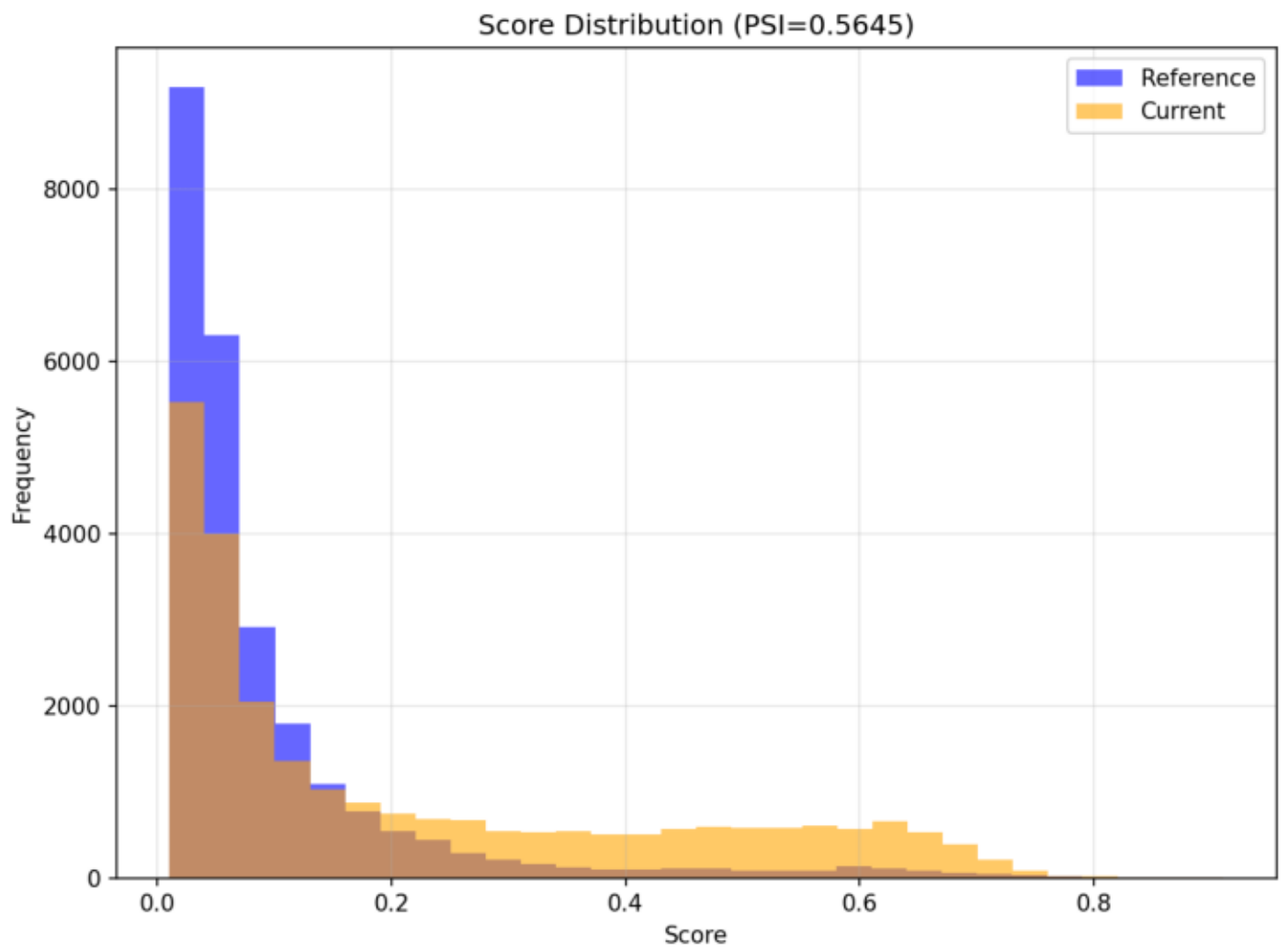
fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

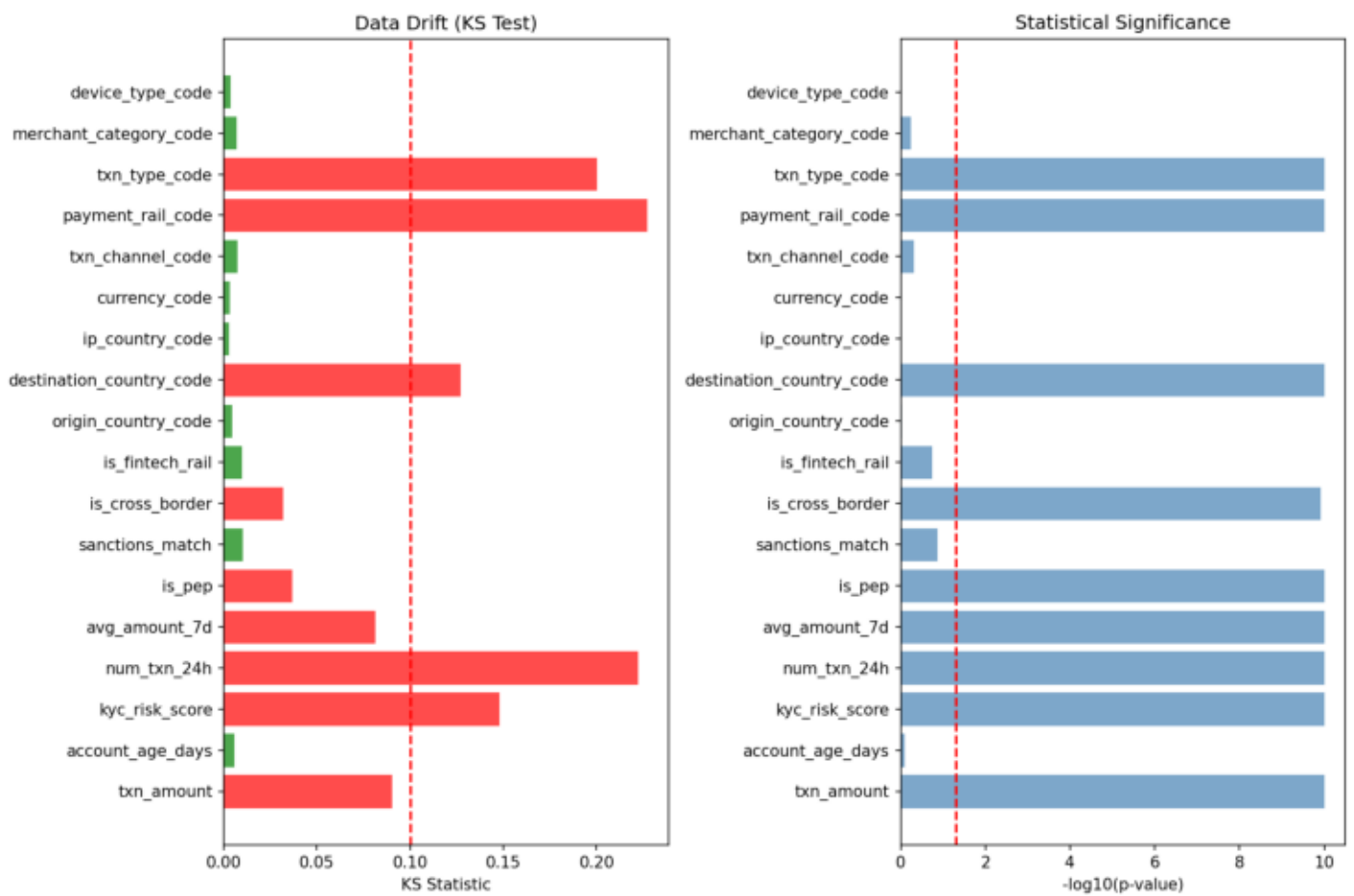
Overall - 3) Model Stability

- psi: 0.5645
- cv_auc_roc_mean: 0.8471
- cv_auc_roc_std: 0.0082
- cv_auc_pr_mean: 0.5350
- cv_auc_pr_std: 0.0155
- bootstrap_auc_roc_mean: 0.8219
- bootstrap_auc_roc_ci_lower: 0.8154
- bootstrap_auc_roc_ci_upper: 0.8279
- concept_drift_detected: False
- concept_drift_score: 0.0058

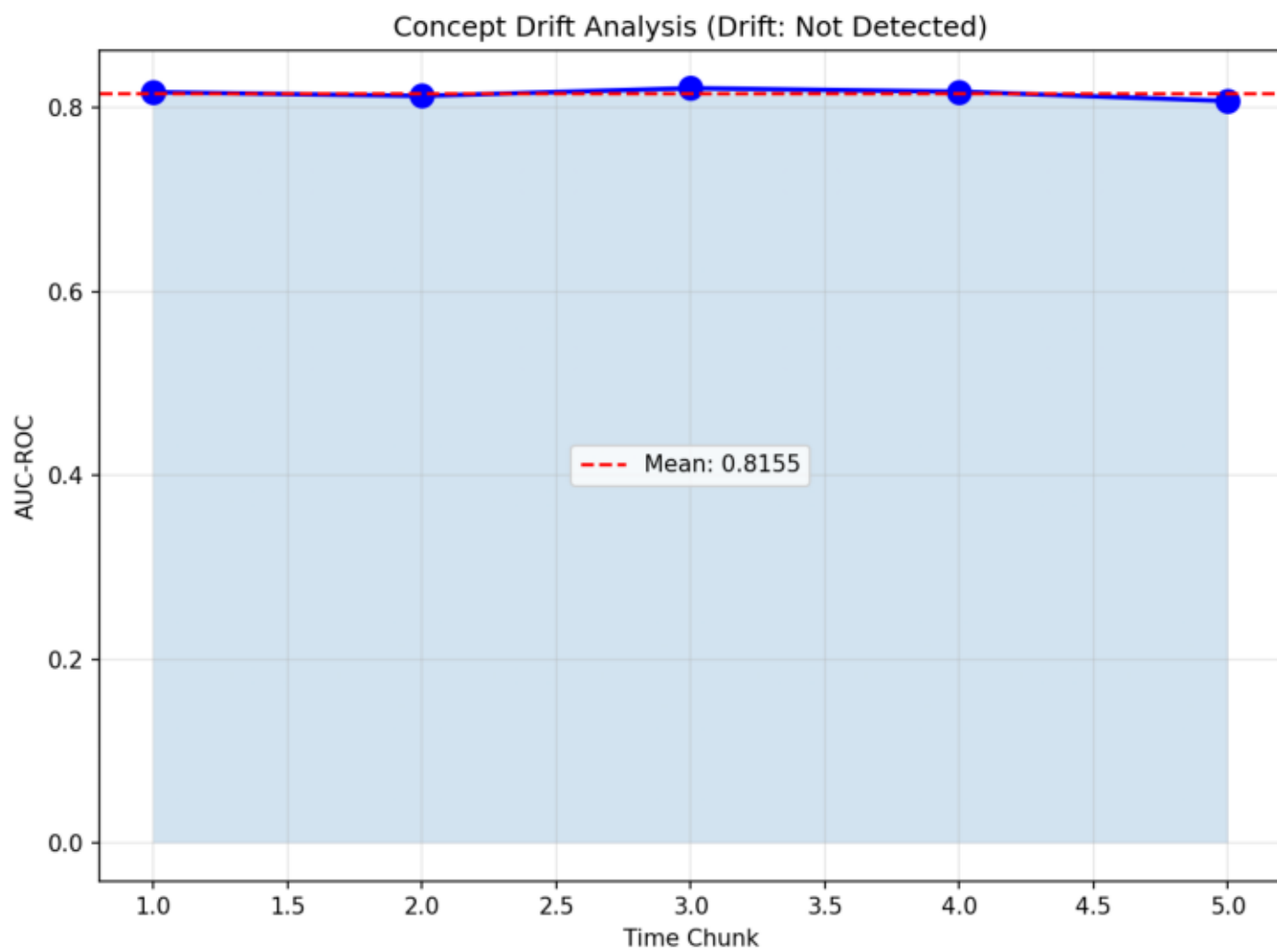
Overall - 3) Model Stability - Explanations

- PSI is 0.5645, indicating material score distribution shift between reference and current data.
- CV AUC-ROC is 0.8471 ± 0.0082 (stable across folds).
- CV AUC-PR is 0.5350 ± 0.0155 (moderate across folds).
- Bootstrap AUC-ROC mean is 0.8219 with 95% CI [0.8154, 0.8279] (width 0.0125).
- No concept drift detected (score 0.0058), suggesting stable performance over time.
- Data drift flagged 9/18 numeric features; top drift signals: ['payment_rail_code', 'num_txn_24h', 'txn_type_code'].

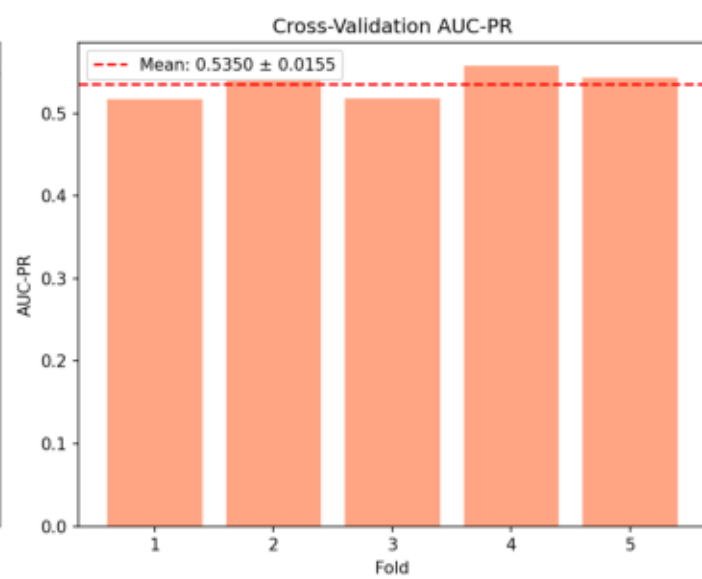
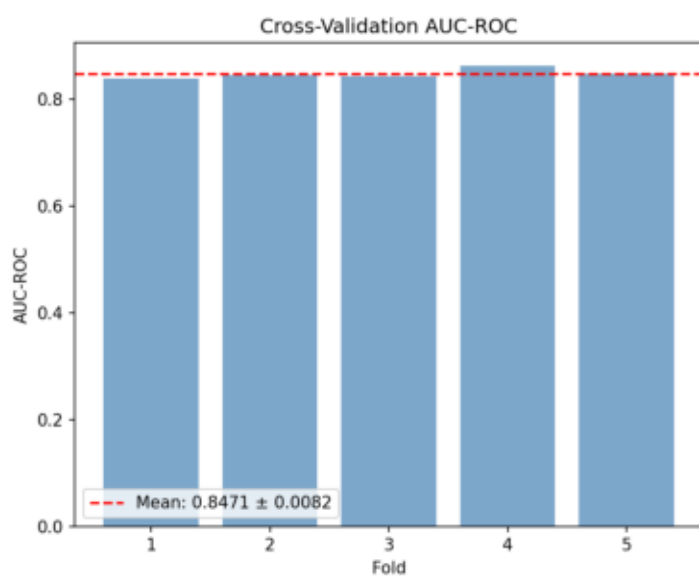




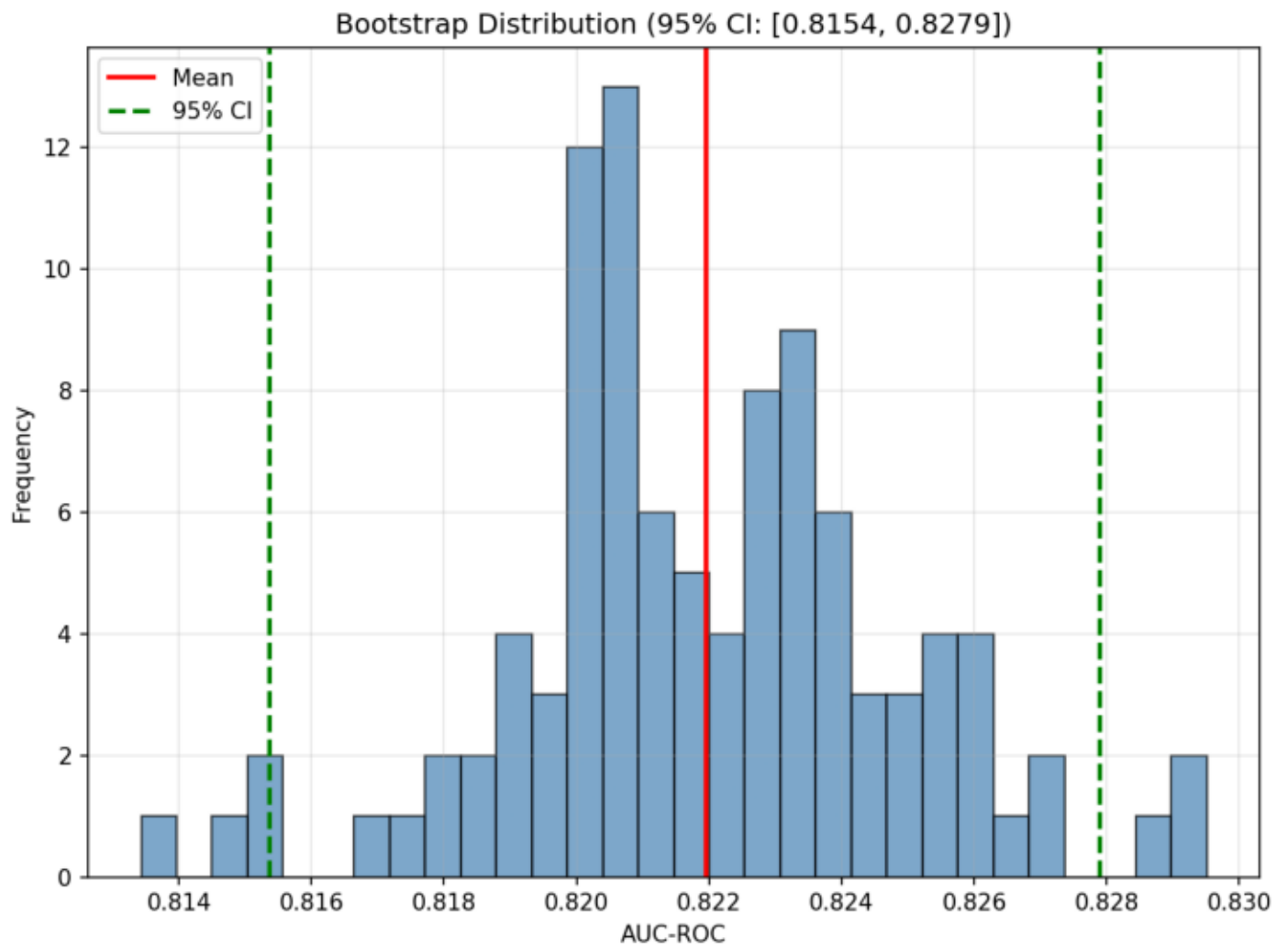
data_drift_heatmap: 9/18 features show drift; red bars highlight flagged features.



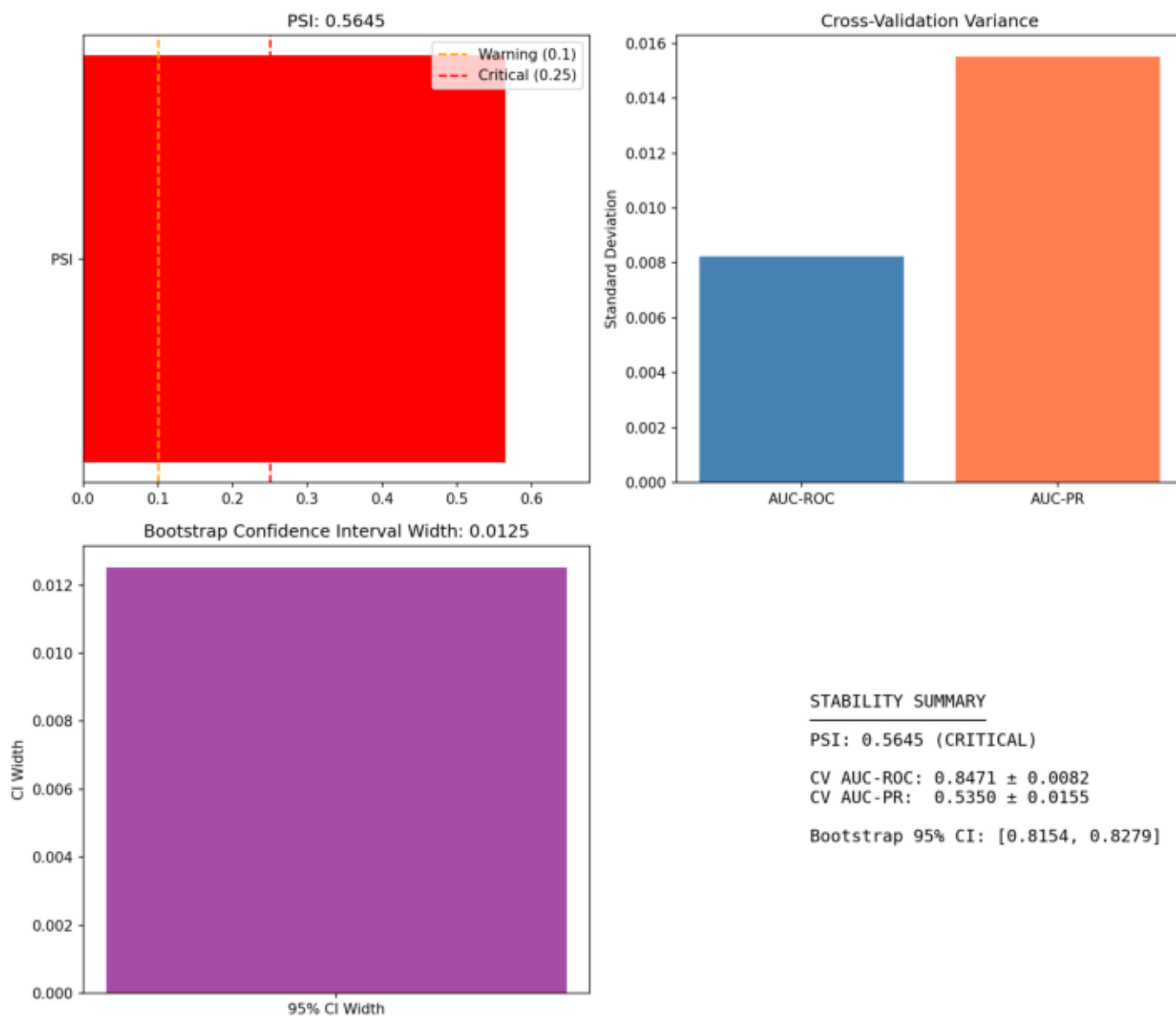
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8471 and PR 0.5350.



bootstrap_distribution: Bootstrap CI [0.8154, 0.8279] reflects performance stability.



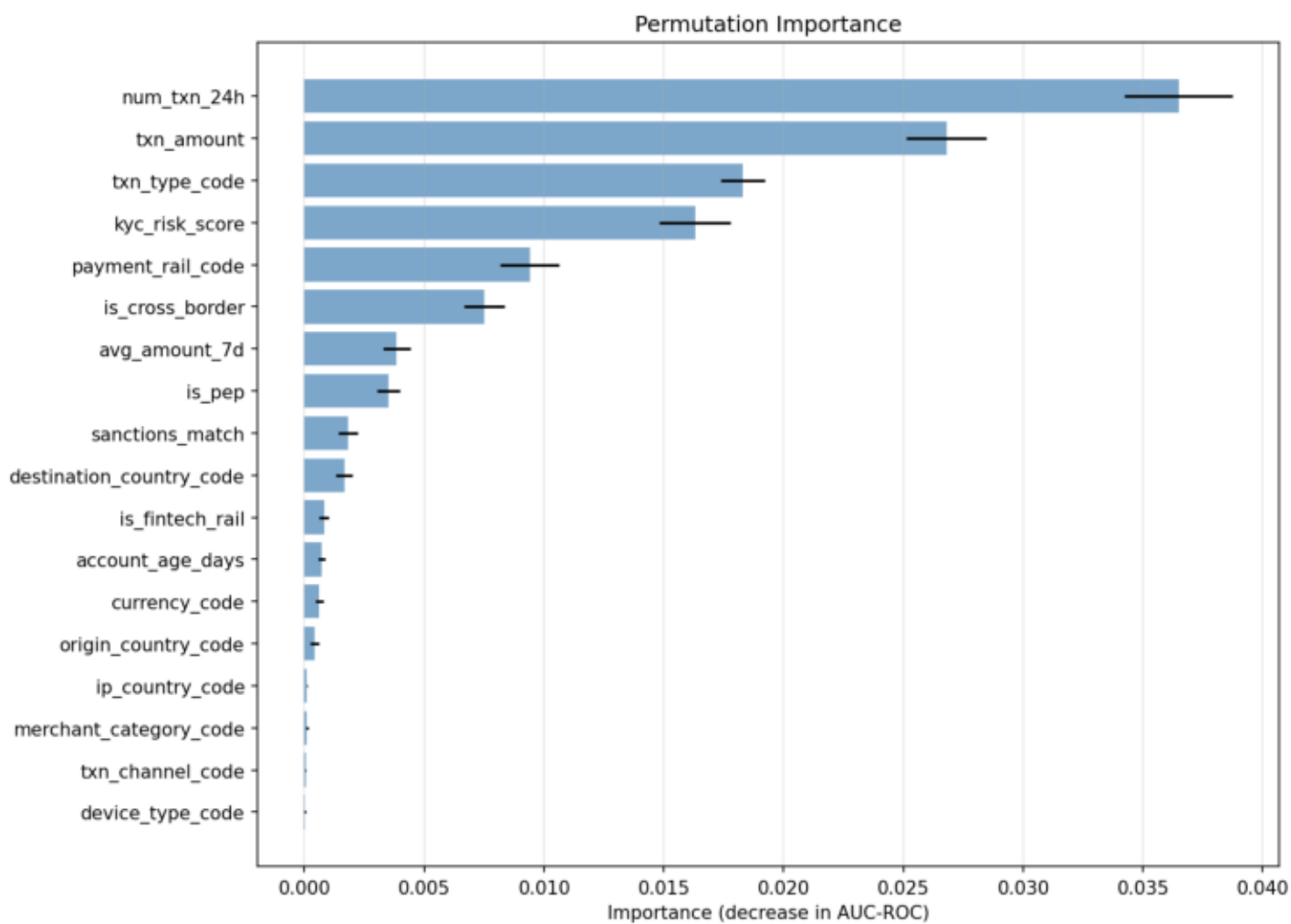
stability_summary: Dashboard summarizes PSI, CV variance, and bootstrap CI width.

Overall - 4) Model Interpretability

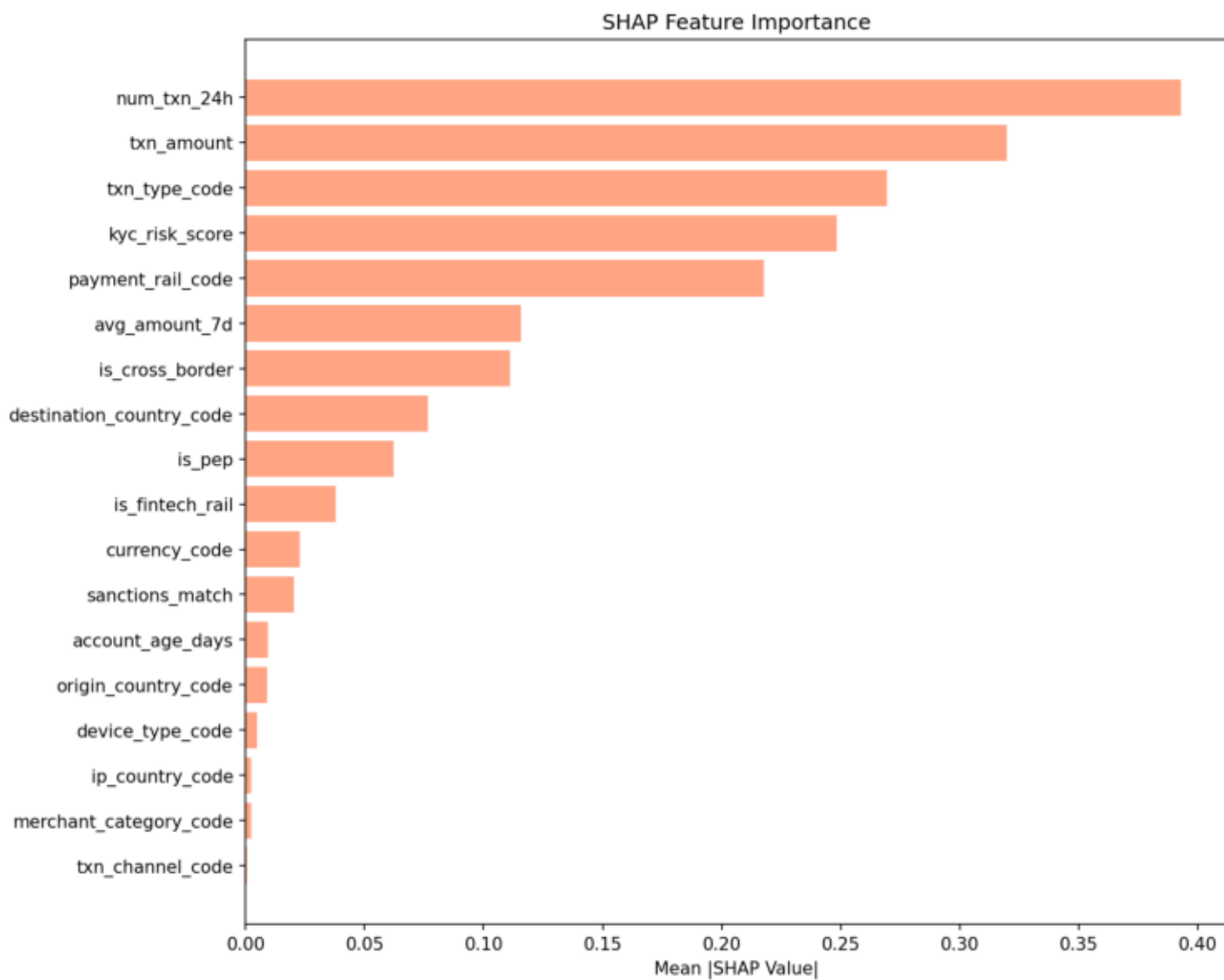
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'shap', 'lime']
- perm_top_features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code']...
- shap_top_features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code']...
- lime_instances: 3
- pdp_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score']

Overall - 4) Model Interpretability - Explanations

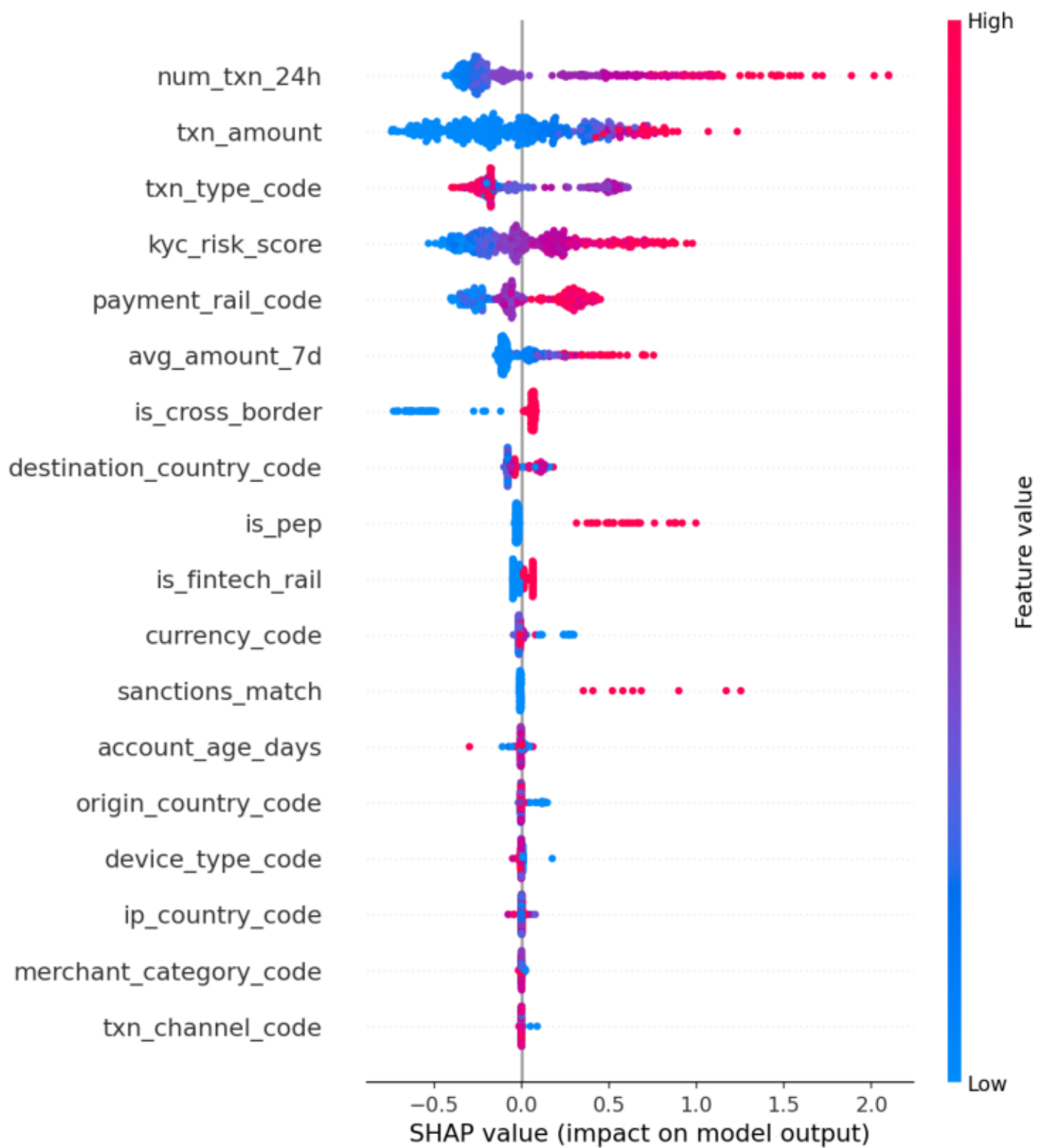
- Permutation importance highlights: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code'].
- SHAP top features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code'].
- LIME generated local explanations for 3 instances; top contributors for instance_12: ['sanctions_match <= 0.00', 'is_pep <= 0.00', 'payment_rail_code <= 3.00', '46.35 < txn_amount <= 128.25', '27.66 < kyc_risk_score <= 38.86'].
- PDP plots show average effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'origin_country_code'].
- ICE plots show individual-level effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score'].



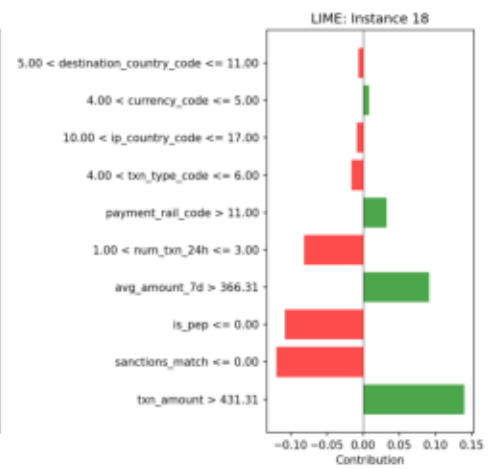
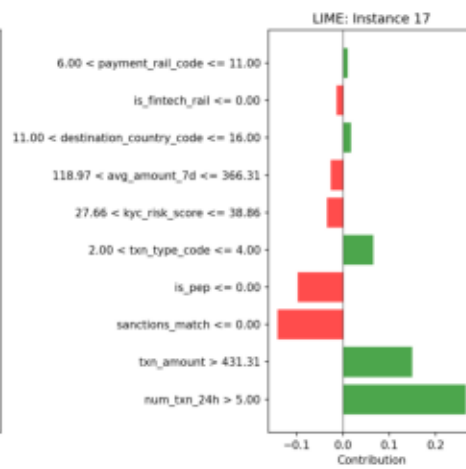
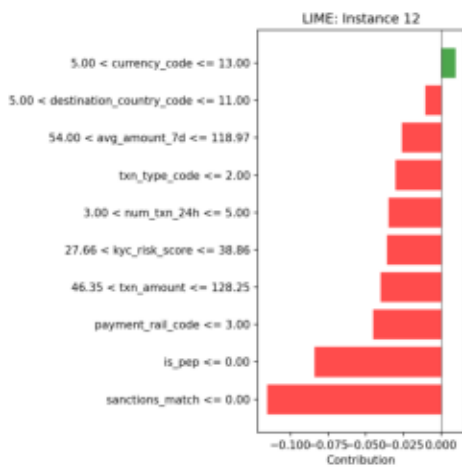
permutation_importance: Bars show how much AUC drops when each feature is permuted.



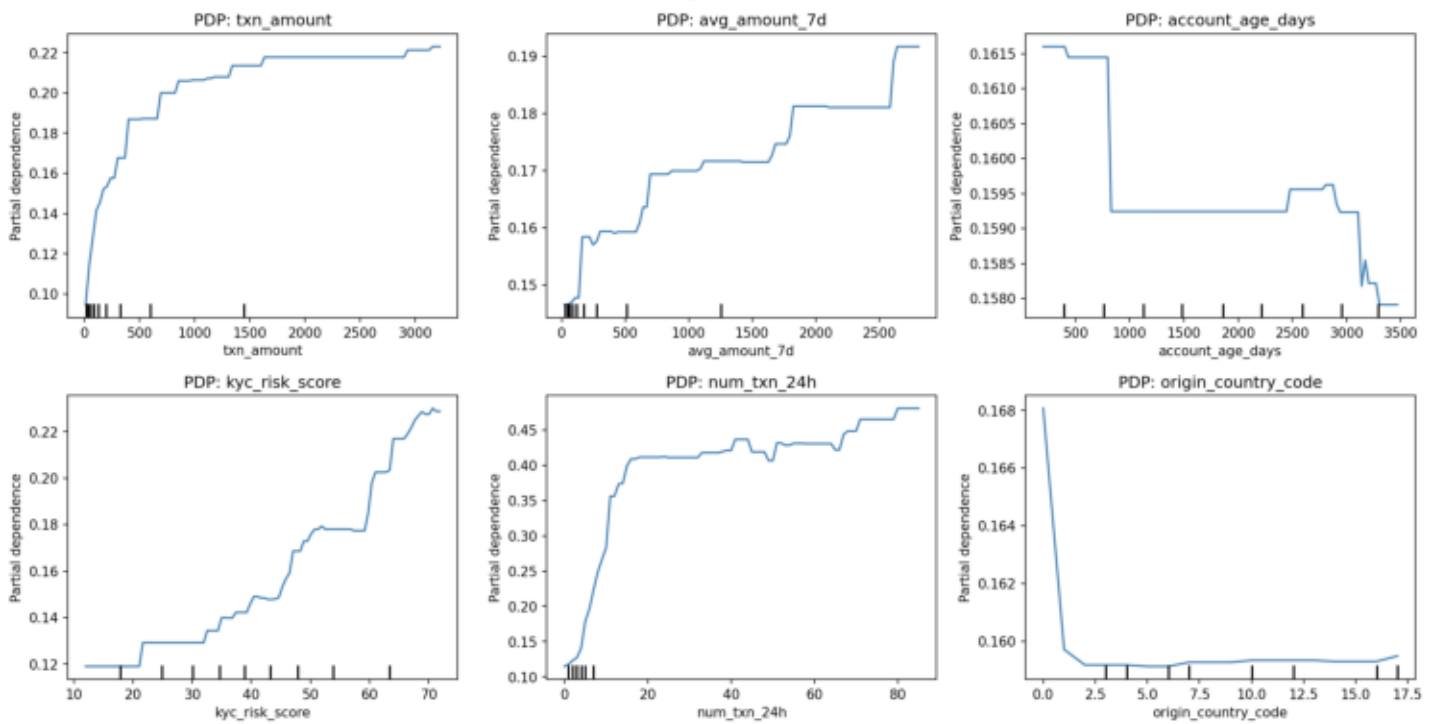
shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.



shap_beeswarm: SHAP beeswarm shows both impact size and direction per feature.

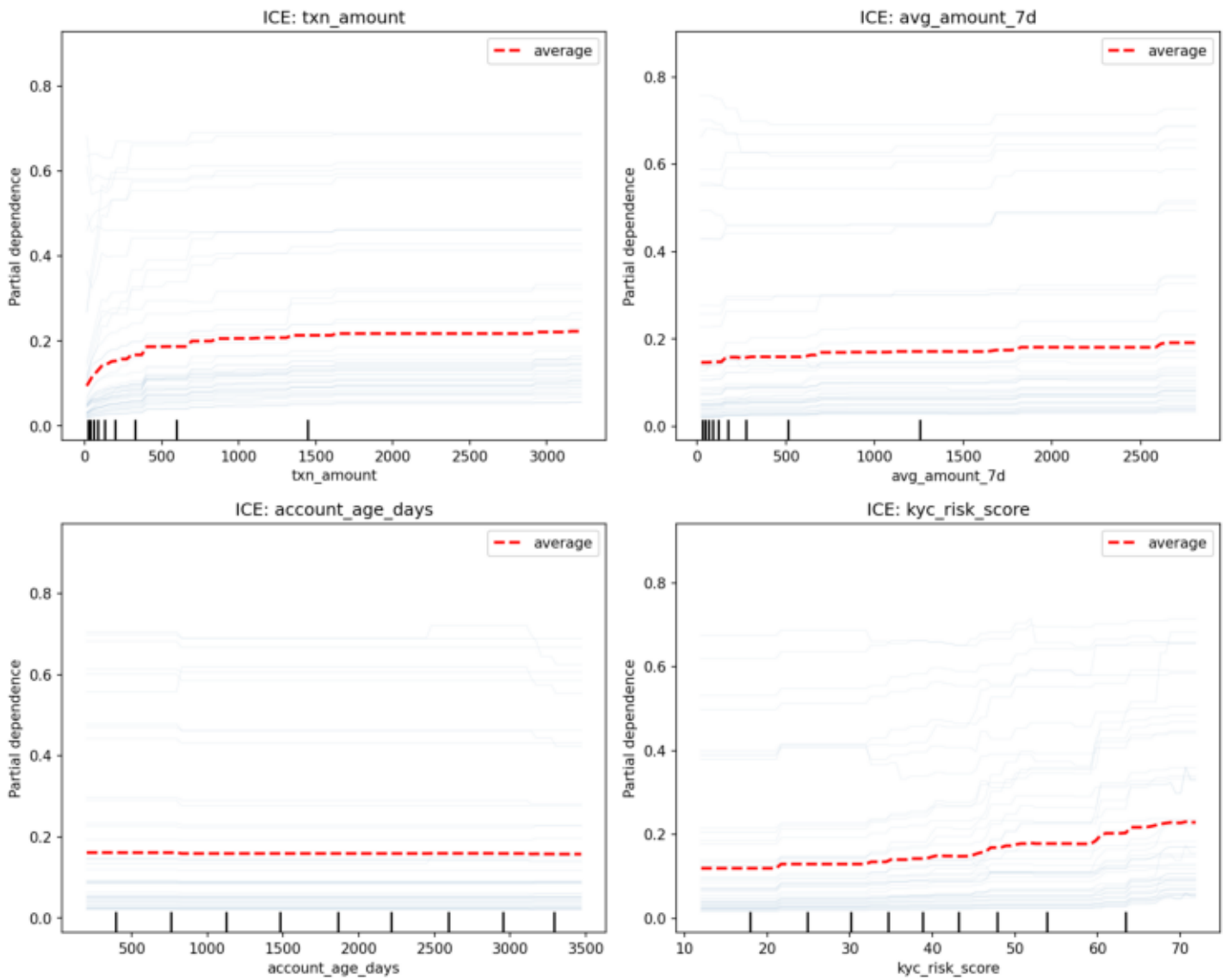


Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



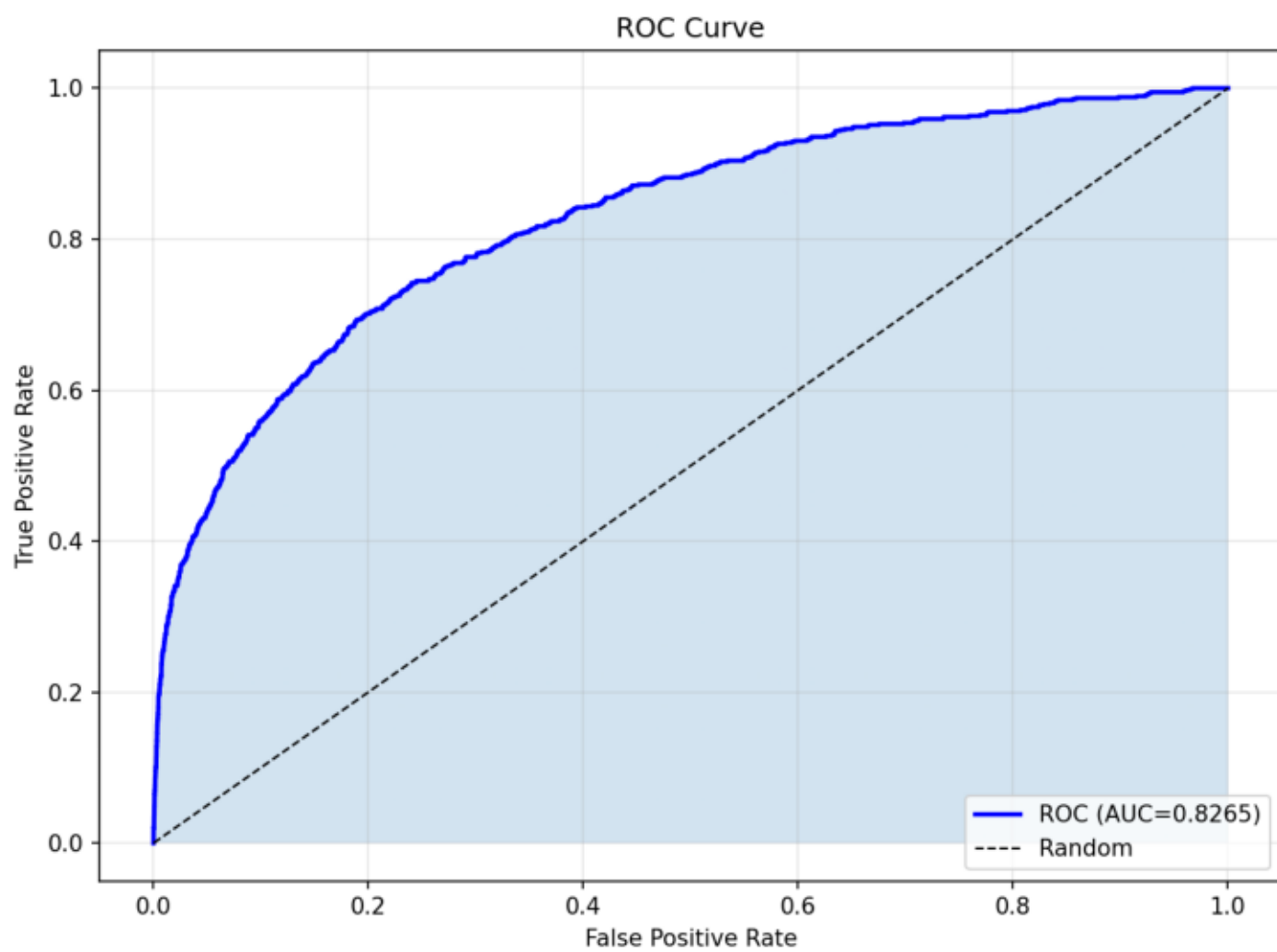
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q1 - 1) Model Effectiveness

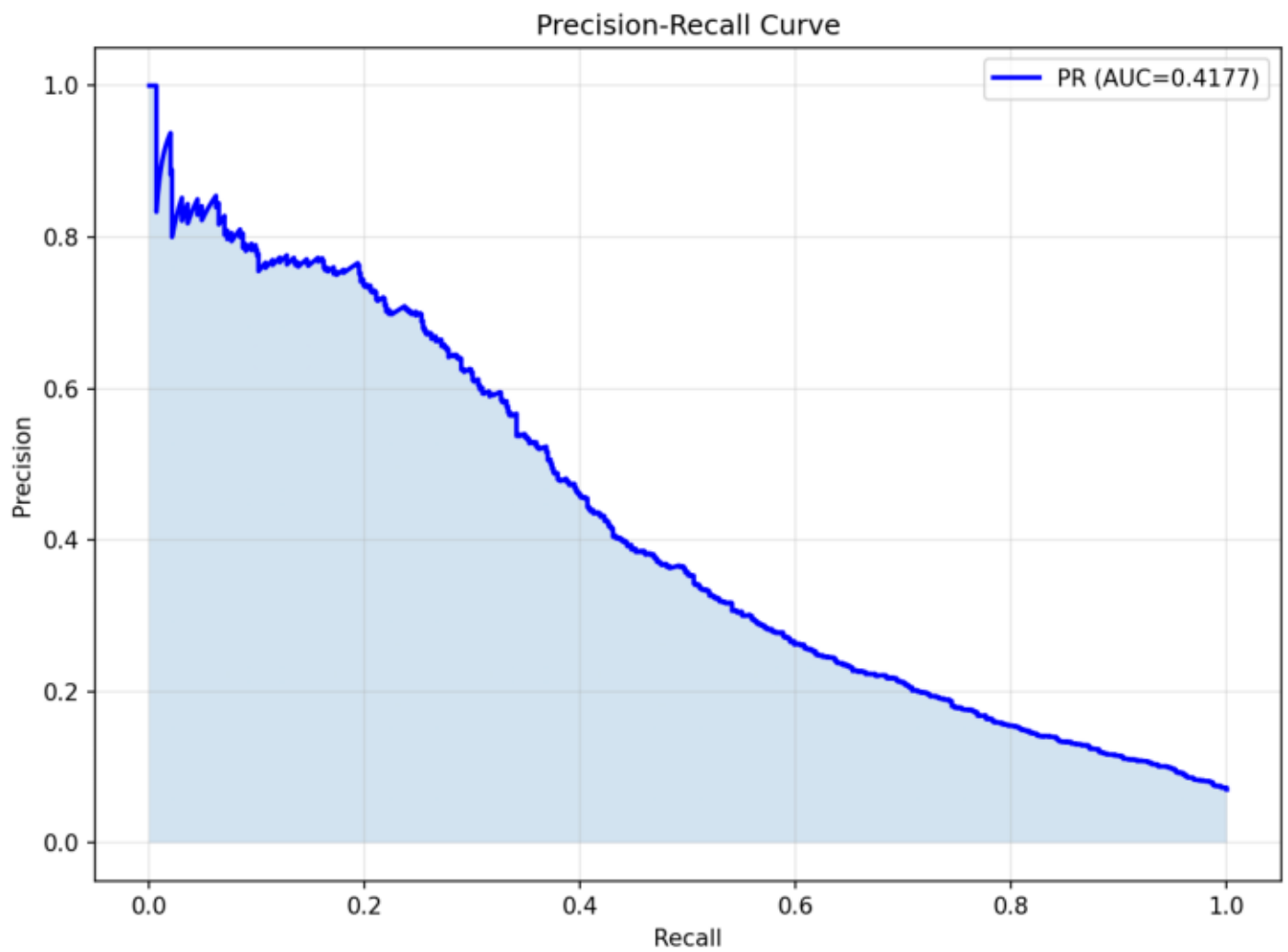
- auc_roc: 0.8265
- auc_pr: 0.4177
- precision: 0.7004
- recall: 0.2276
- f1: 0.3436
- ks_statistic: 0.5010
- ks_threshold: 0.1100
- confusion_matrix.TN: 9967
- confusion_matrix.FP: 74
- confusion_matrix.FN: 587
- confusion_matrix.TP: 173
- precision_at_k.10: 0.9000
- precision_at_k.50: 0.8400
- precision_at_k.100: 0.7700
- precision_at_k.200: 0.7450
- precision_at_k.500: 0.5340
- recall_at_k.10: 0.0118
- recall_at_k.50: 0.0553
- recall_at_k.100: 0.1013
- recall_at_k.200: 0.1961
- recall_at_k.500: 0.3513

value_date_quarter=2024Q1 - 1) Model Effectiveness - Explanations

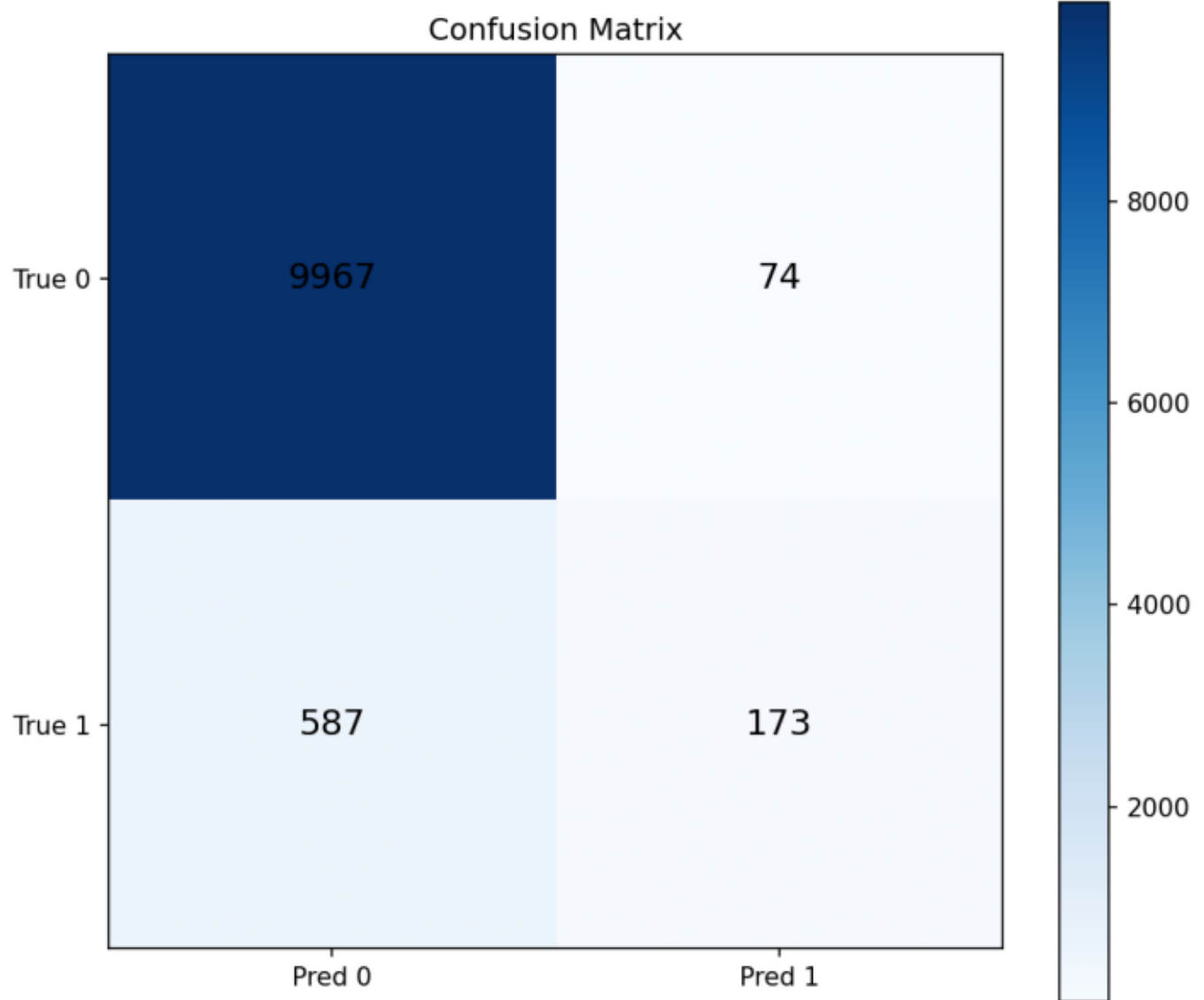
- Positive rate is 7.04%; AUC-ROC is 0.8265 (strong ranking).
- AUC-PR is 0.4177, above the base rate 7.04%, indicating meaningful lift.
- At threshold 0.50, precision=0.7004, recall=0.2276, F1=0.3436 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=173, FP=74, FN=587, TN=9967 (FPR=0.74%, FNR=77.24%).
- KS is 0.5010 at threshold 0.11, indicating very strong separation.
- Top-10: precision=0.9000, recall=0.0118 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.5340, recall=0.3513 shows how much coverage you get with a larger queue.



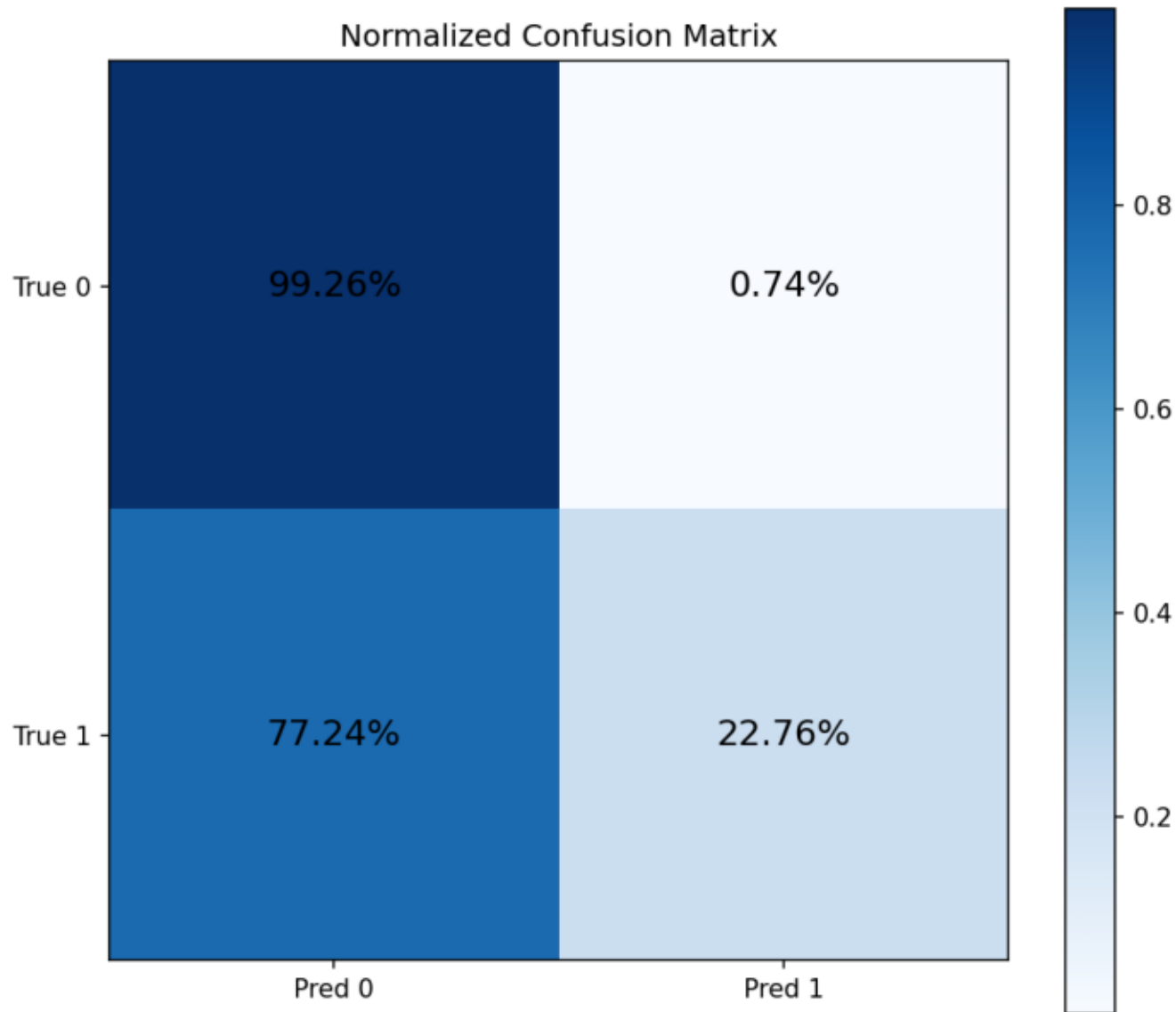
roc_curve: AUC-ROC=0.8265; curve above diagonal indicates strong separation.



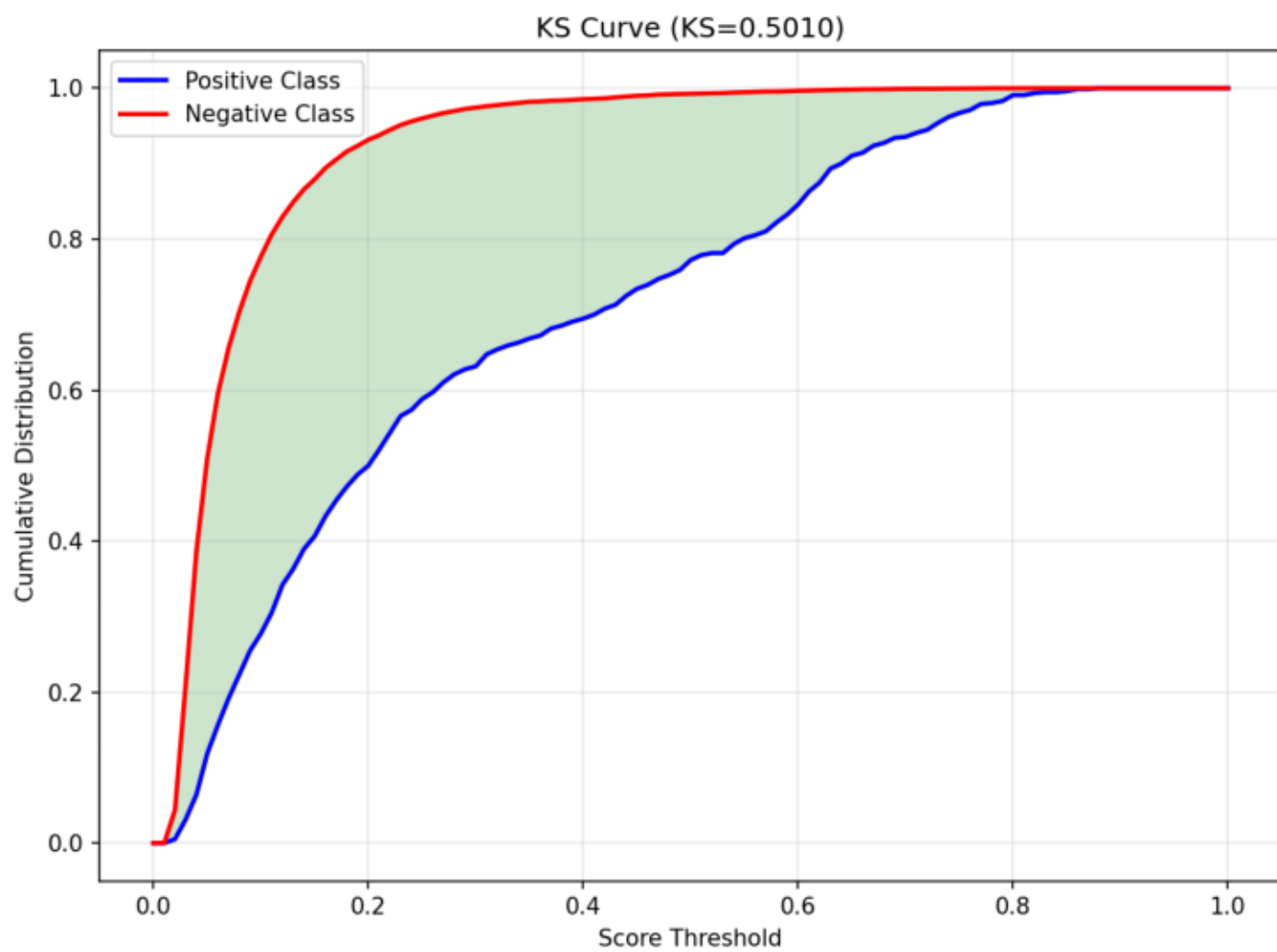
pr_curve: AUC-PR=0.4177 vs baseline 7.04%; higher curve implies better precision at recall.



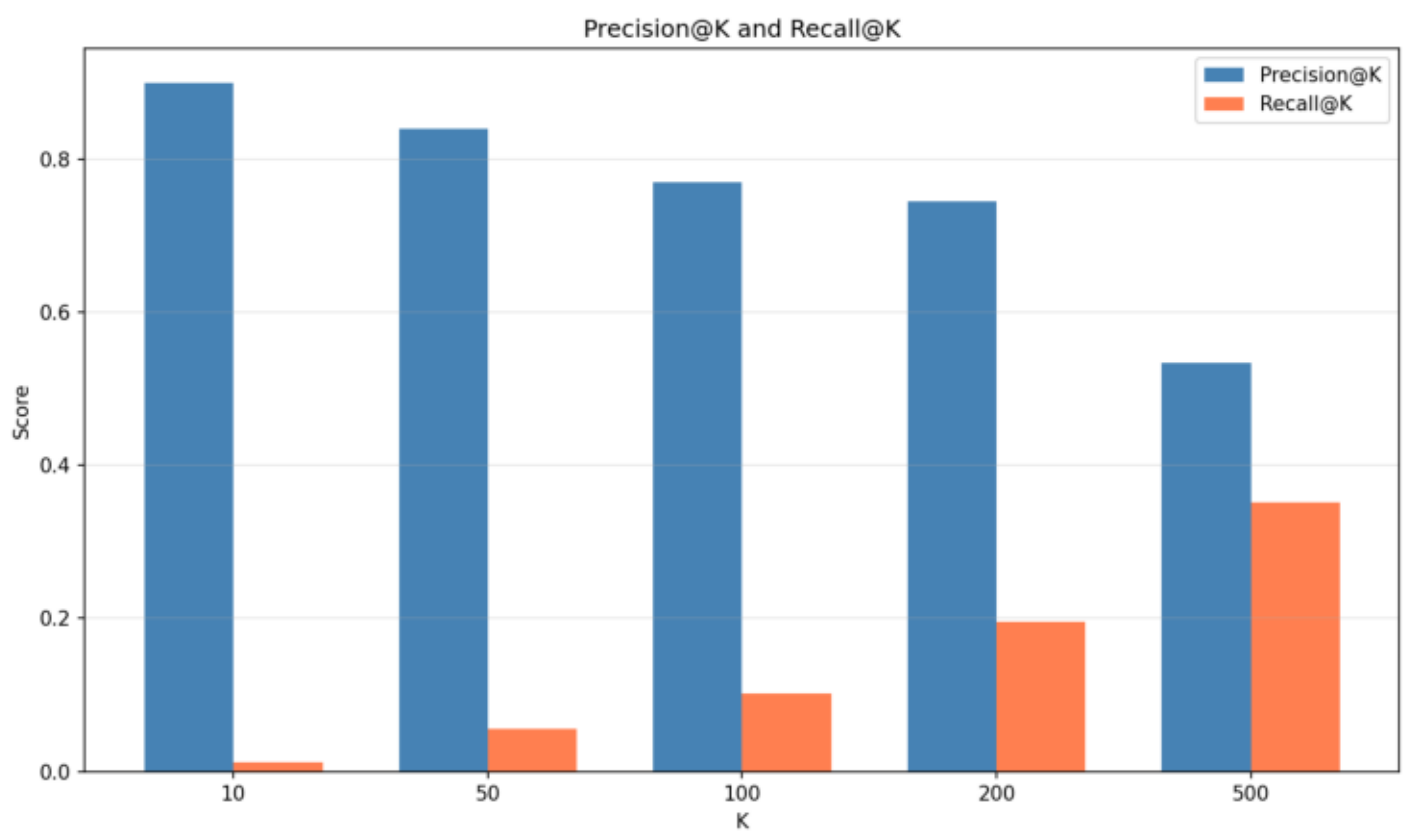
confusion_matrix: At threshold 0.50: TP=173, FP=74, FN=587, TN=9967.



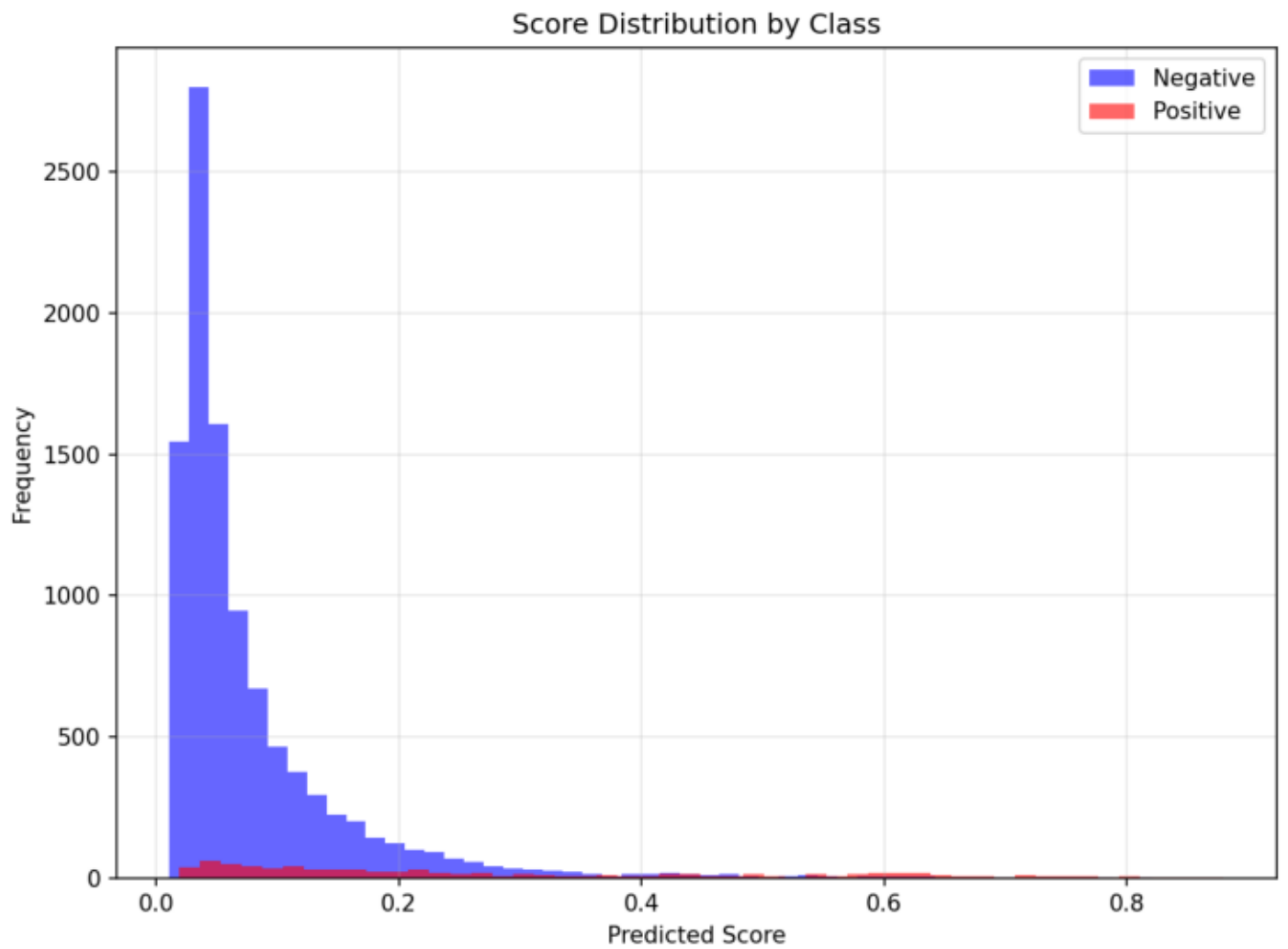
confusion_matrix_norm: Normalized rates: TPR=22.76%, TNR=99.26%.



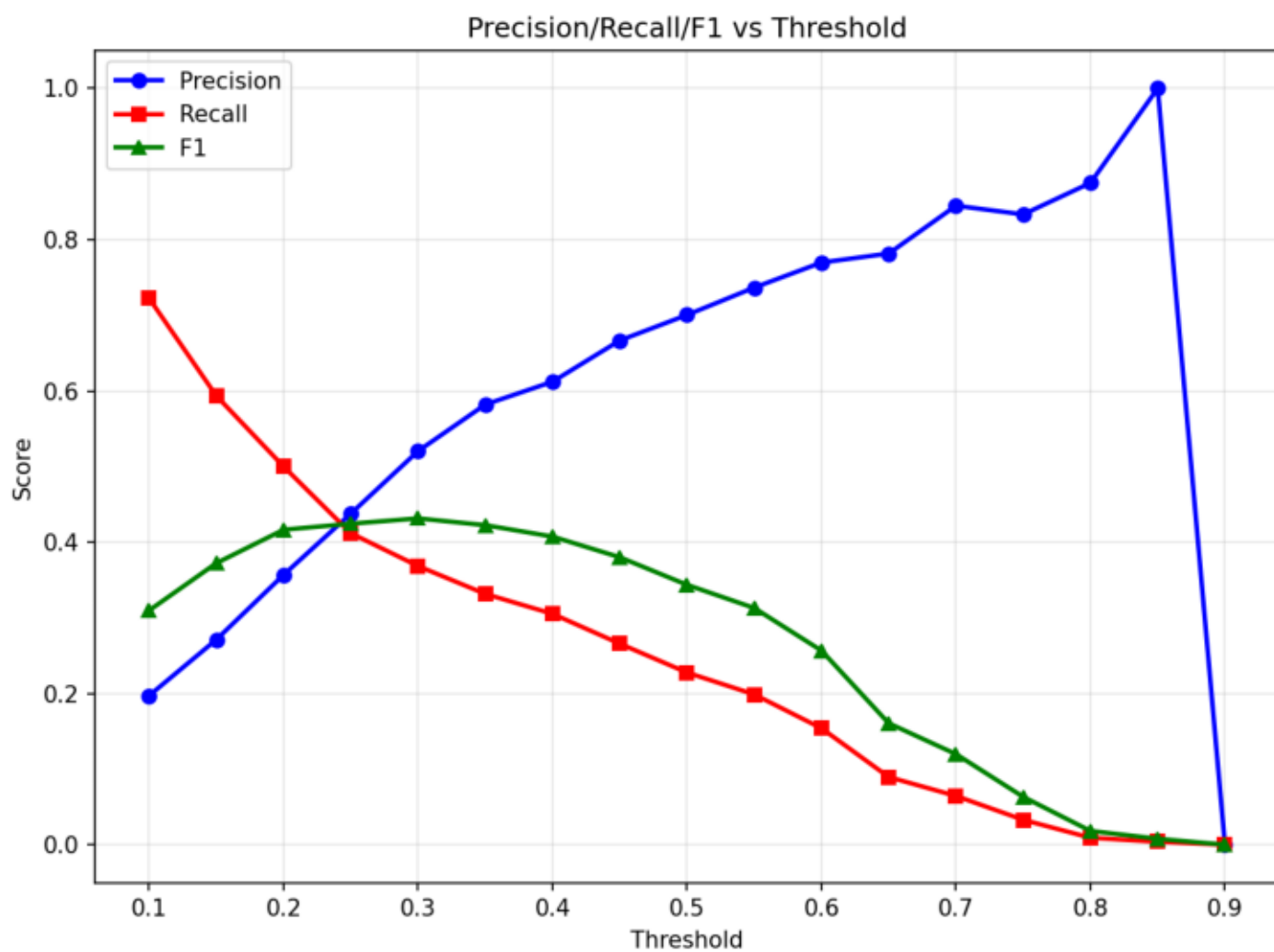
ks_curve: Maximum separation KS=0.5010 at threshold 0.11.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.283, negative=0.078 (larger gap is better).



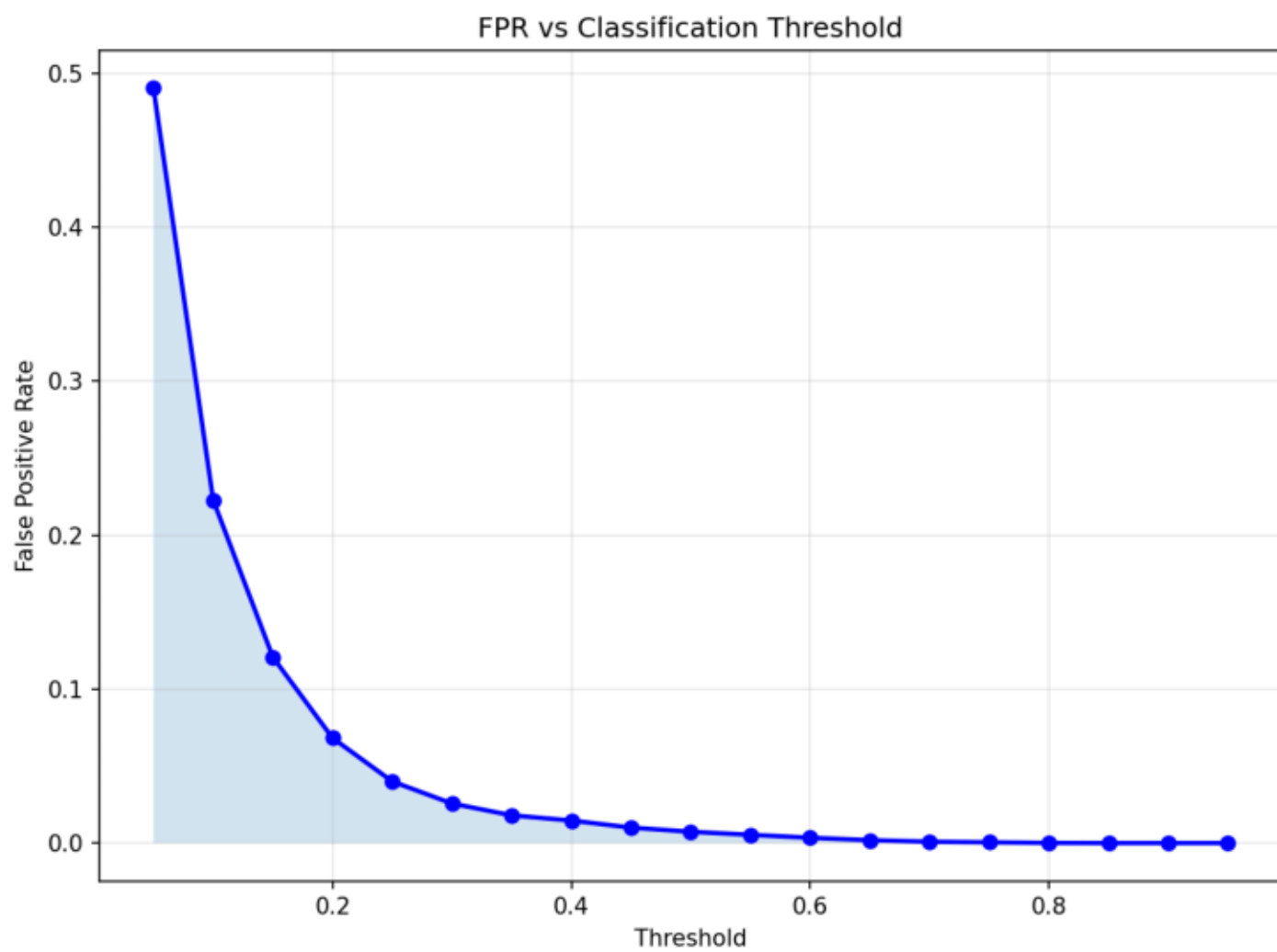
threshold_analysis: Best F1 in tested grid is 0.4314 at threshold 0.30.

value_date_quarter=2024Q1 - 2) Model Efficiency

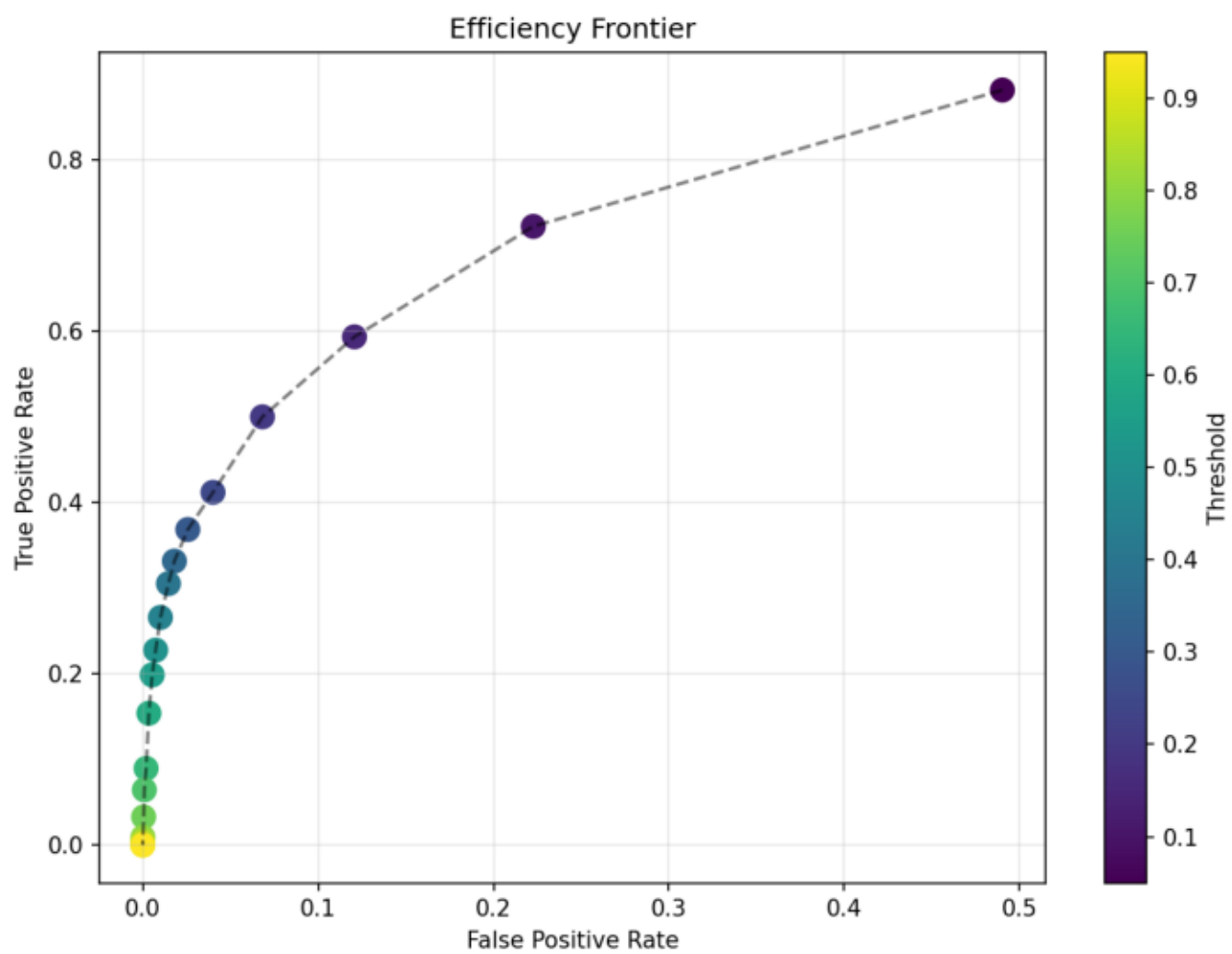
- fpr: 0.0074
- tn: 9967
- fp: 74
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.4902
- fpr_at_thresholds.t_0.10: 0.2227
- fpr_at_thresholds.t_0.15: 0.1207
- fpr_at_thresholds.t_0.20: 0.0682
- fpr_at_thresholds.t_0.25: 0.0400
- fpr_at_thresholds.t_0.30: 0.0257
- fpr_at_thresholds.t_0.35: 0.0180
- fpr_at_thresholds.t_0.40: 0.0146
- fpr_at_thresholds.t_0.45: 0.0101
- fpr_at_thresholds.t_0.50: 0.0074
- fpr_at_thresholds.t_0.55: 0.0054
- fpr_at_thresholds.t_0.60: 0.0035
- fpr_at_thresholds.t_0.65: 0.0019
- fpr_at_thresholds.t_0.70: 0.0009
- fpr_at_thresholds.t_0.75: 0.0005
- fpr_at_thresholds.t_0.80: 0.0001
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q1 - 2) Model Efficiency - Explanations

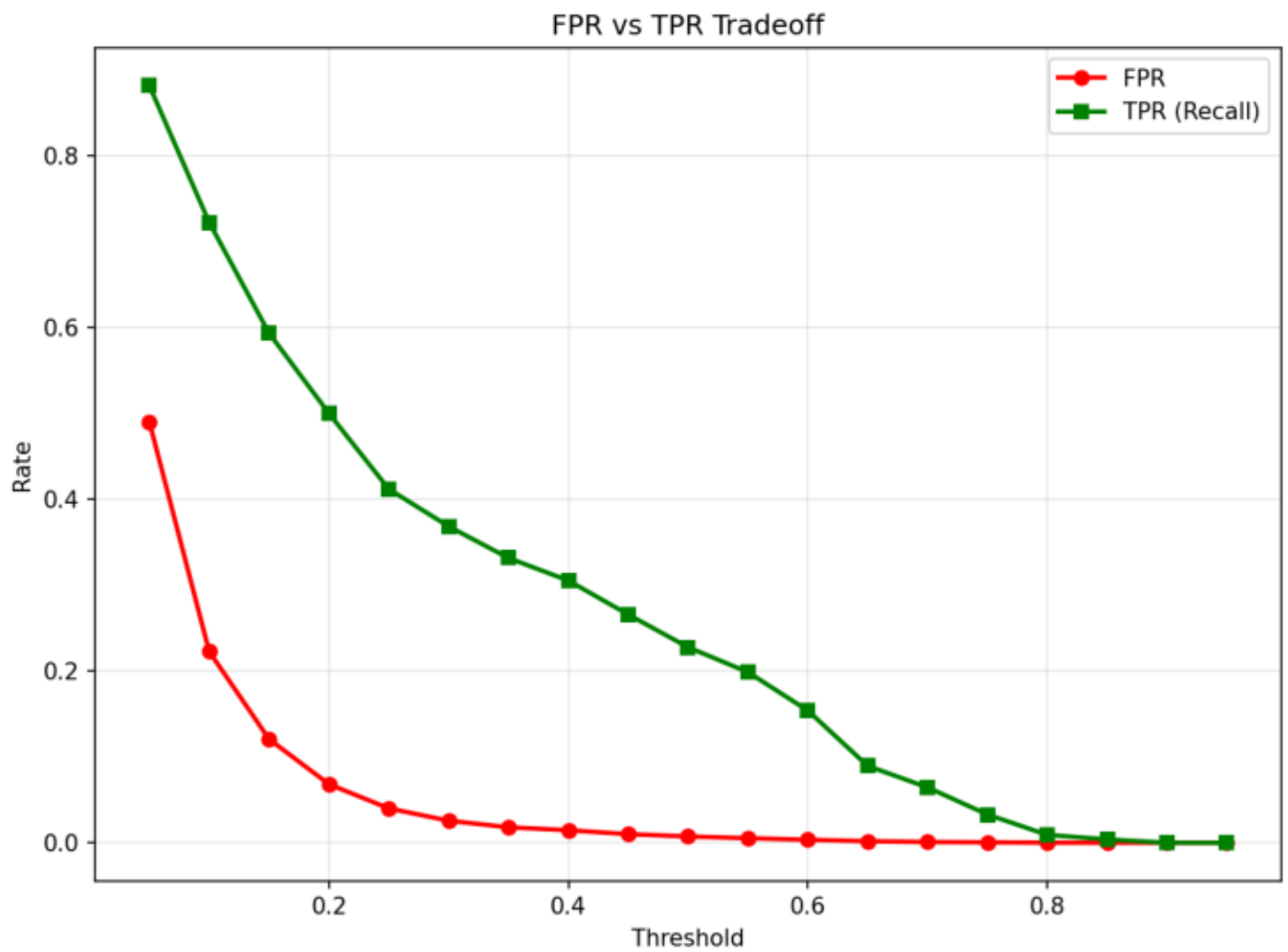
- At threshold 0.50, FPR=0.0074 (low); FP=74 out of 10041 negatives.
- At the same threshold, TPR=0.2276 with TP=173 and FN=587, showing the capture rate of positives.
- A threshold near 0.25 yields FPR $\approx$ 0.0400 with TPR $\approx$ 0.4118 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

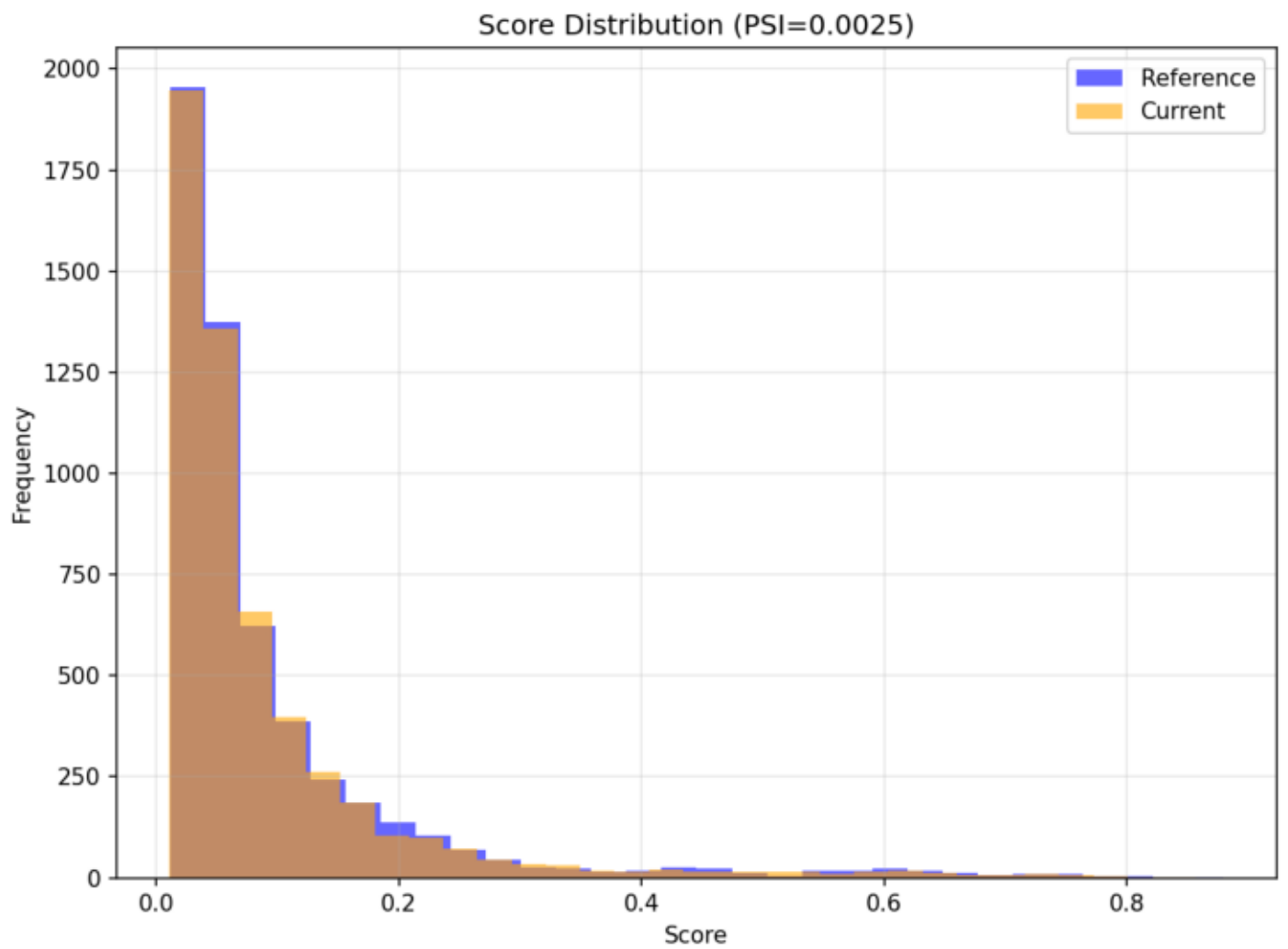


value_date_quarter=2024Q1 - 3) Model Stability

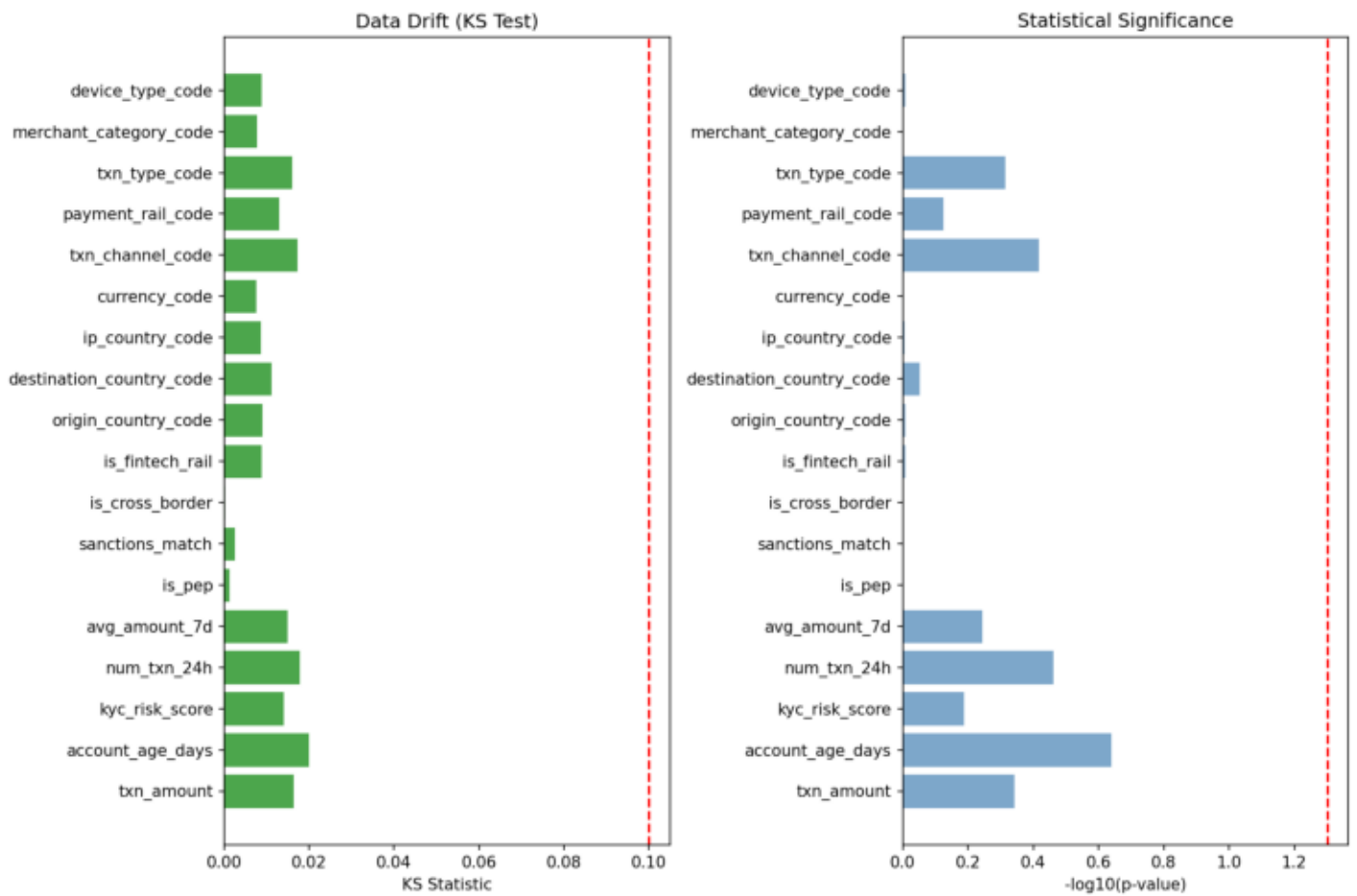
- psi: 0.0025
- cv_auc_roc_mean: 0.8262
- cv_auc_roc_std: 0.0077
- cv_auc_pr_mean: 0.4196
- cv_auc_pr_std: 0.0141
- bootstrap_auc_roc_mean: 0.8177
- bootstrap_auc_roc_ci_lower: 0.7946
- bootstrap_auc_roc_ci_upper: 0.8393
- concept_drift_detected: False
- concept_drift_score: 0.0308

value_date_quarter=2024Q1 - 3) Model Stability - Explanations

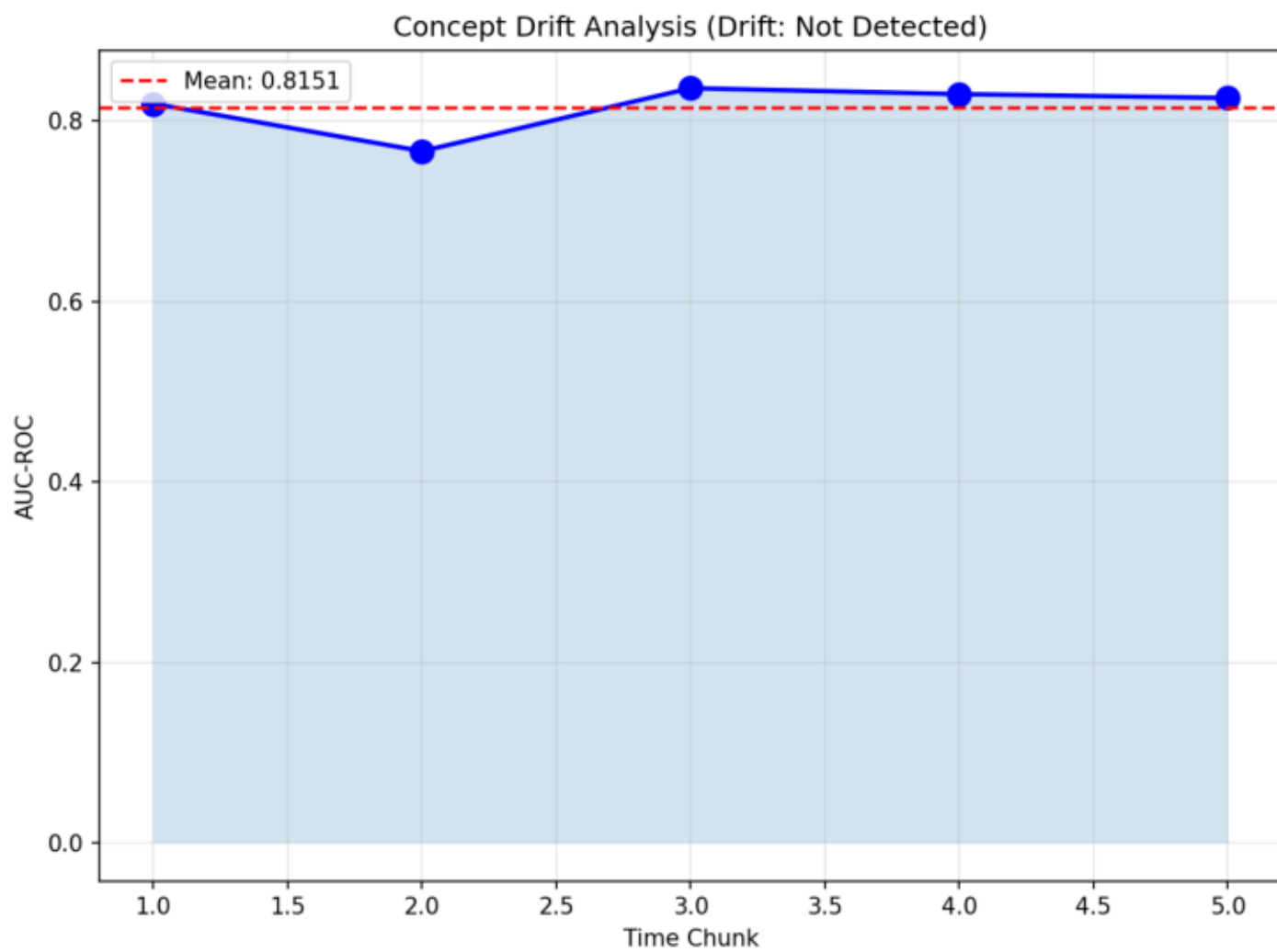
- PSI is 0.0025, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8262 ± 0.0077 (stable across folds).
- CV AUC-PR is 0.4196 ± 0.0141 (moderate across folds).
- Bootstrap AUC-ROC mean is 0.8177 with 95% CI [0.7946, 0.8393] (width 0.0446).
- No concept drift detected (score 0.0308), suggesting stable performance over time.
- Data drift flagged 0/18 numeric features; top drift signals: ['account_age_days', 'num_txn_24h', 'txn_channel_code'].



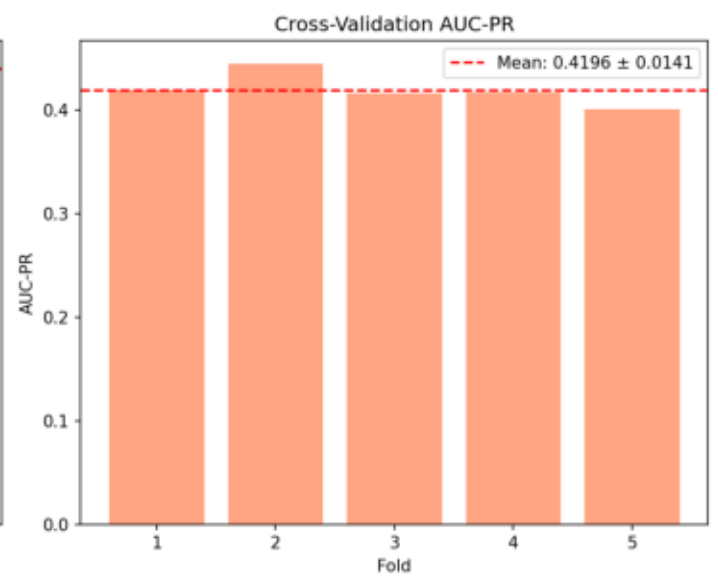
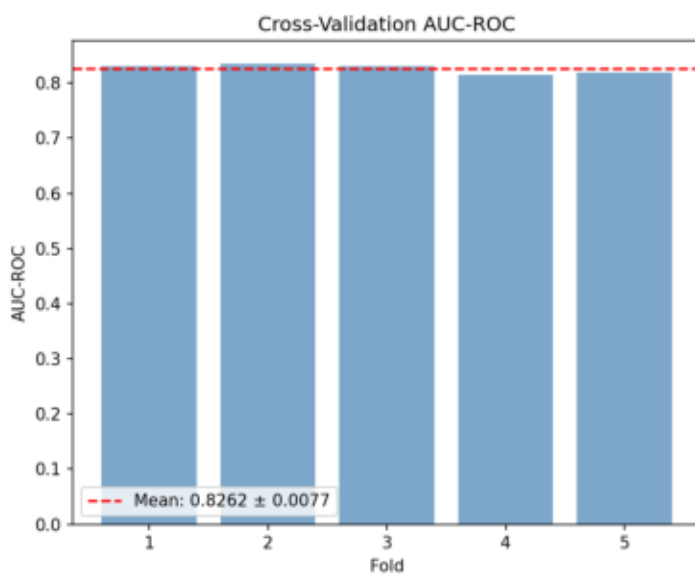
psi_distribution: Score distributions overlap consistent with PSI=0.0025.



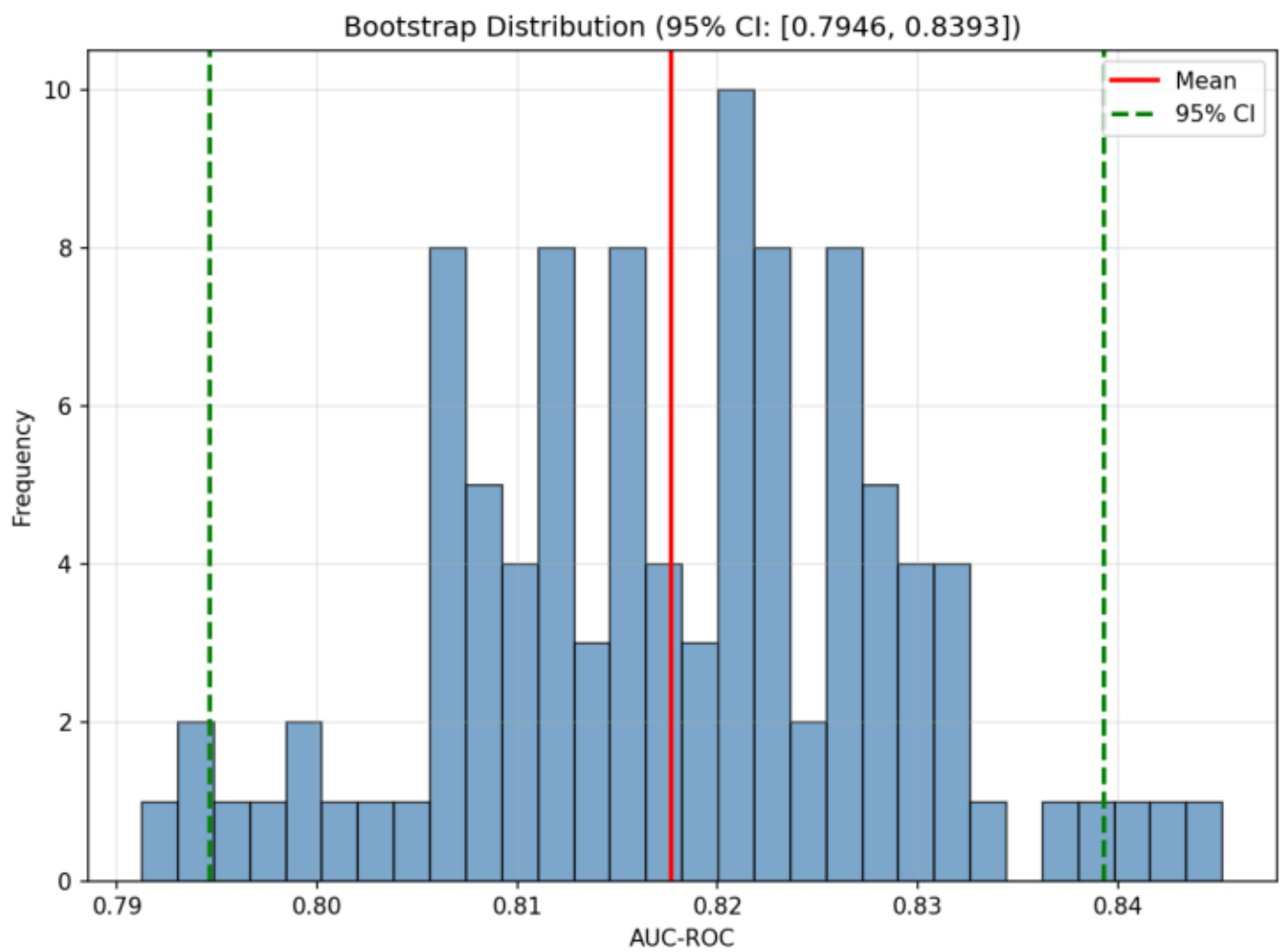
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.



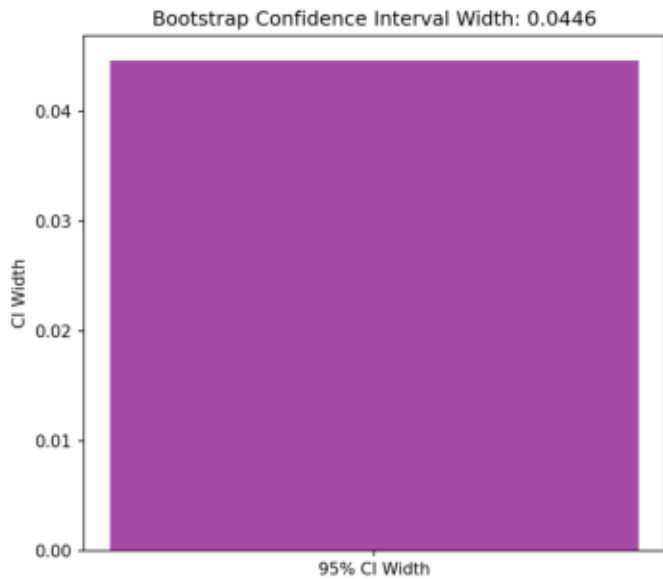
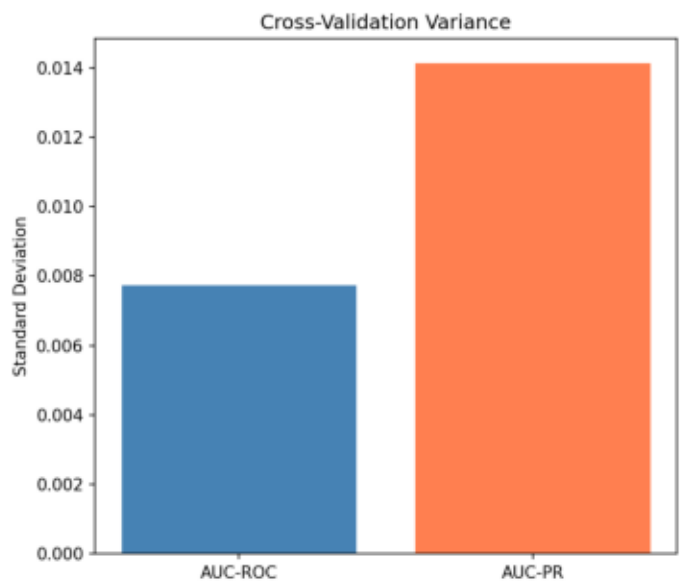
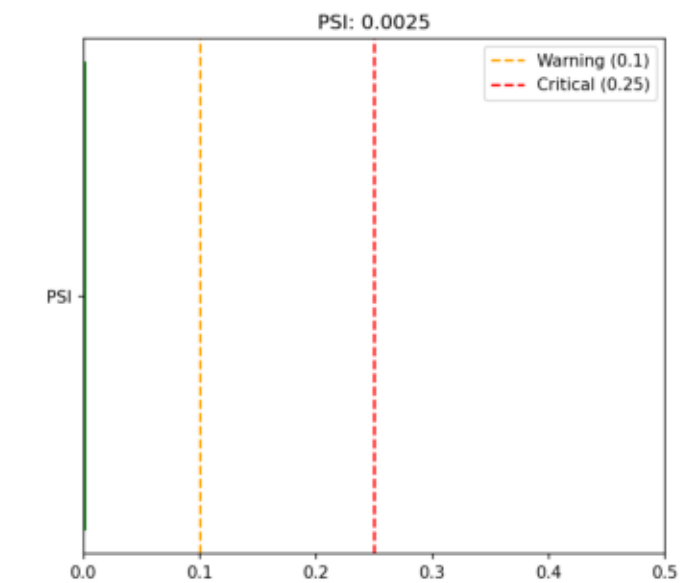
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8262 and PR 0.4196.



bootstrap_distribution: Bootstrap CI [0.7946, 0.8393] reflects performance stability.



STABILITY SUMMARY

PSI: 0.0025 (OK)

CV AUC-ROC: 0.8262 ± 0.0077

CV AUC-PR: 0.4196 ± 0.0141

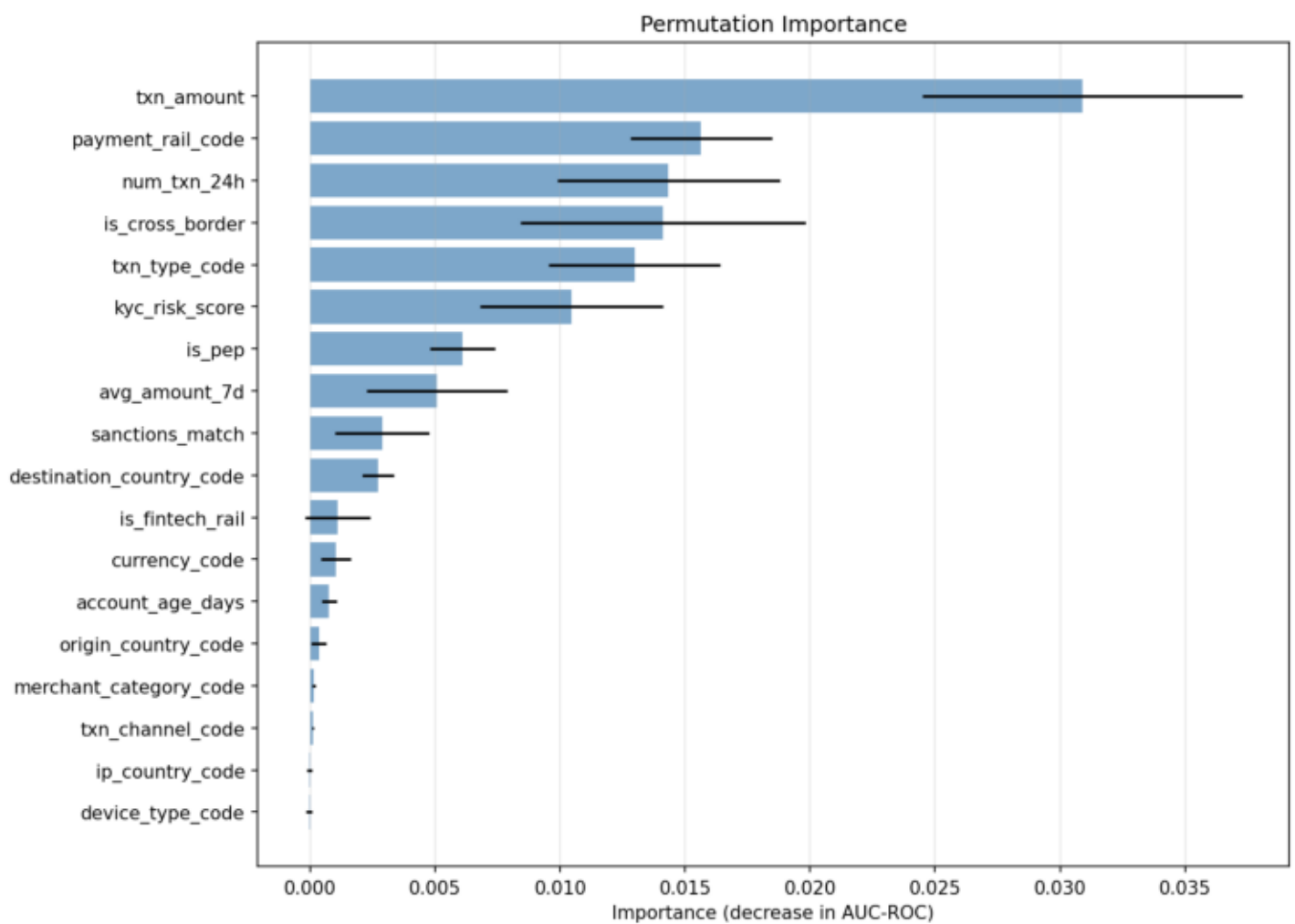
Bootstrap 95% CI: [0.7946, 0.8393]

value_date_quarter=2024Q1 - 4) Model Interpretability

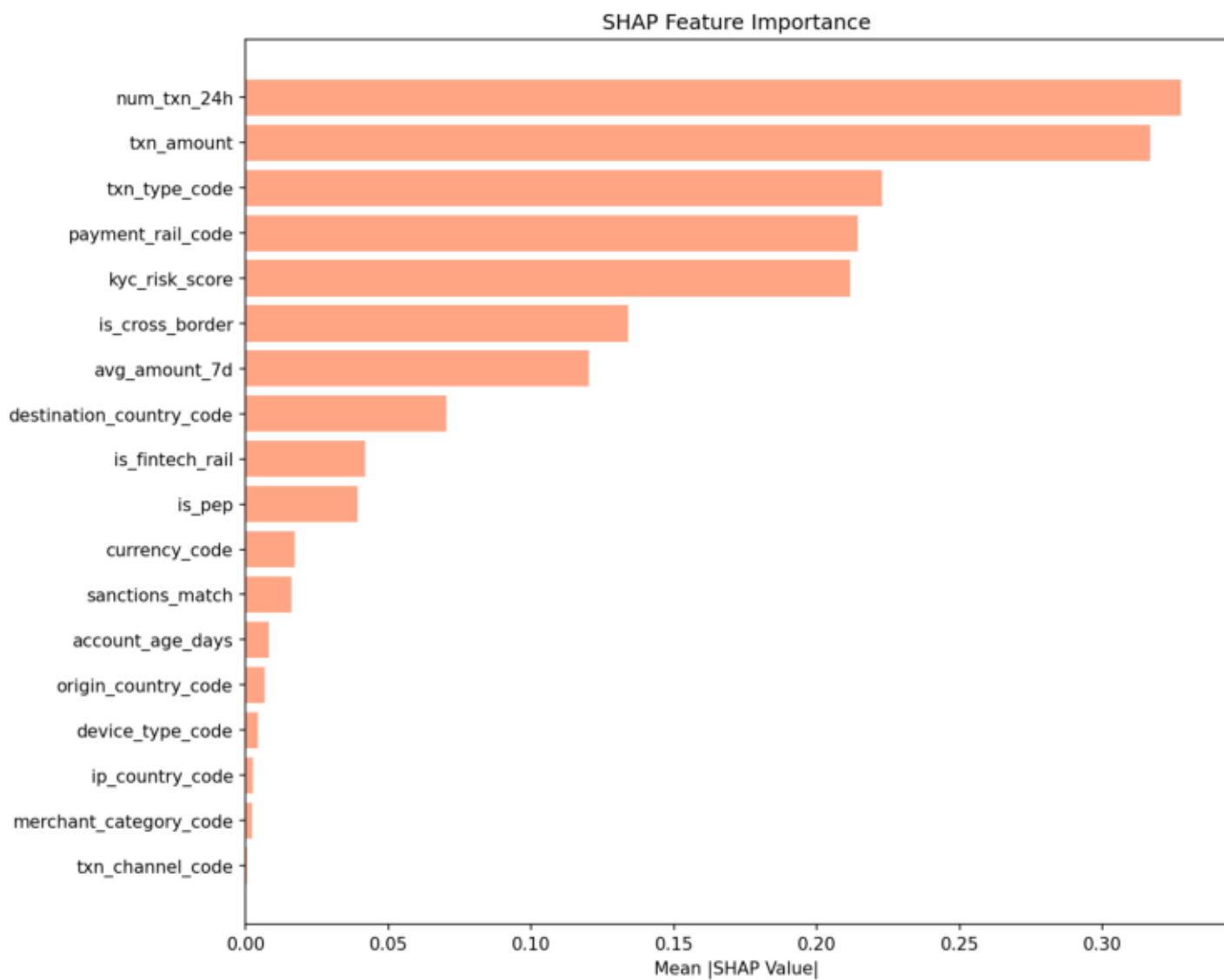
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'shap', 'lime']
- perm_top_features: ['txn_amount', 'payment_rail_code', 'num_txn_24h', 'is_cross_border', 'txn_type_code']...
- shap_top_features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'payment_rail_code', 'kyc_risk_score']...
- lime_instances: 3
- pdp_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q1 - 4) Model Interpretability - Explanations

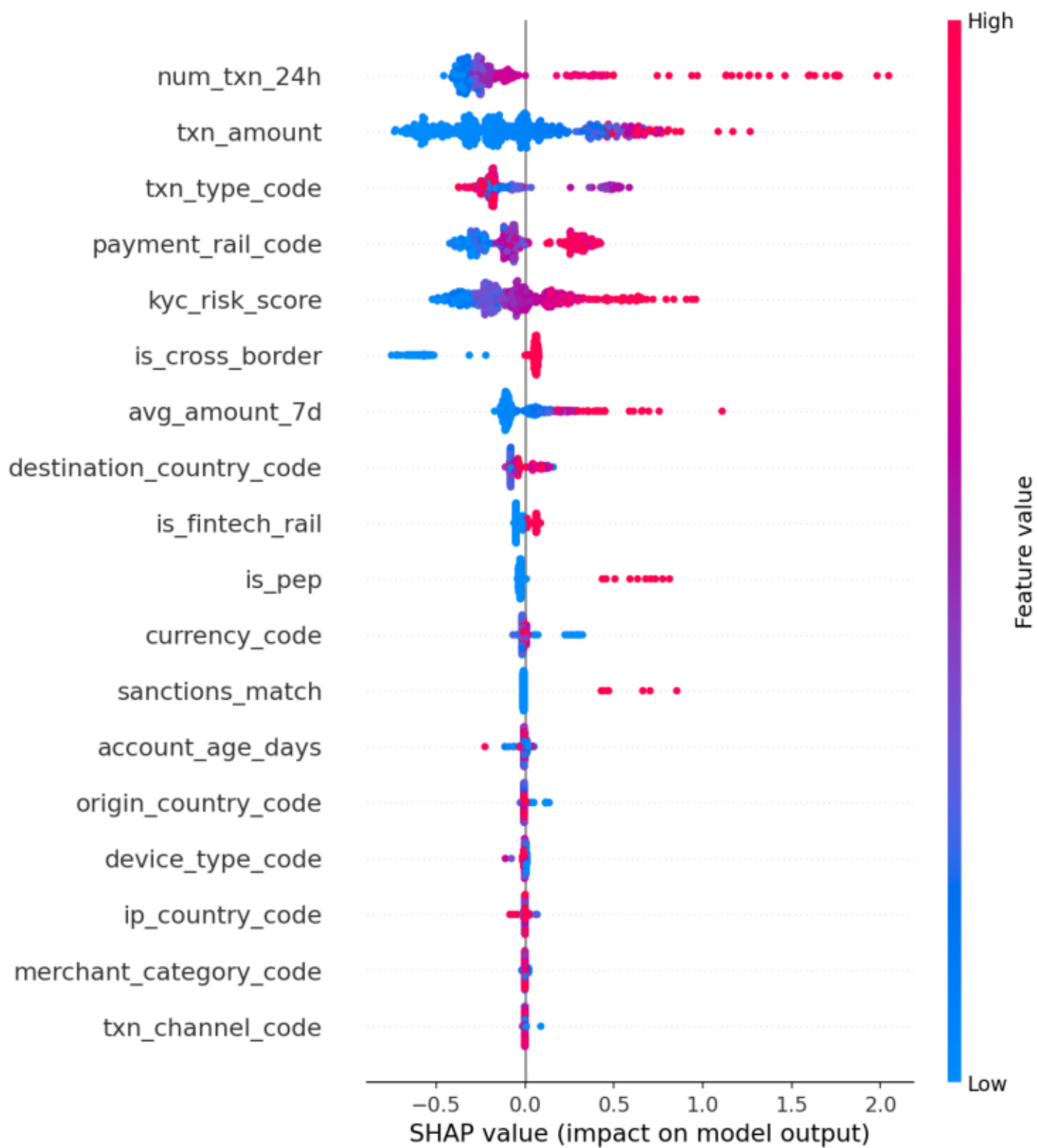
- Permutation importance highlights: ['txn_amount', 'payment_rail_code', 'num_txn_24h', 'is_cross_border', 'txn_type_code'].
- SHAP top features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'payment_rail_code', 'kyc_risk_score'].
- LIME generated local explanations for 3 instances; top contributors for instance_0: ['sanctions_match <= 0.00', 'is_pep <= 0.00', 'num_txn_24h <= 1.00', 'kyc_risk_score <= 25.47', '5.00 < txn_type_code <= 7.00'].
- PDP plots show average effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'ip_country_code'].
- ICE plots show individual-level effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score'].

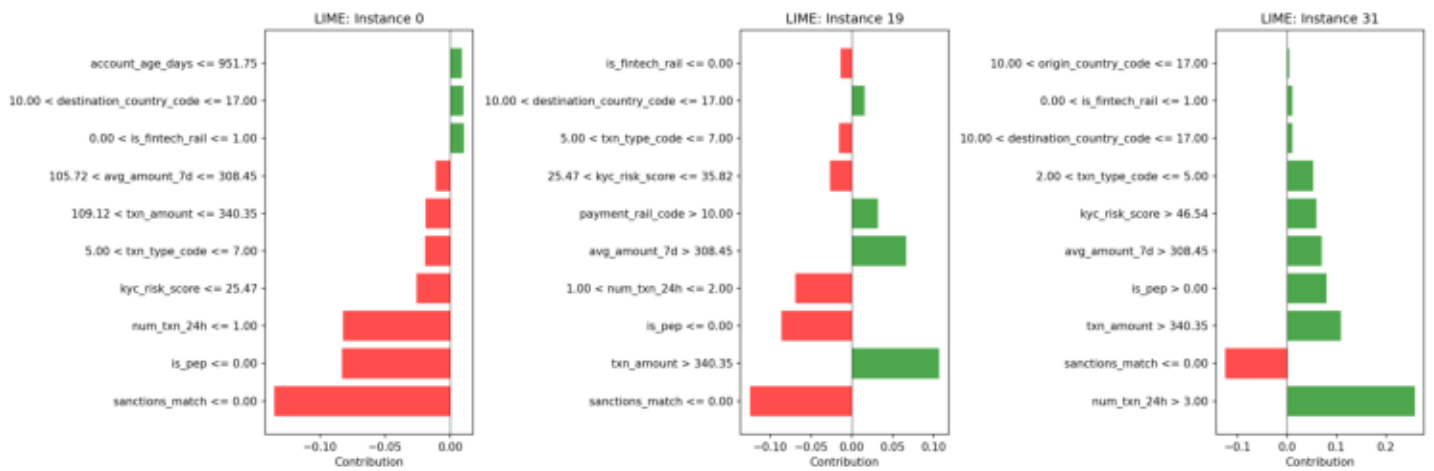


permutation_importance: Bars show how much AUC drops when each feature is permuted.



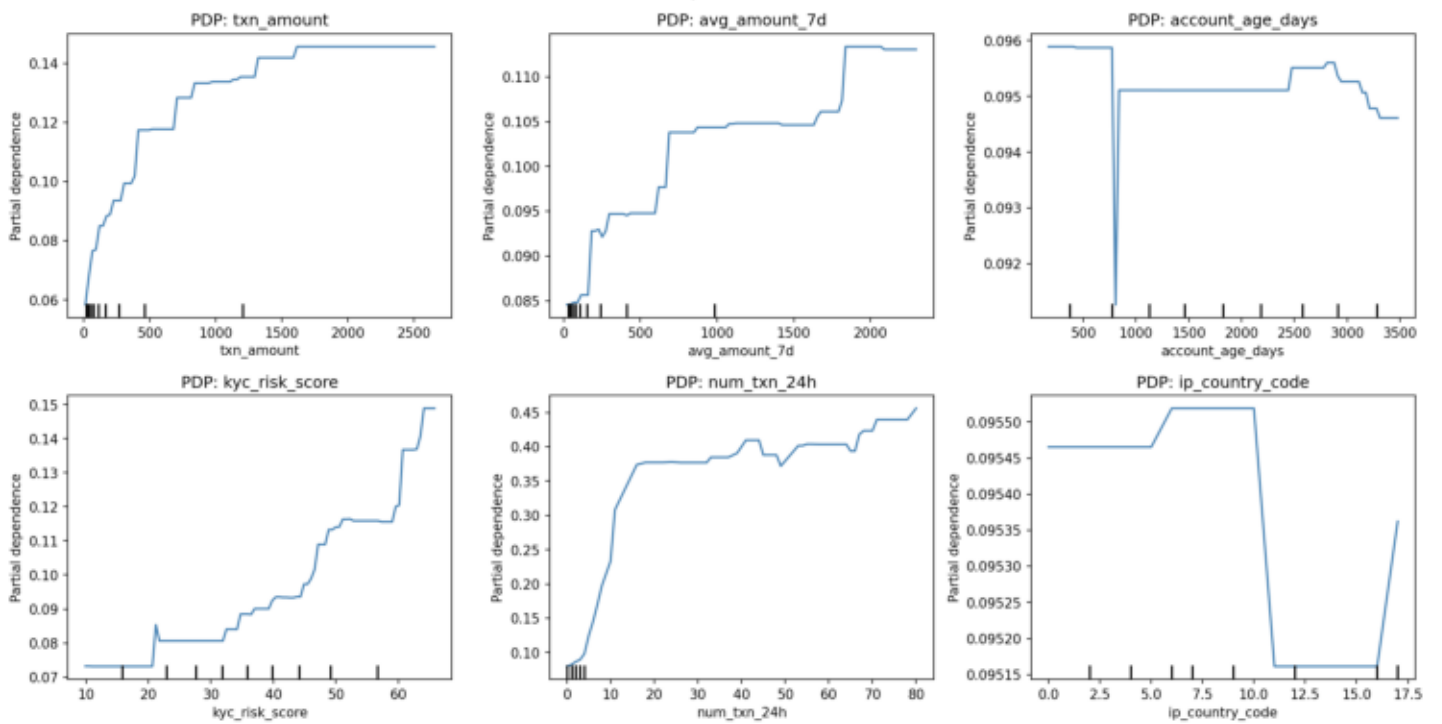
shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.





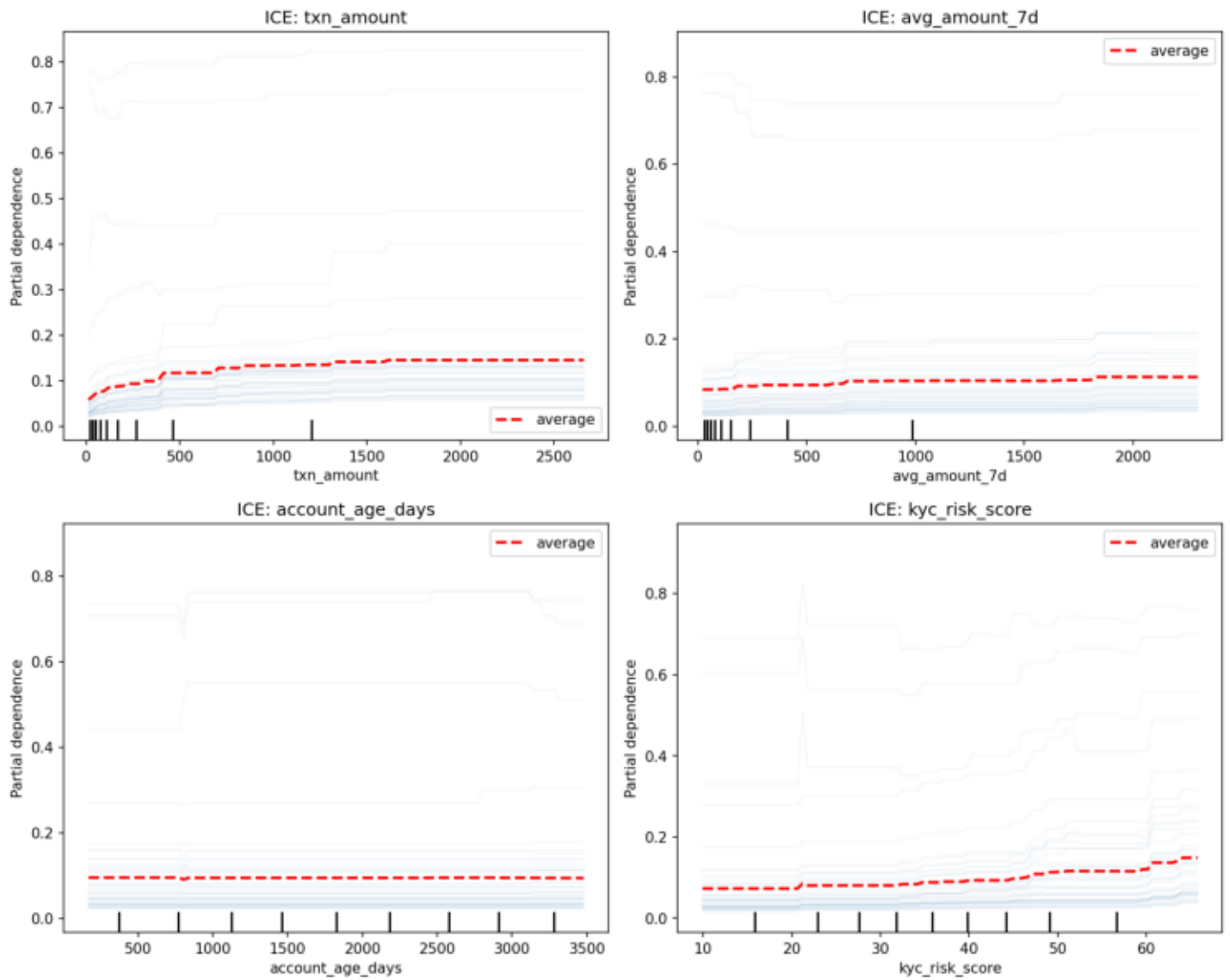
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



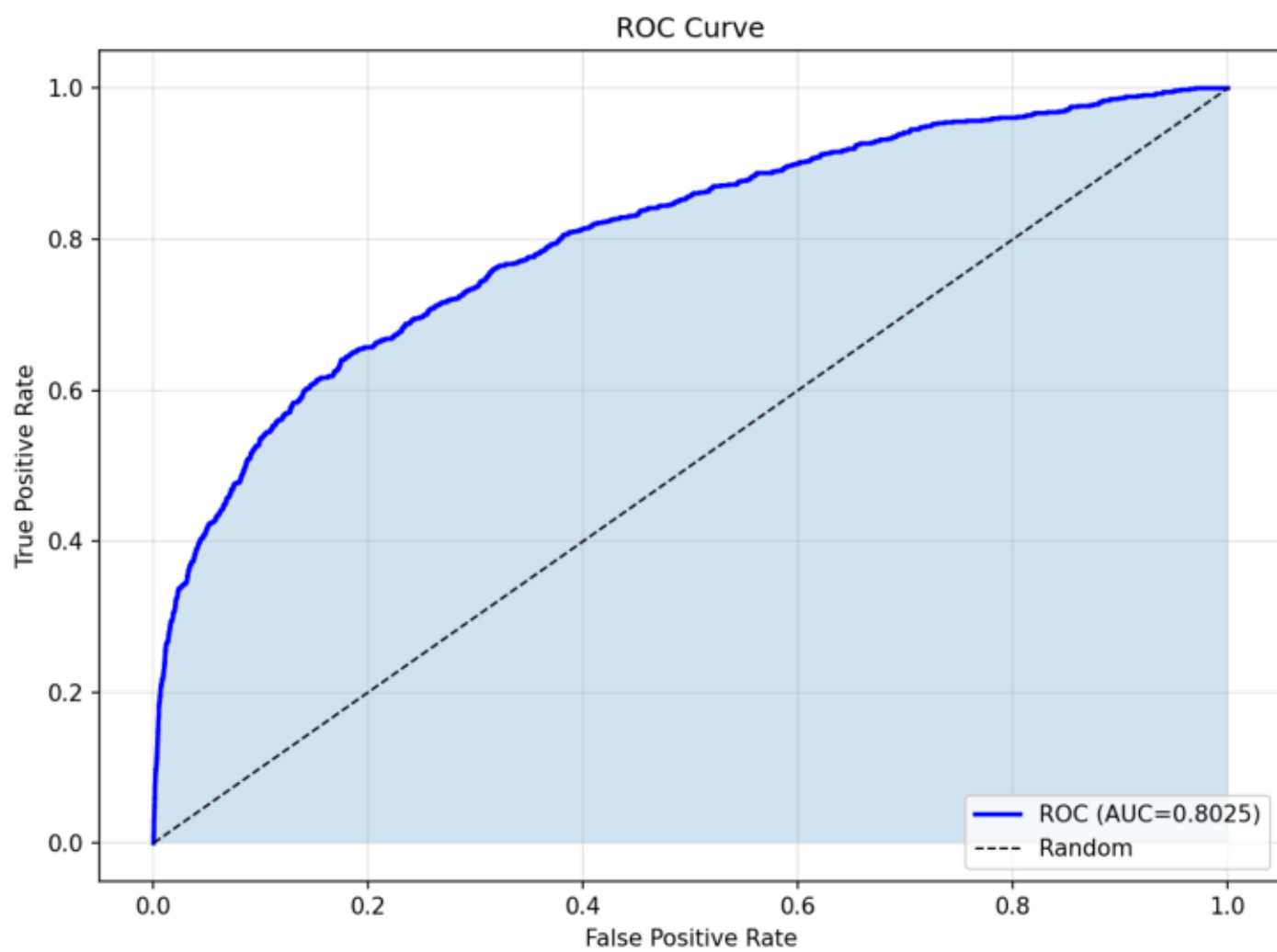
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q2 - 1) Model Effectiveness

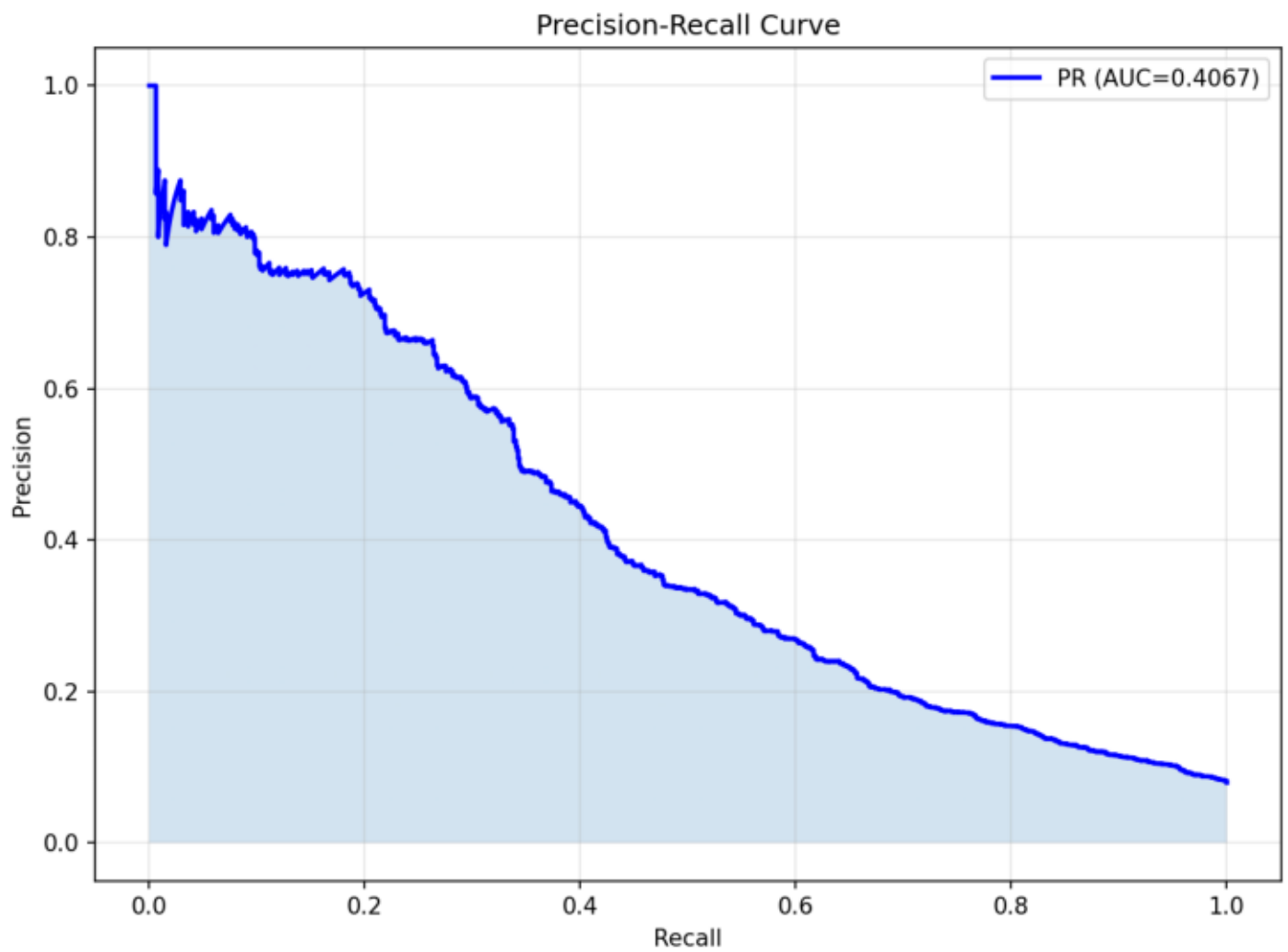
- auc_roc: 0.8025
- auc_pr: 0.4067
- precision: 0.6795
- recall: 0.2186
- f1: 0.3307
- ks_statistic: 0.4591
- ks_threshold: 0.1200
- confusion_matrix.TN: 11136
- confusion_matrix.FP: 100
- confusion_matrix.FN: 758
- confusion_matrix.TP: 212
- precision_at_k.10: 0.8000
- precision_at_k.50: 0.8200
- precision_at_k.100: 0.8100
- precision_at_k.200: 0.7500
- precision_at_k.500: 0.5880
- recall_at_k.10: 0.0082
- recall_at_k.50: 0.0423
- recall_at_k.100: 0.0835
- recall_at_k.200: 0.1546
- recall_at_k.500: 0.3031

value_date_quarter=2024Q2 - 1) Model Effectiveness - Explanations

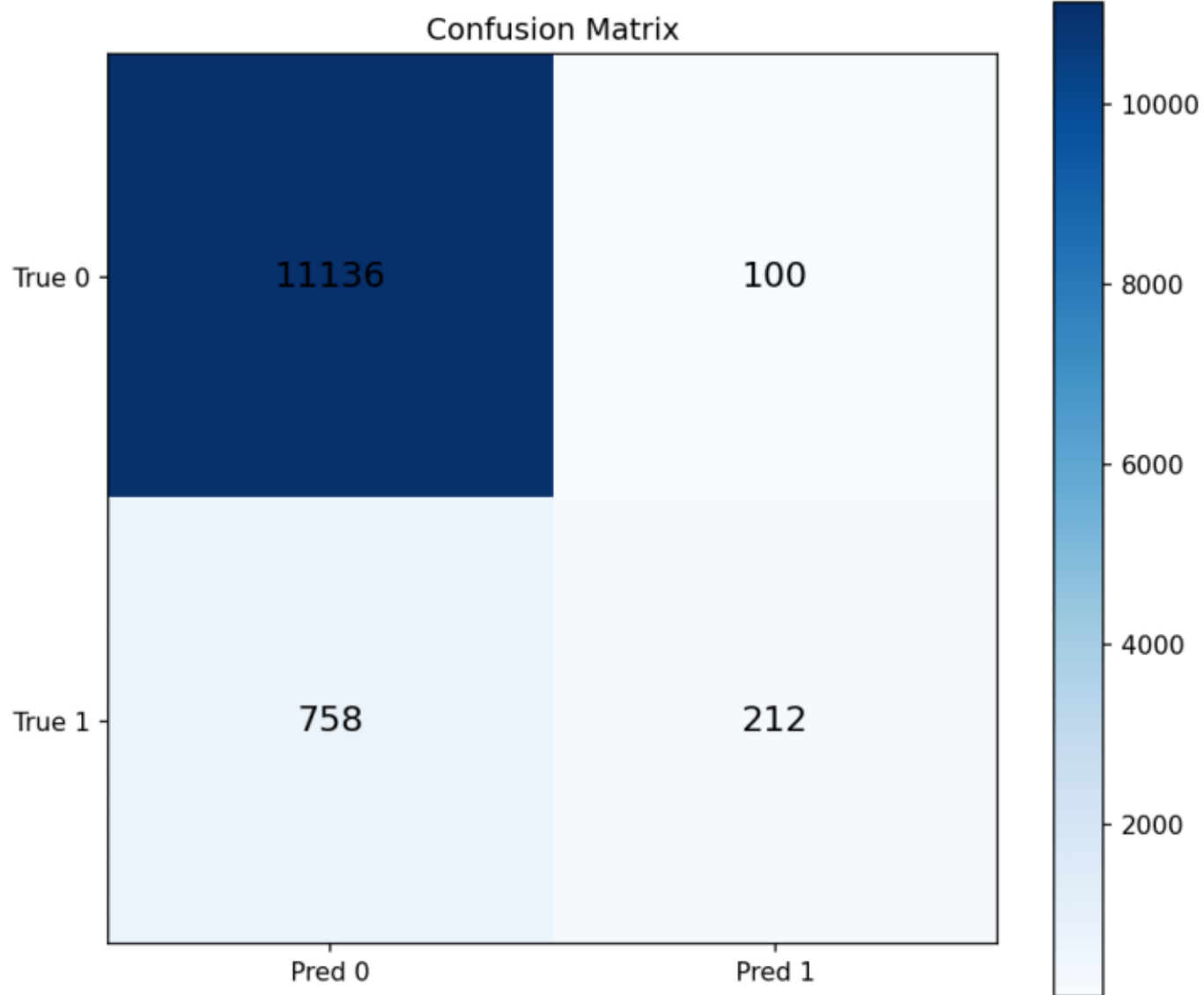
- Positive rate is 7.95%; AUC-ROC is 0.8025 (strong ranking).
- AUC-PR is 0.4067, above the base rate 7.95%, indicating meaningful lift.
- At threshold 0.50, precision=0.6795, recall=0.2186, F1=0.3307 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=212, FP=100, FN=758, TN=11136 (FPR=0.89%, FNR=78.14%).
- KS is 0.4591 at threshold 0.12, indicating strong separation.
- Top-10: precision=0.8000, recall=0.0082 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.5880, recall=0.3031 shows how much coverage you get with a larger queue.



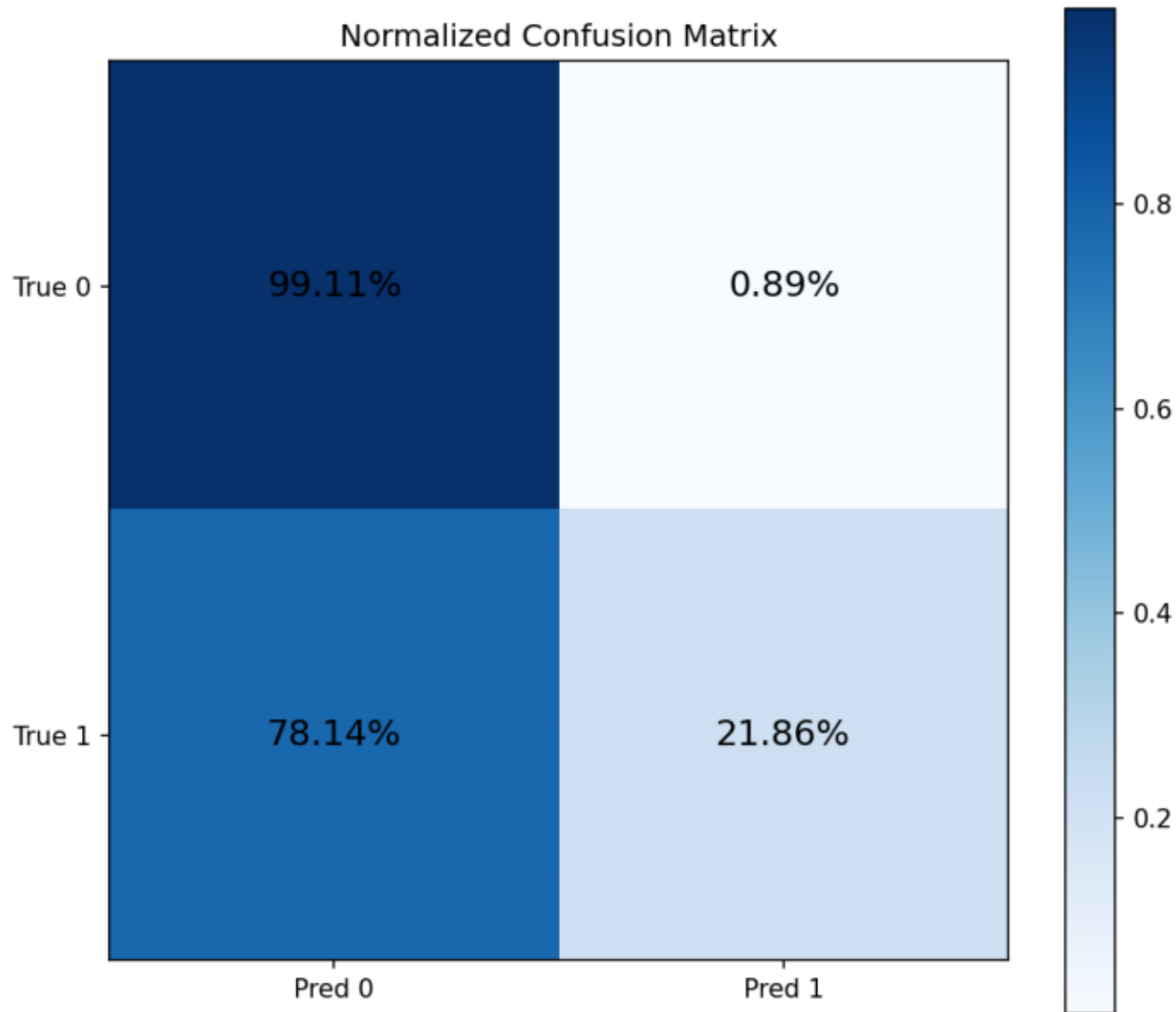
roc_curve: AUC-ROC=0.8025; curve above diagonal indicates strong separation.



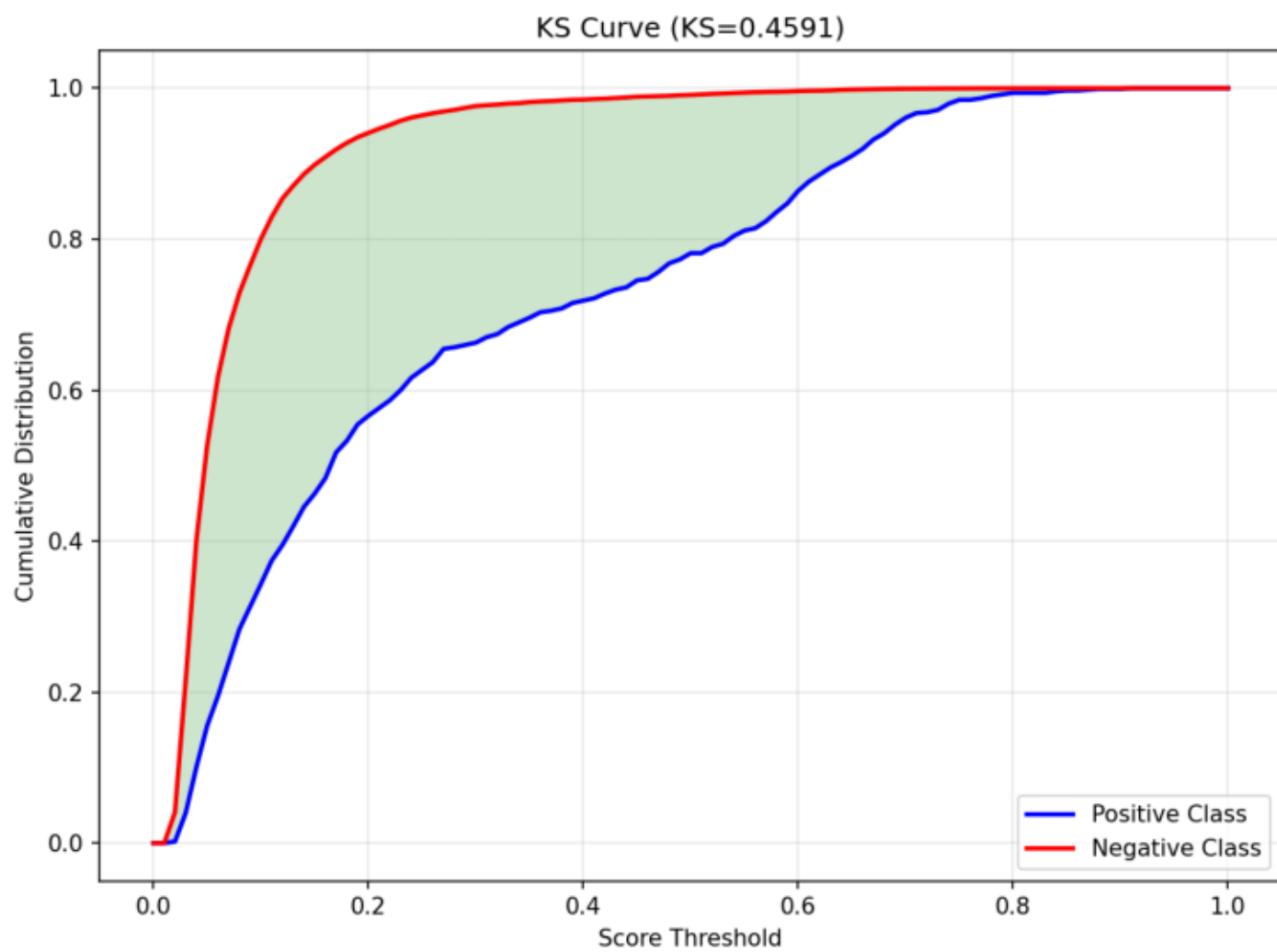
pr_curve: AUC-PR=0.4067 vs baseline 7.95%; higher curve implies better precision at recall.



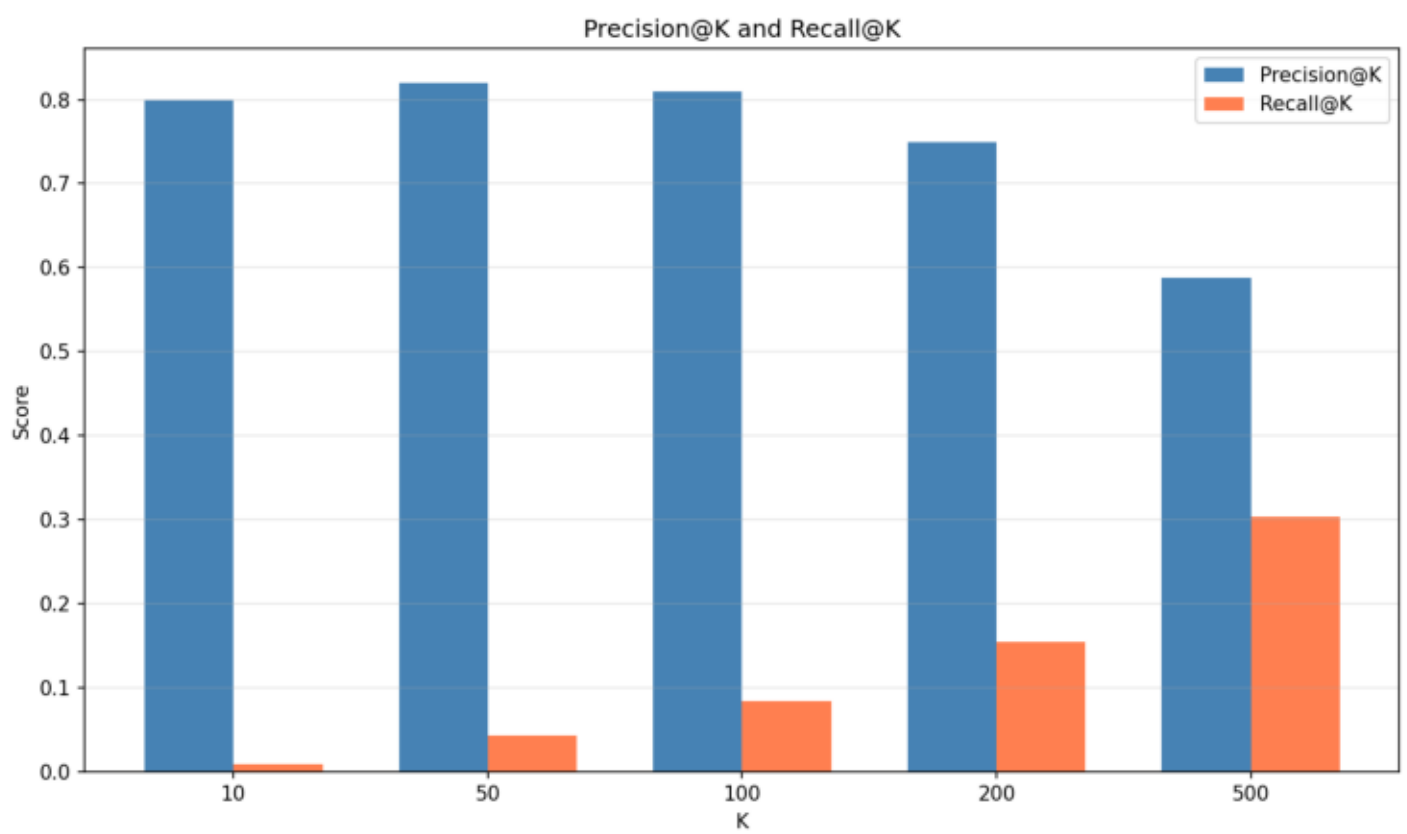
confusion_matrix: At threshold 0.50: TP=212, FP=100, FN=758, TN=11136.



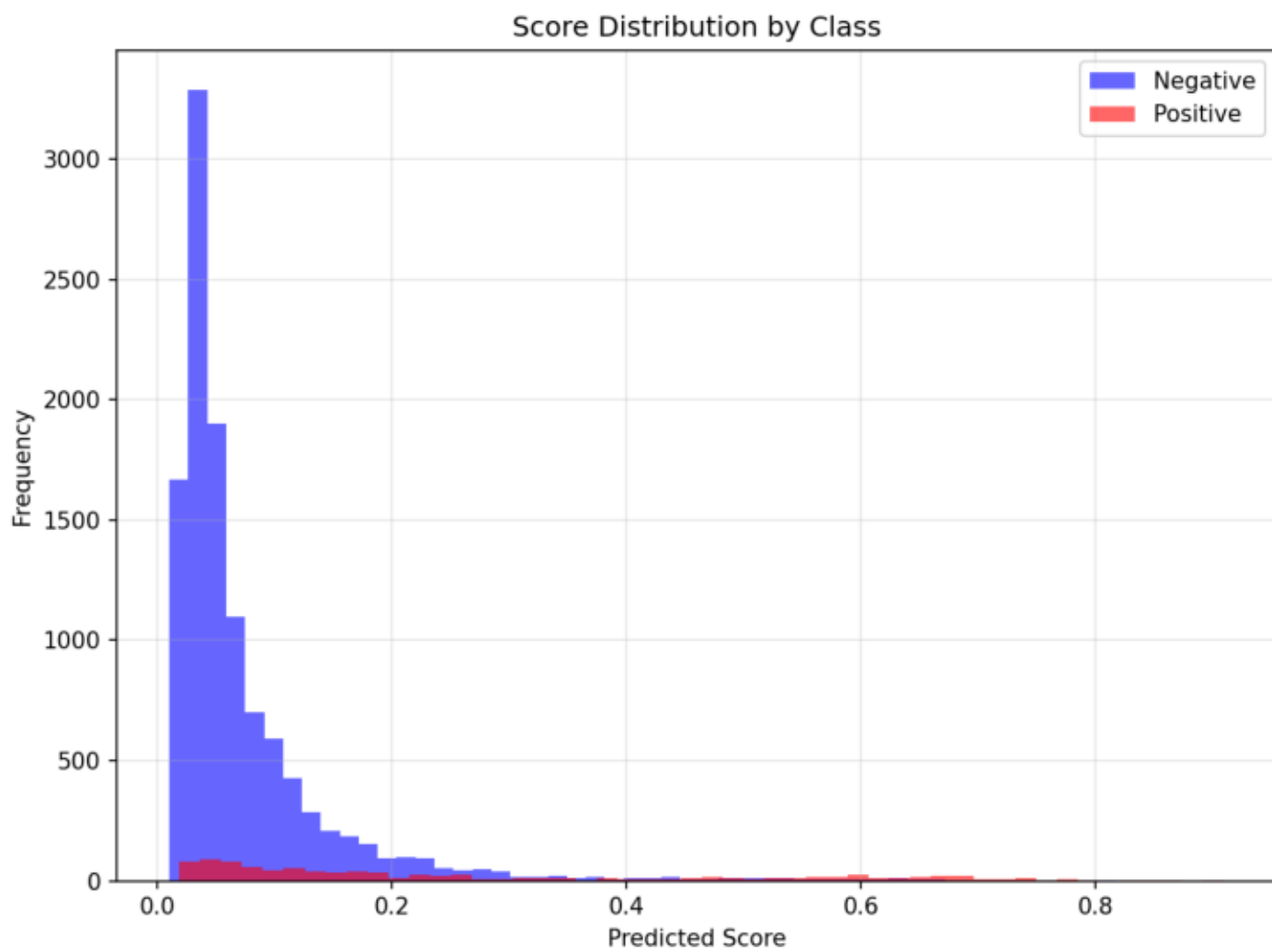
confusion_matrix_norm: Normalized rates: TPR=21.86%, TNR=99.11%.



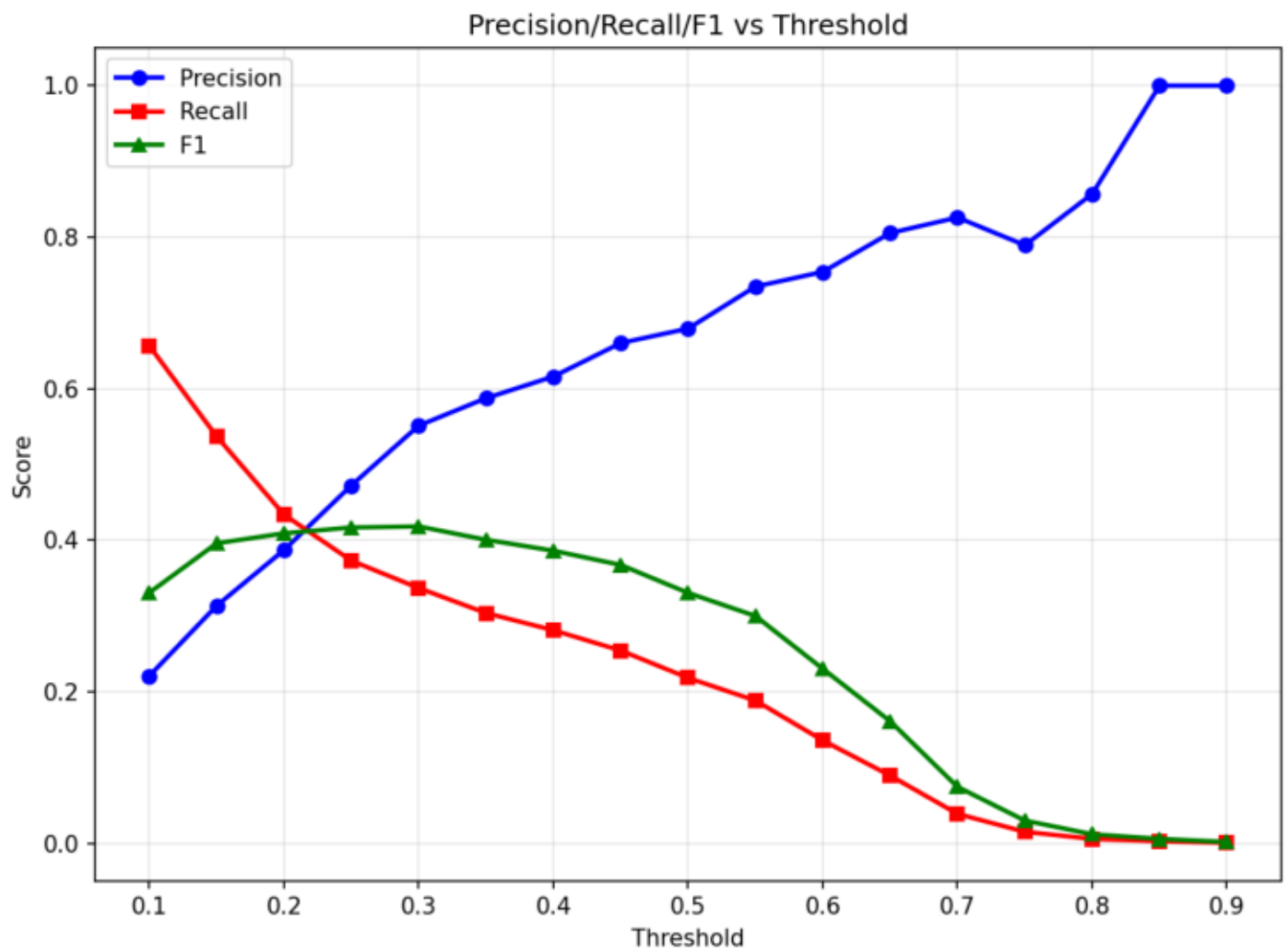
ks_curve: Maximum separation KS=0.4591 at threshold 0.12.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.262, negative=0.075 (larger gap is better).



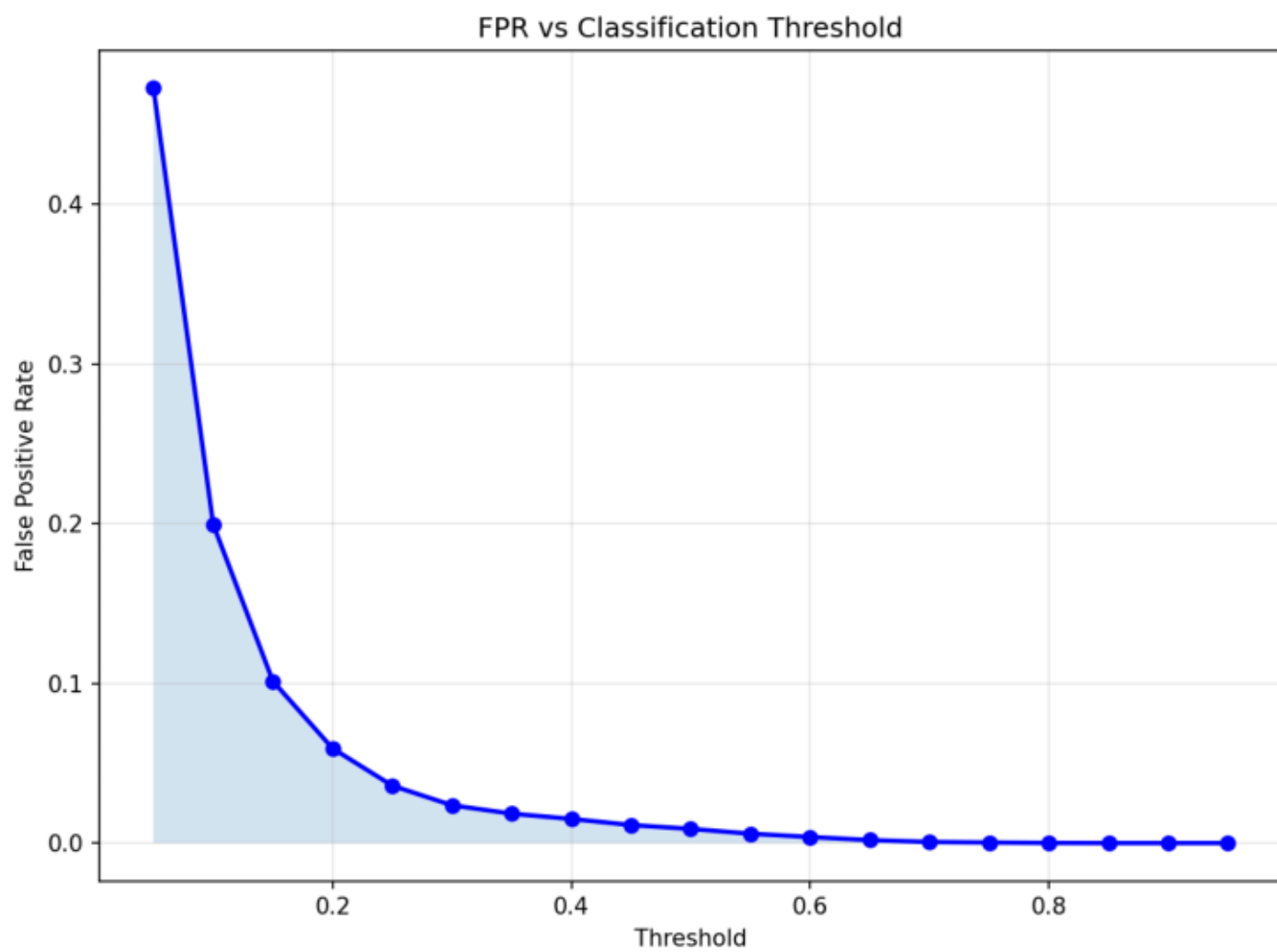
threshold_analysis: Best F1 in tested grid is 0.4184 at threshold 0.30.

value_date_quarter=2024Q2 - 2) Model Efficiency

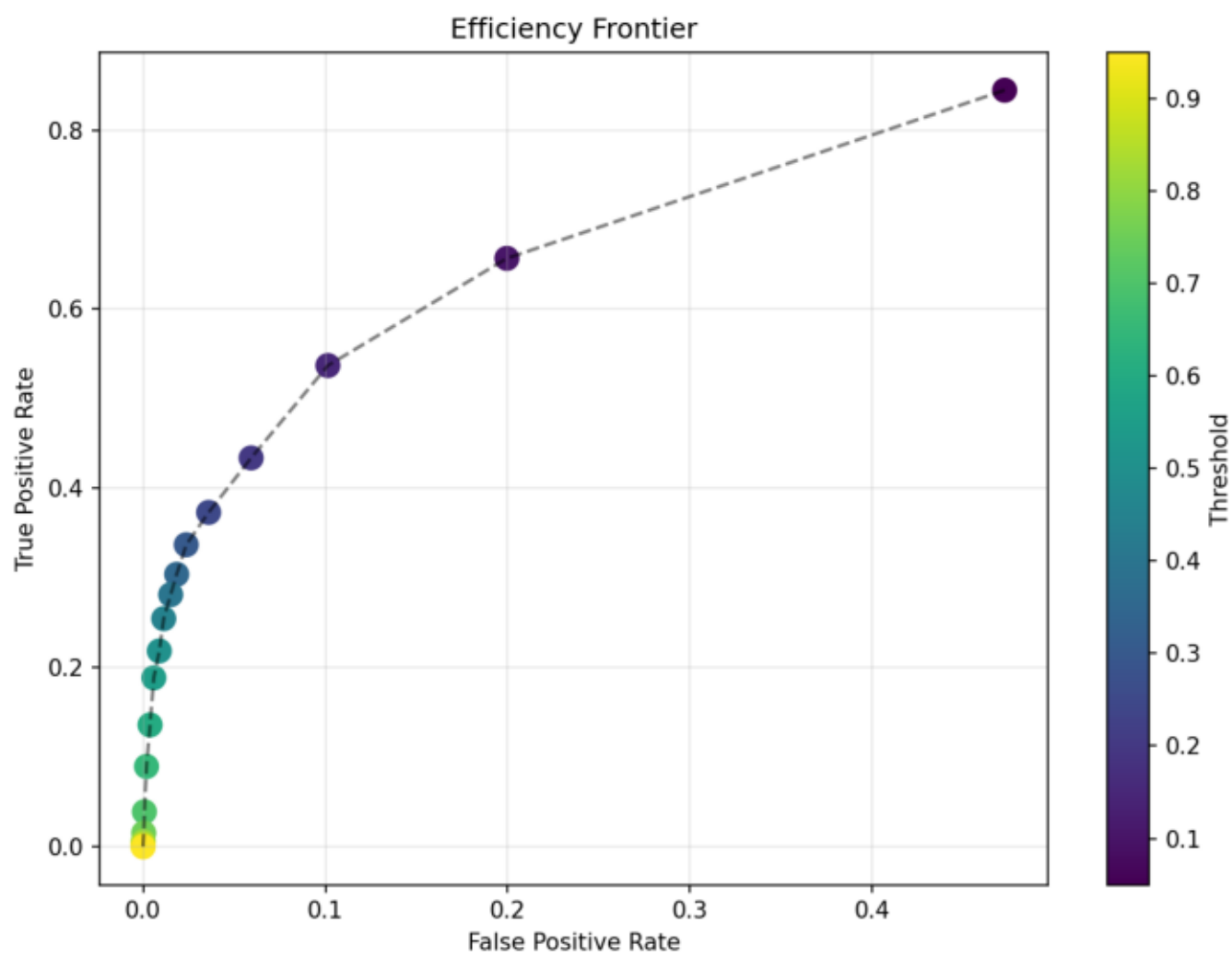
- fpr: 0.0089
- tn: 11136
- fp: 100
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.4726
- fpr_at_thresholds.t_0.10: 0.1995
- fpr_at_thresholds.t_0.15: 0.1014
- fpr_at_thresholds.t_0.20: 0.0593
- fpr_at_thresholds.t_0.25: 0.0360
- fpr_at_thresholds.t_0.30: 0.0237
- fpr_at_thresholds.t_0.35: 0.0184
- fpr_at_thresholds.t_0.40: 0.0151
- fpr_at_thresholds.t_0.45: 0.0113
- fpr_at_thresholds.t_0.50: 0.0089
- fpr_at_thresholds.t_0.55: 0.0059
- fpr_at_thresholds.t_0.60: 0.0038
- fpr_at_thresholds.t_0.65: 0.0019
- fpr_at_thresholds.t_0.70: 0.0007
- fpr_at_thresholds.t_0.75: 0.0004
- fpr_at_thresholds.t_0.80: 0.0001
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q2 - 2) Model Efficiency - Explanations

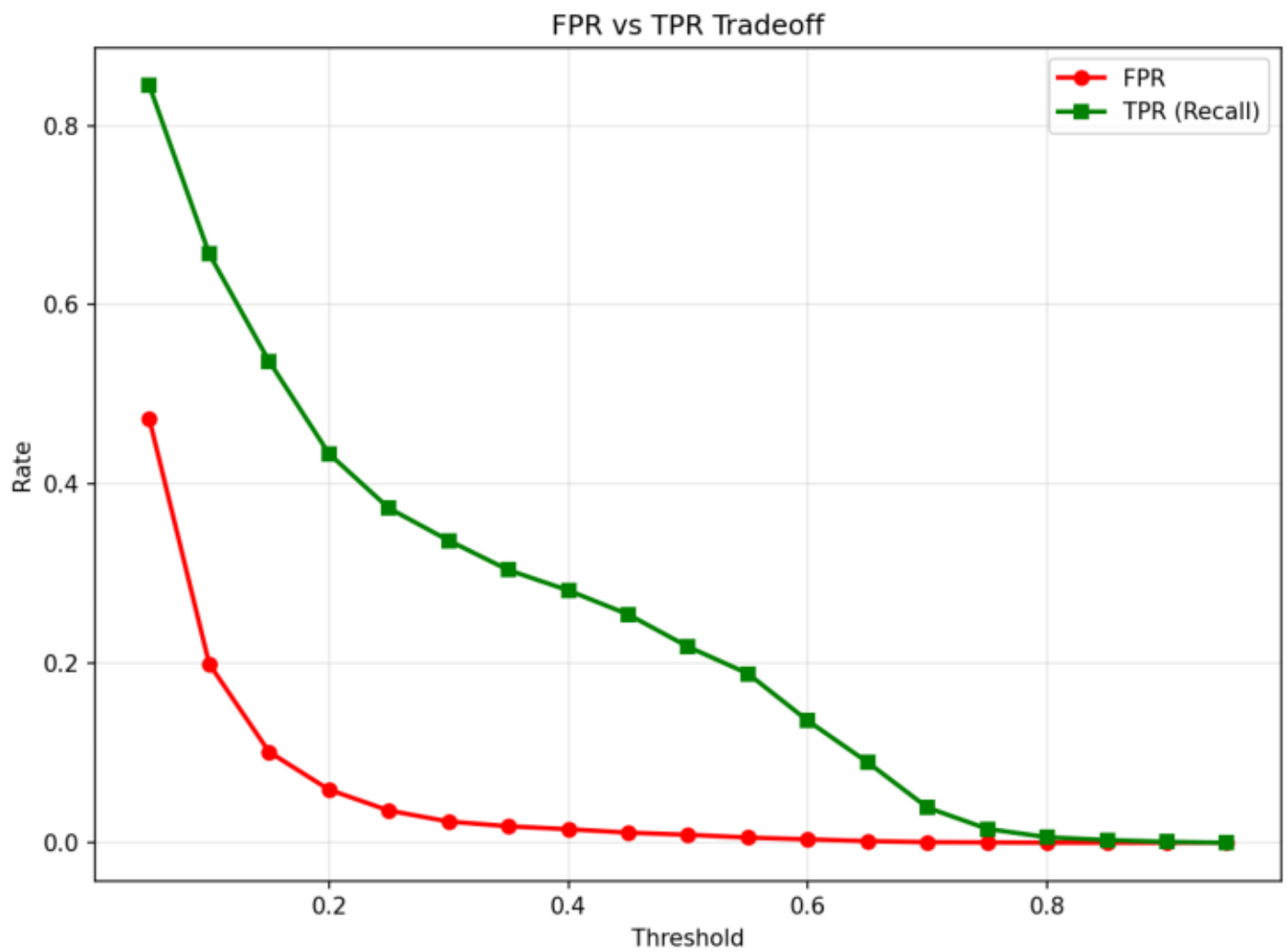
- At threshold 0.50, FPR=0.0089 (low); FP=100 out of 11236 negatives.
- At the same threshold, TPR=0.2186 with TP=212 and FN=758, showing the capture rate of positives.
- A threshold near 0.20 yields FPR $\approx$ 0.0593 with TPR $\approx$ 0.4340 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.



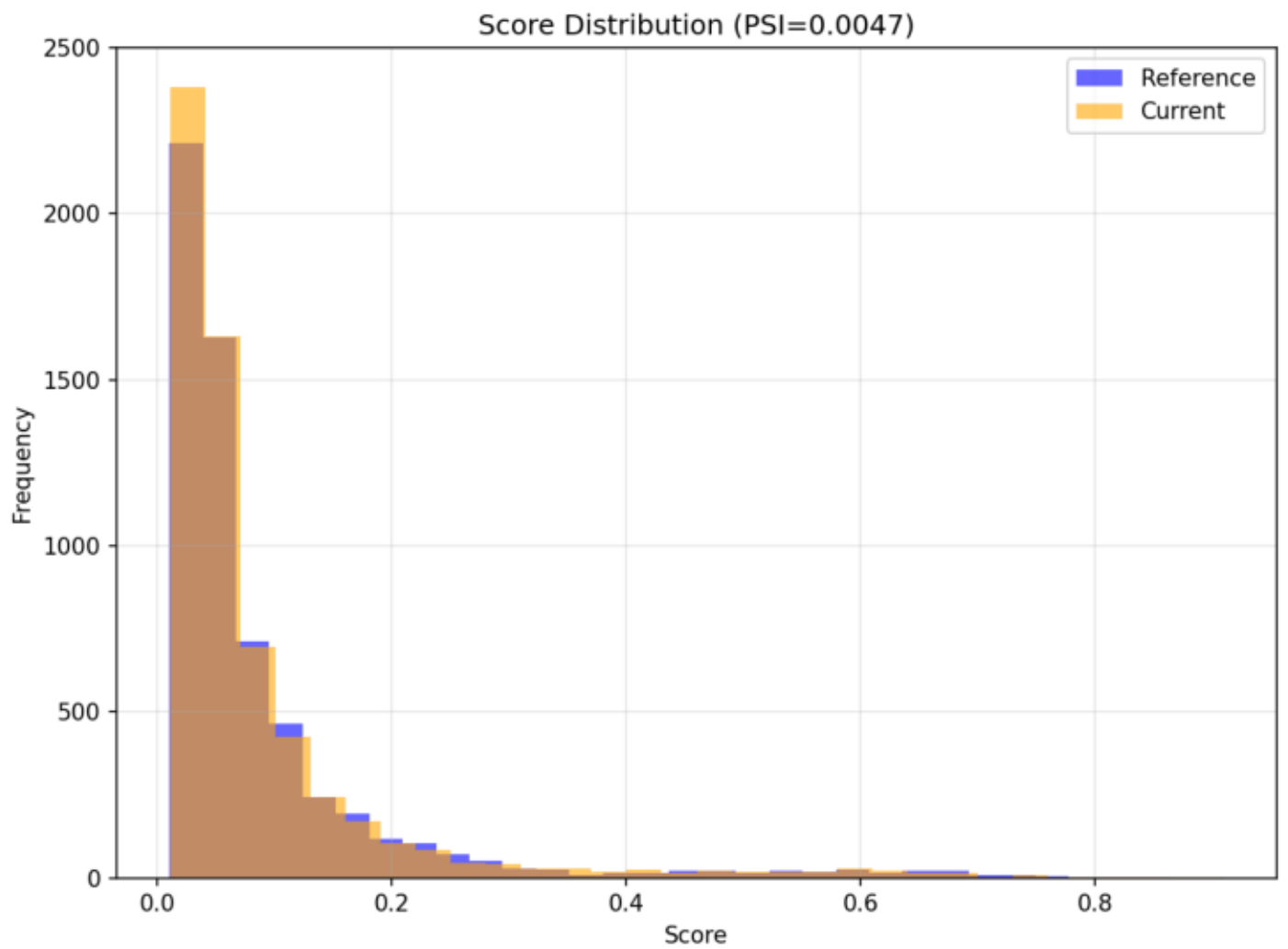
fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

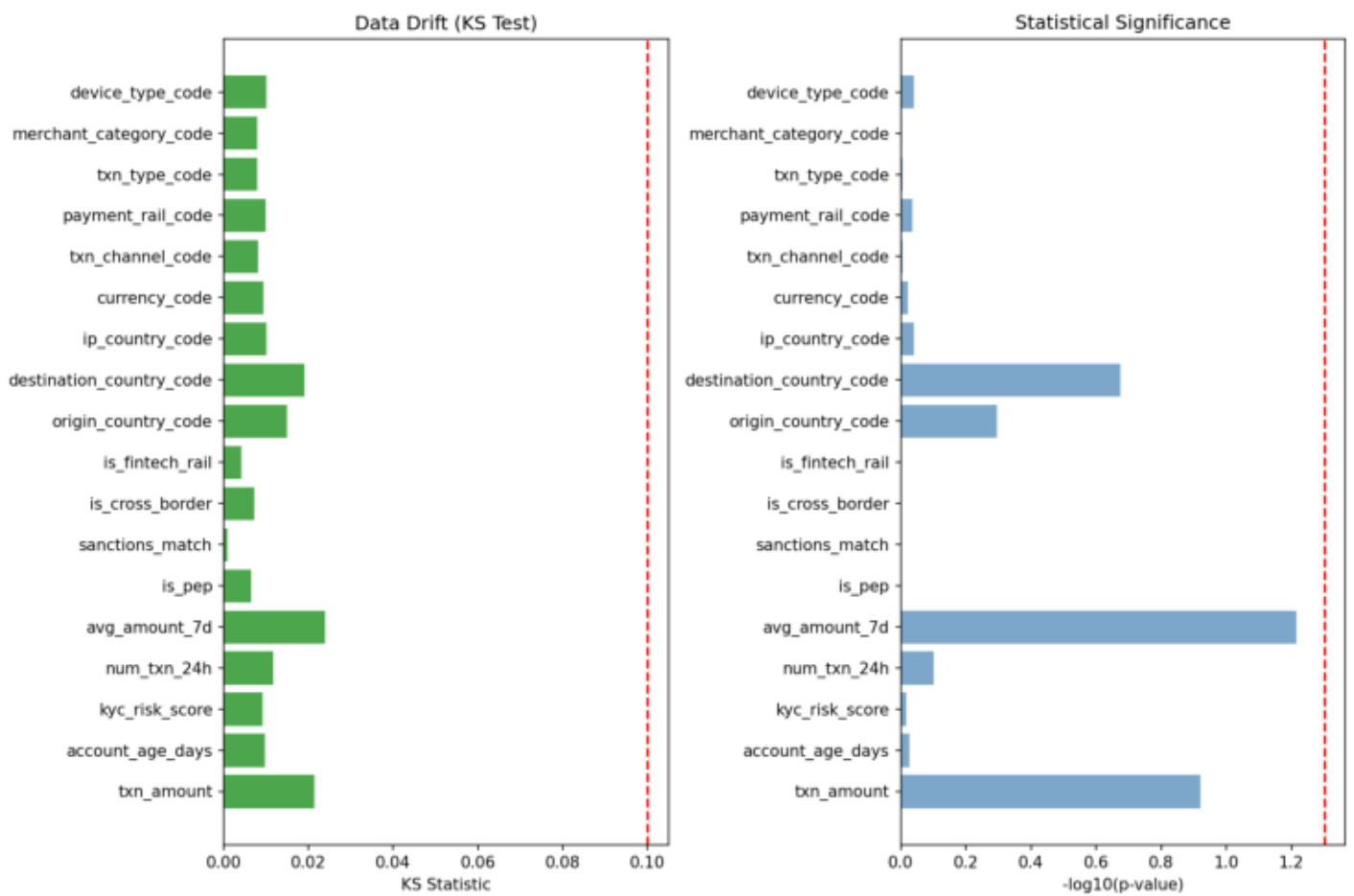
value_date_quarter=2024Q2 - 3) Model Stability

- psi: 0.0047
- cv_auc_roc_mean: 0.8025
- cv_auc_roc_std: 0.0174
- cv_auc_pr_mean: 0.4091
- cv_auc_pr_std: 0.0321
- bootstrap_auc_roc_mean: 0.7998
- bootstrap_auc_roc_ci_lower: 0.7788
- bootstrap_auc_roc_ci_upper: 0.8200
- concept_drift_detected: False
- concept_drift_score: 0.0177

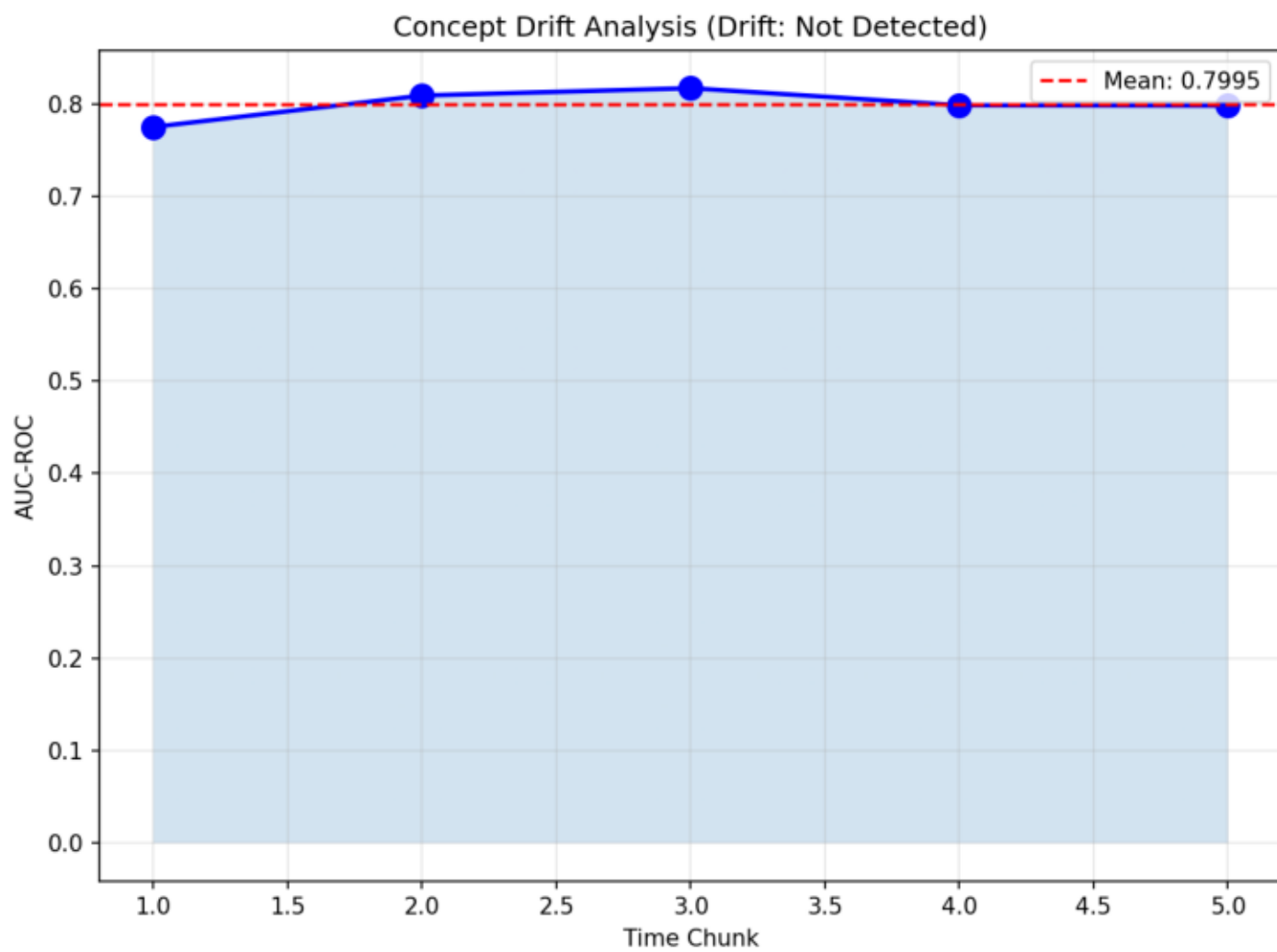
value_date_quarter=2024Q2 - 3) Model Stability - Explanations

- PSI is 0.0047, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8025 ± 0.0174 (moderate across folds).
- CV AUC-PR is 0.4091 ± 0.0321 (variable across folds).
- Bootstrap AUC-ROC mean is 0.7998 with 95% CI [0.7788, 0.8200] (width 0.0412).
- No concept drift detected (score 0.0177), suggesting stable performance over time.
- Data drift flagged 0/18 numeric features; top drift signals: ['avg_amount_7d', 'txn_amount', 'destination_country_code'].

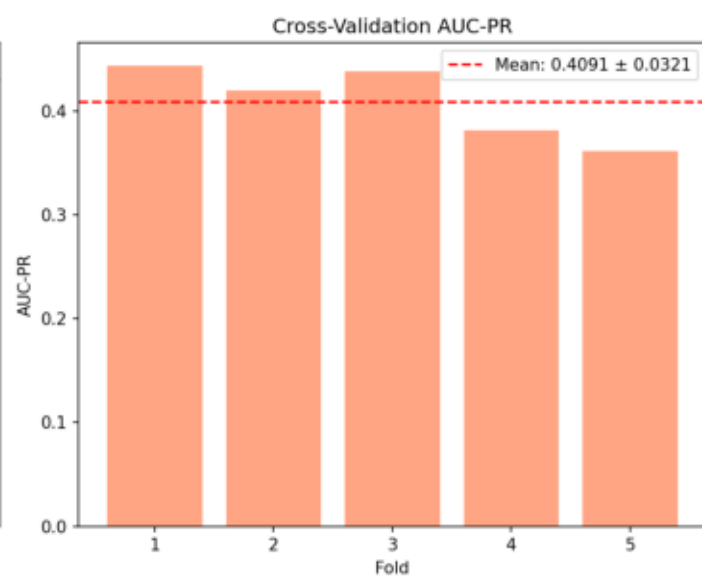
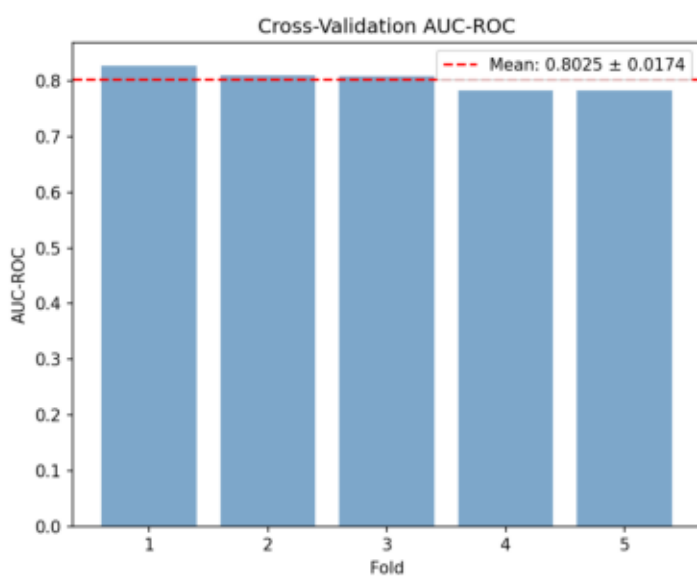




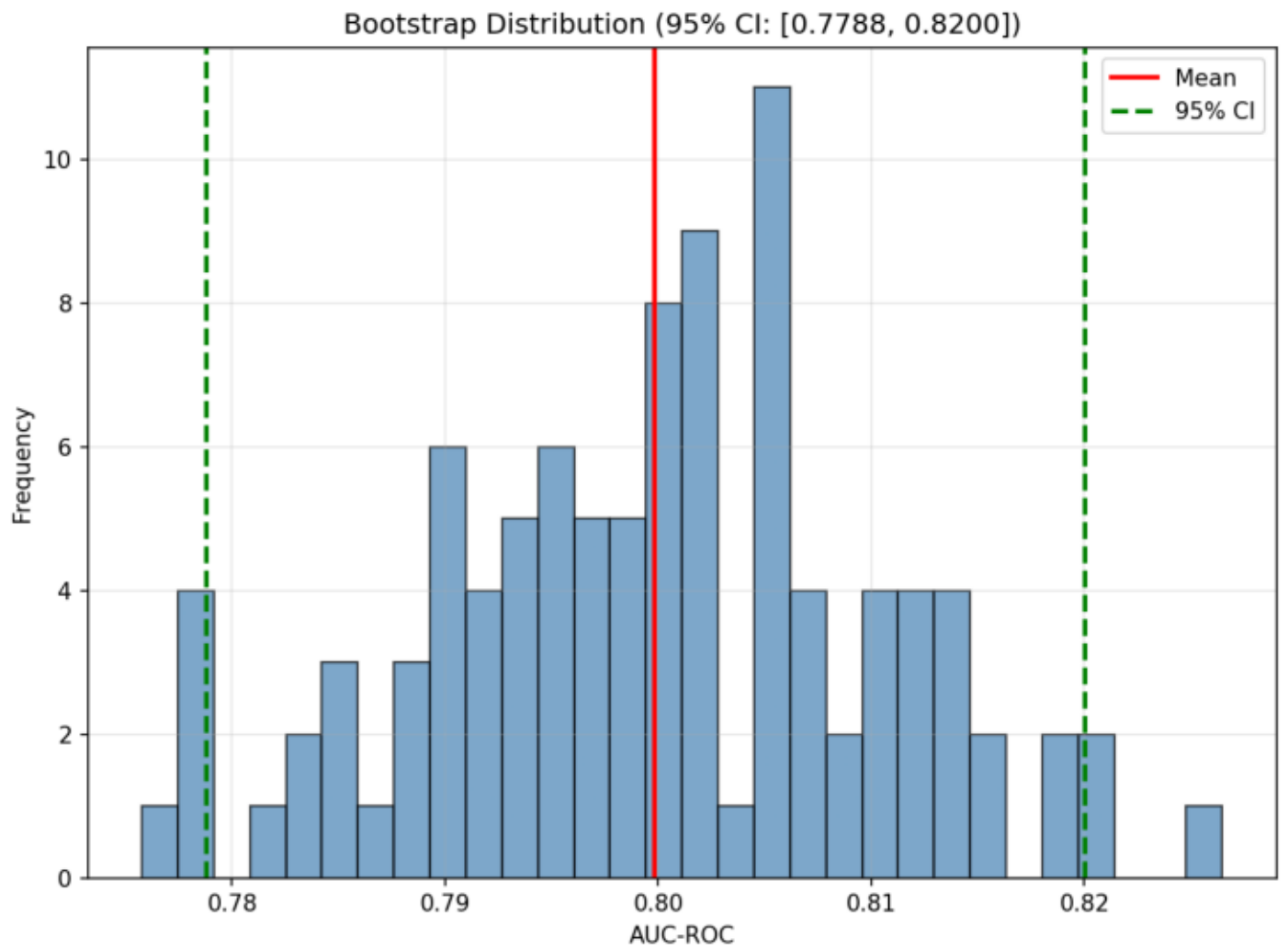
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.



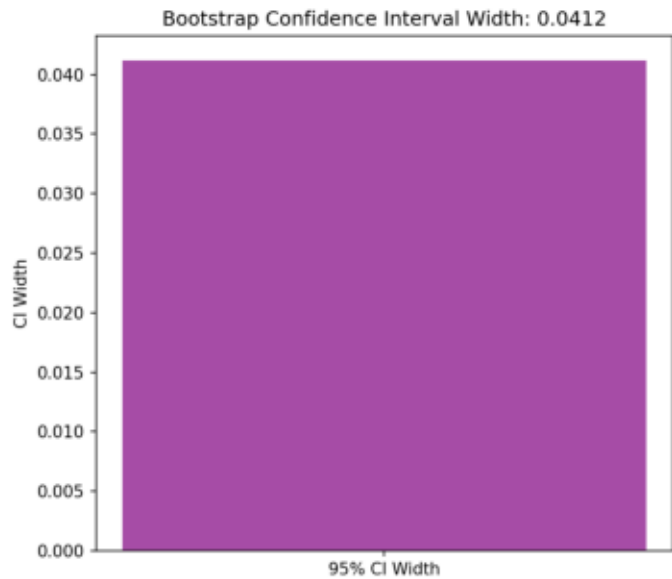
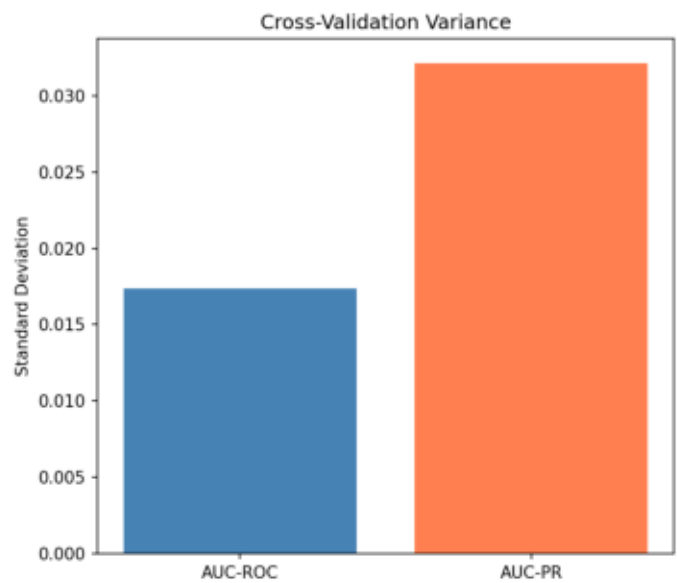
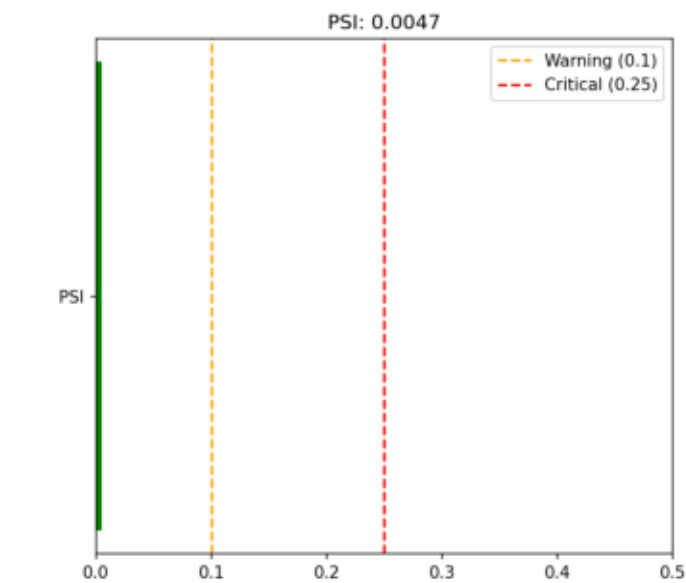
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8025 and PR 0.4091.



bootstrap_distribution: Bootstrap CI [0.7788, 0.8200] reflects performance stability.



STABILITY SUMMARY

PSI: 0.0047 (OK)

CV AUC-ROC: 0.8025 $\pm$ 0.0174

CV AUC-PR: 0.4091 $\pm$ 0.0321

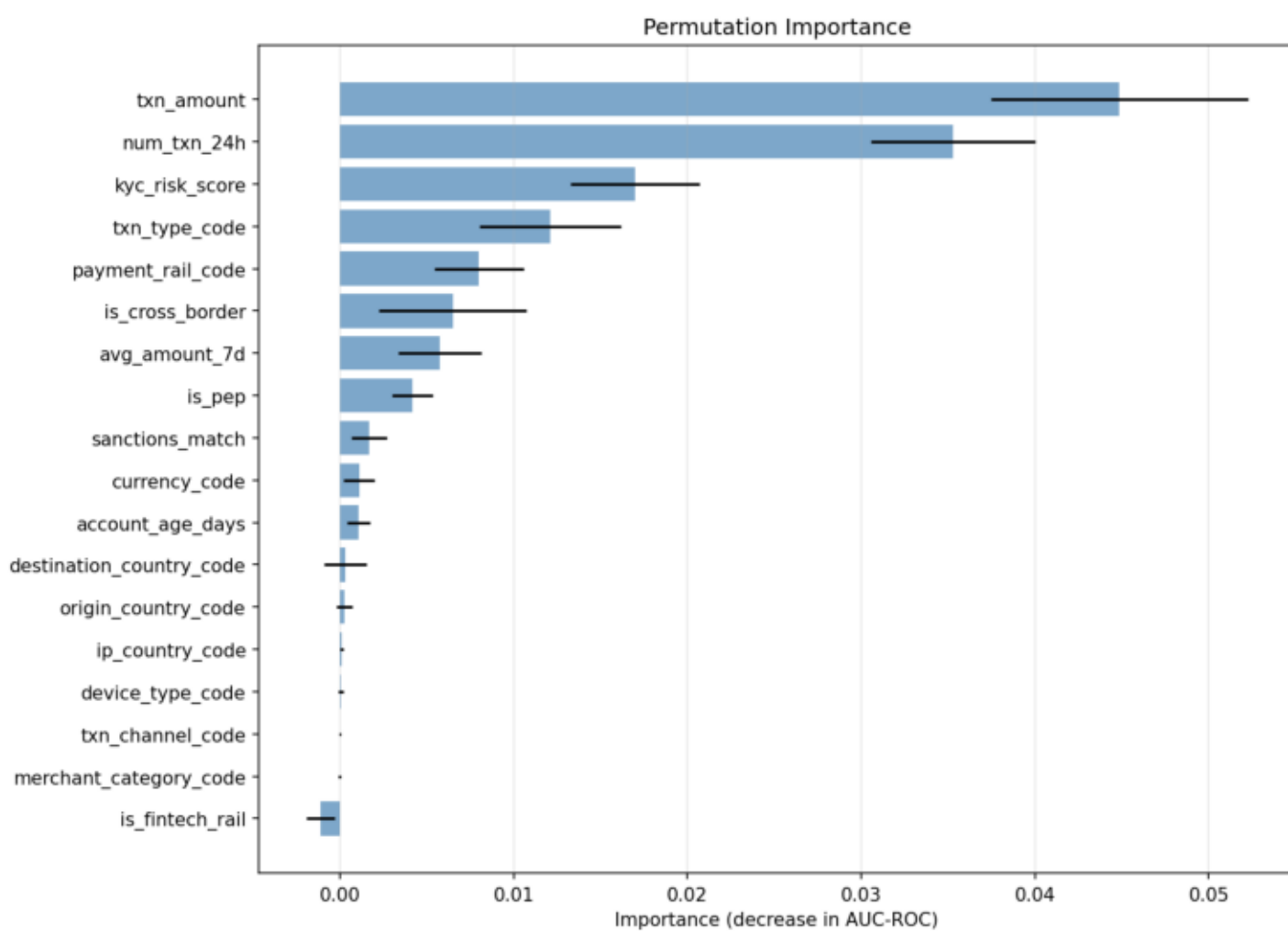
Bootstrap 95% CI: [0.7788, 0.8200]

value_date_quarter=2024Q2 - 4) Model Interpretability

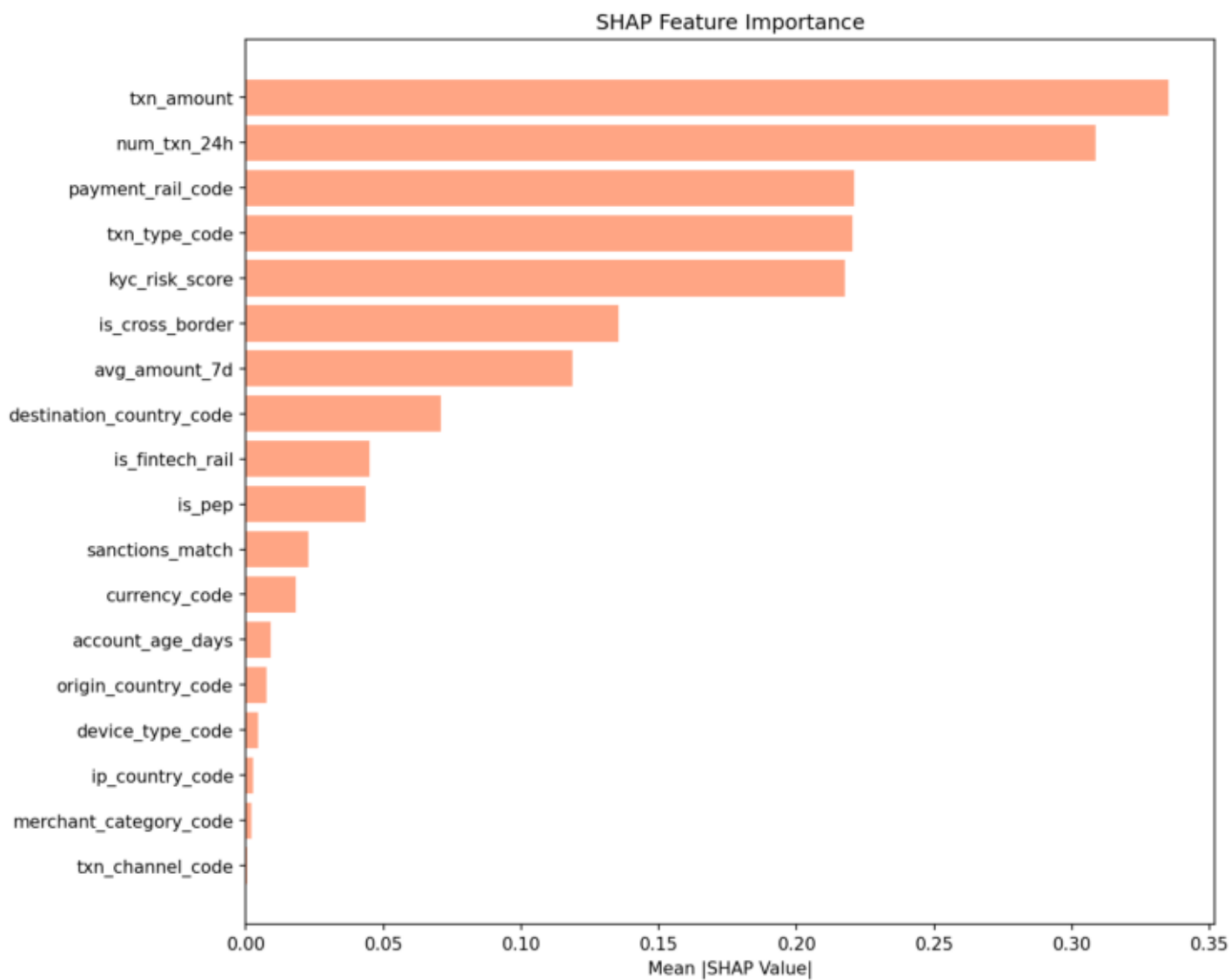
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'shap', 'lime']
- perm_top_features: ['txn_amount', 'num_txn_24h', 'kyc_risk_score', 'txn_type_code', 'payment_rail_code']...
- shap_top_features: ['txn_amount', 'num_txn_24h', 'payment_rail_code', 'txn_type_code', 'kyc_risk_score']...
- lime_instances: 3
- pdp_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q2 - 4) Model Interpretability - Explanations

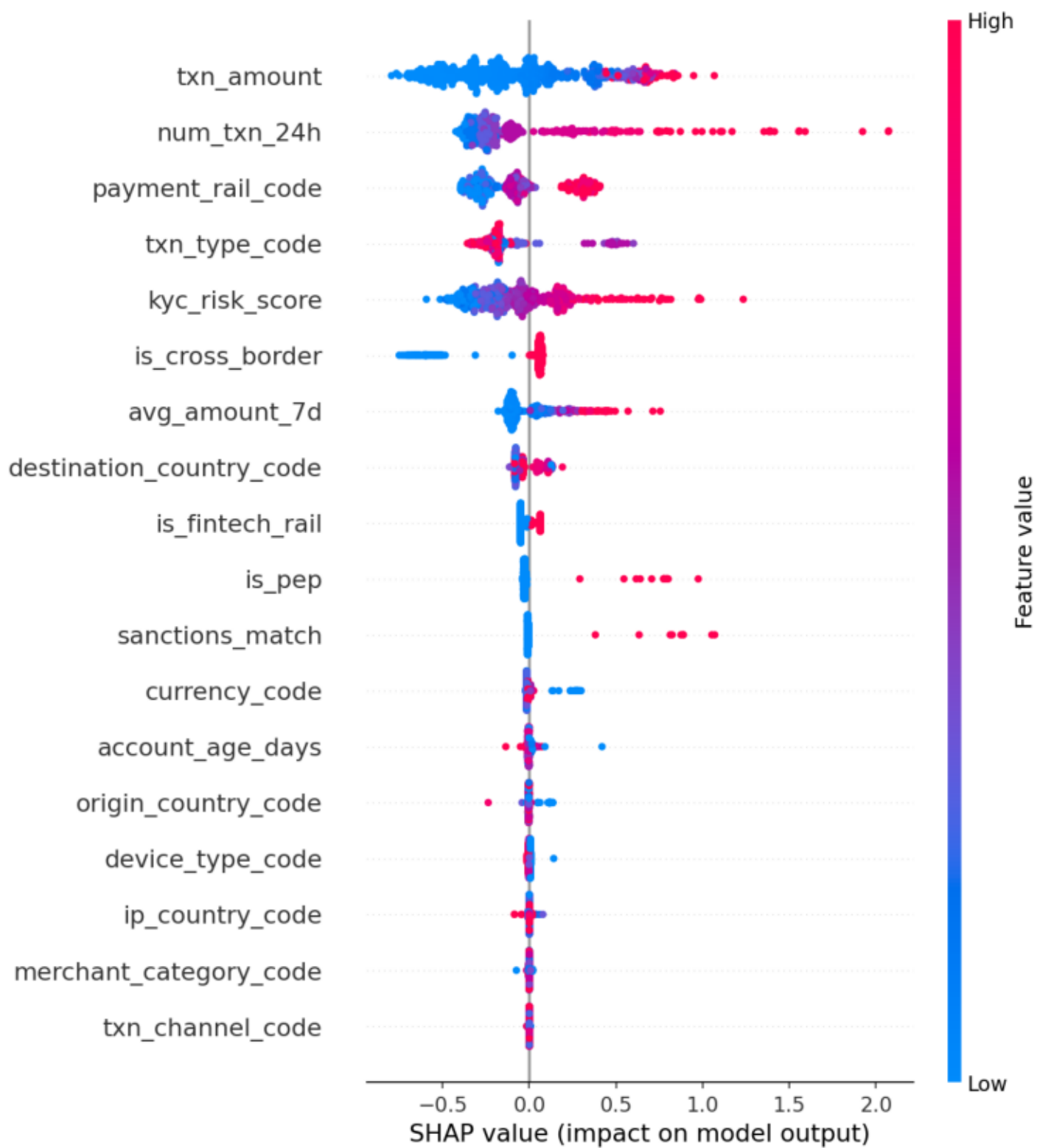
- Permutation importance highlights: ['txn_amount', 'num_txn_24h', 'kyc_risk_score', 'txn_type_code', 'payment_rail_code'].
- SHAP top features: ['txn_amount', 'num_txn_24h', 'payment_rail_code', 'txn_type_code', 'kyc_risk_score'].
- LIME generated local explanations for 3 instances; top contributors for instance_10: ['sanctions_match <= 0.00', 'is_pep > 0.00', 'num_txn_24h <= 1.00', '107.62 < txn_amount <= 365.48', '109.54 < avg_amount_7d <= 303.18'].
- PDP plots show average effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'destination_country_code'].
- ICE plots show individual-level effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score'].

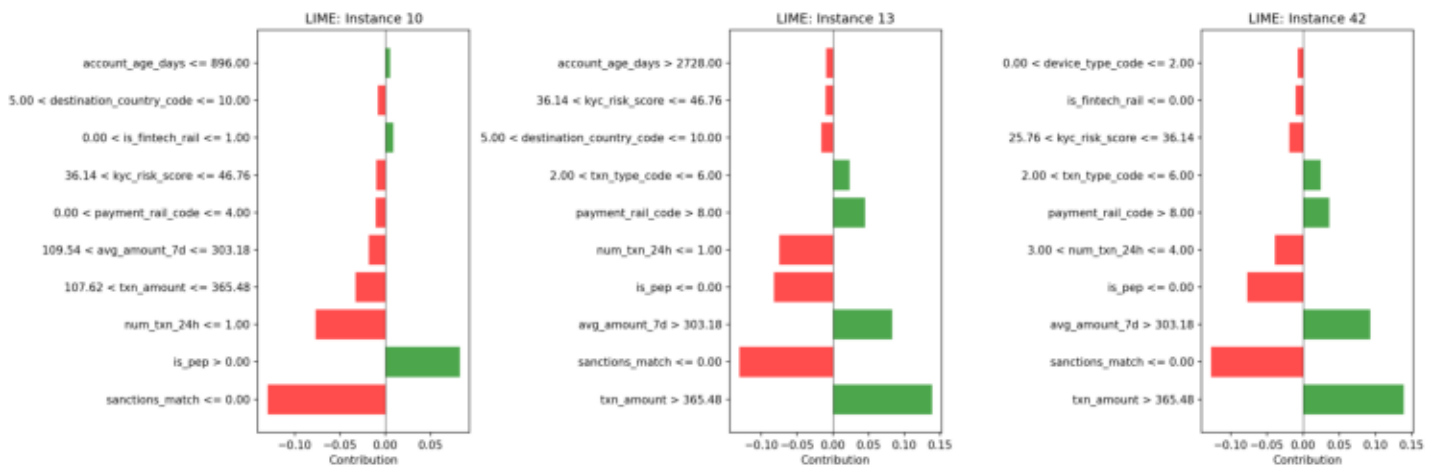


permutation_importance: Bars show how much AUC drops when each feature is permuted.



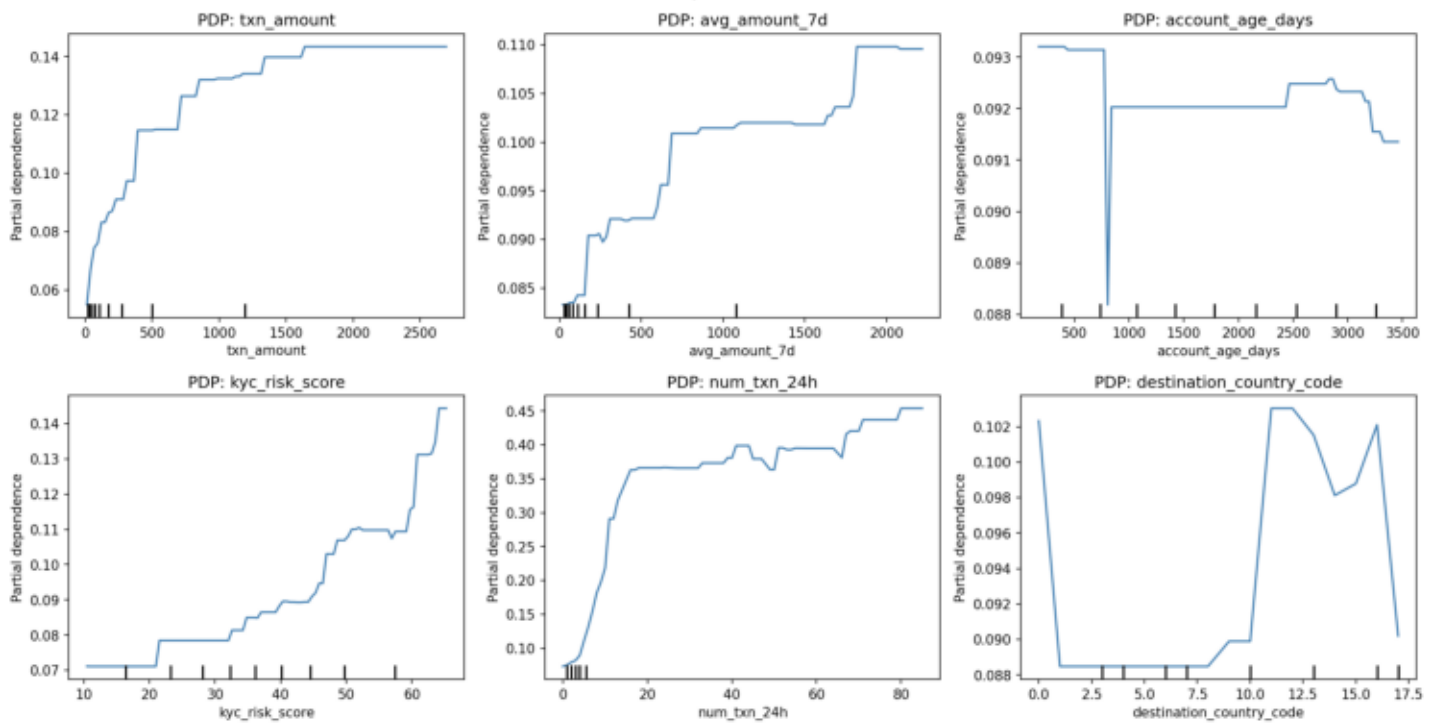
shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.





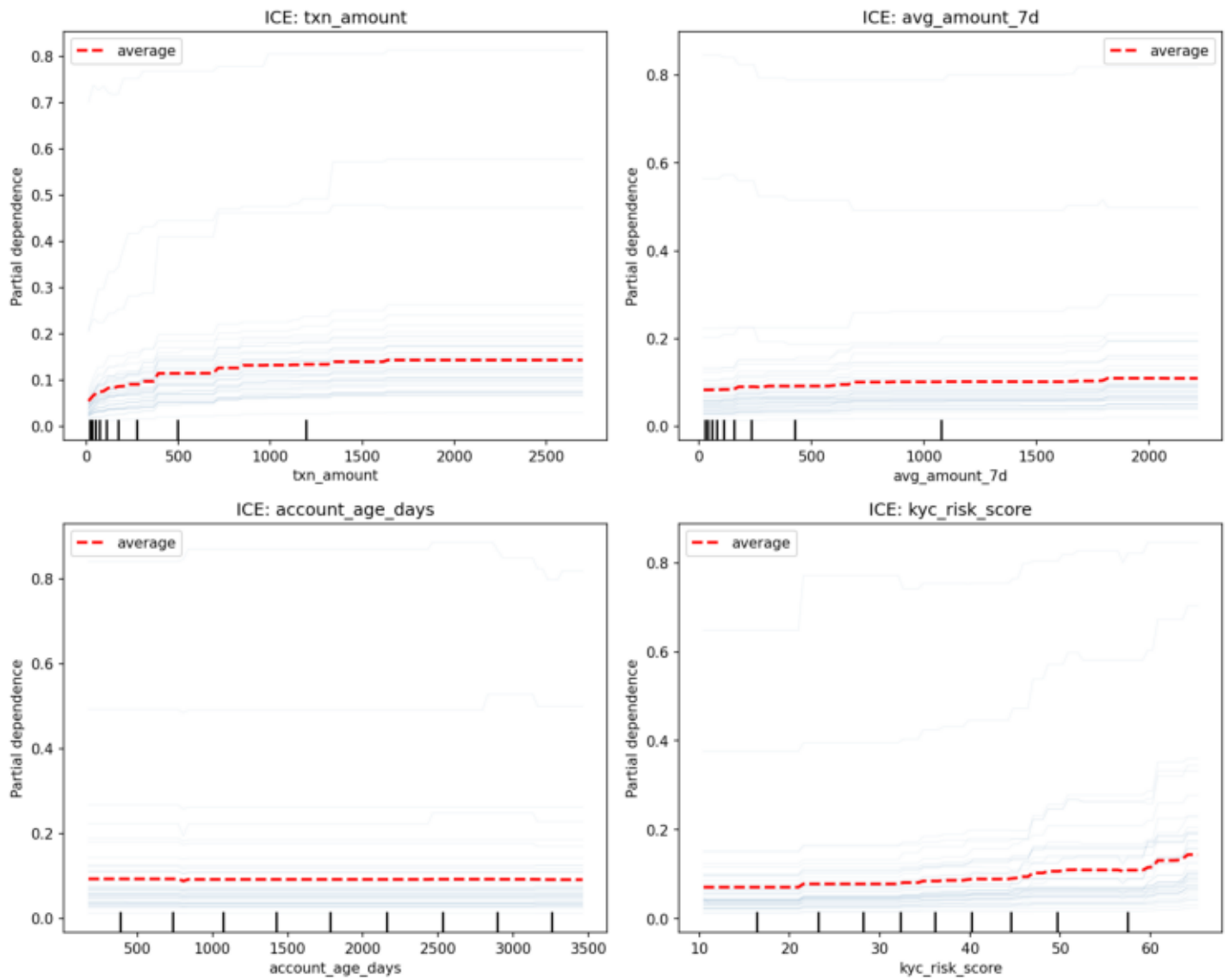
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



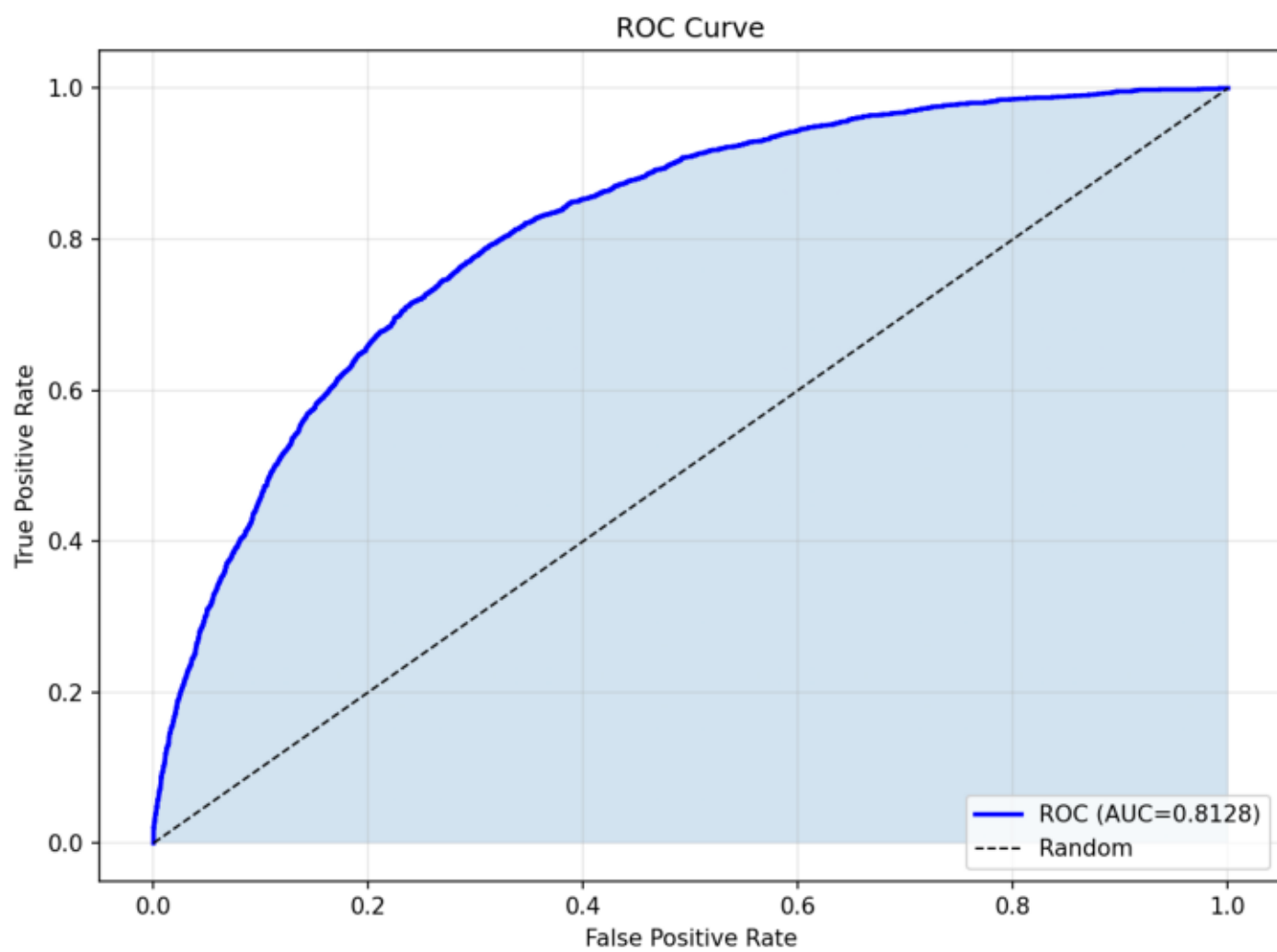
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q3 - 1) Model Effectiveness

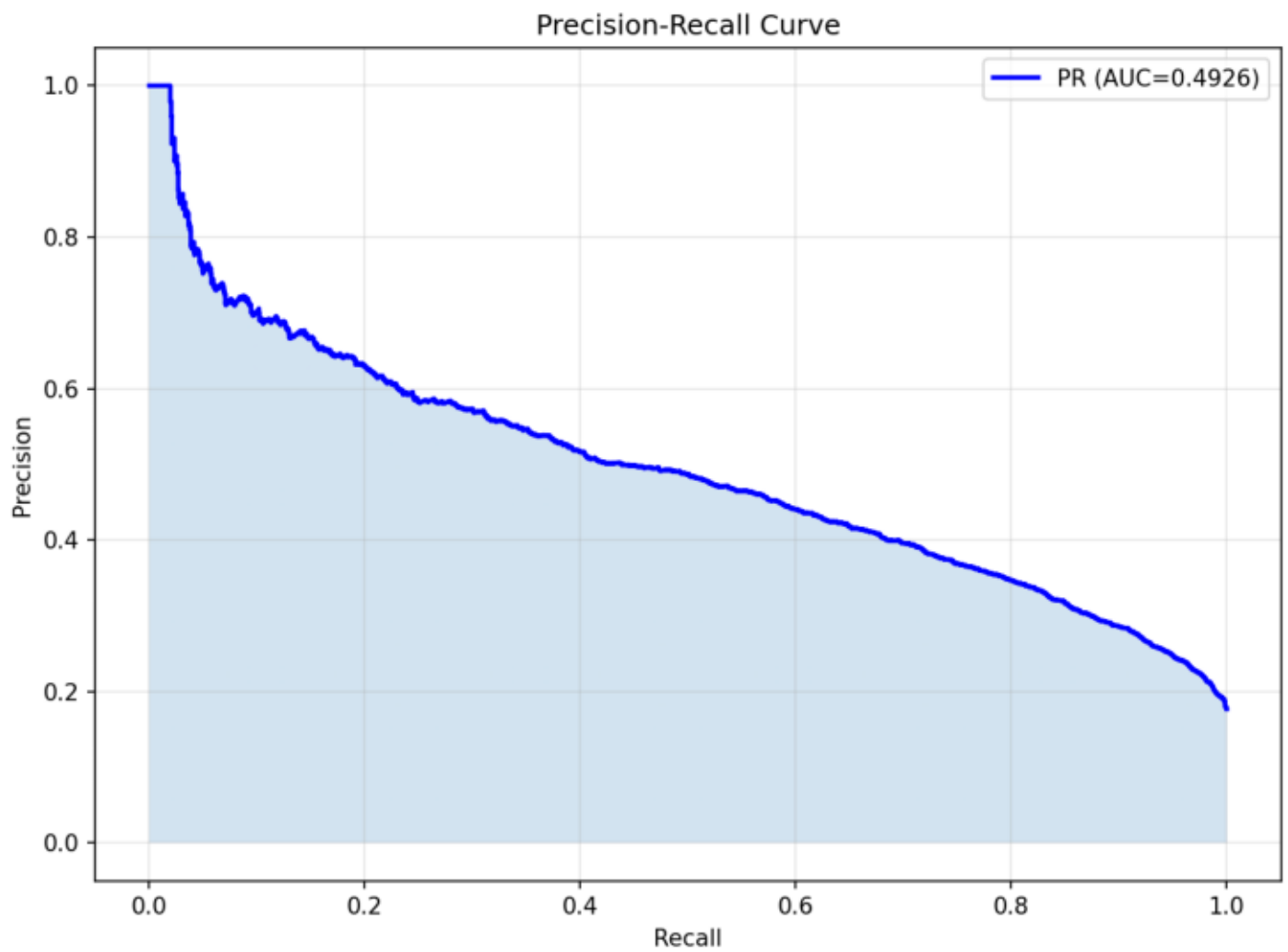
- auc_roc: 0.8128
- auc_pr: 0.4926
- precision: 0.5826
- recall: 0.2717
- f1: 0.3706
- ks_statistic: 0.4777
- ks_threshold: 0.1400
- confusion_matrix.TN: 10316
- confusion_matrix.FP: 450
- confusion_matrix.FN: 1683
- confusion_matrix.TP: 628
- precision_at_k.10: 1.0000
- precision_at_k.50: 0.9600
- precision_at_k.100: 0.8300
- precision_at_k.200: 0.7350
- precision_at_k.500: 0.6720
- recall_at_k.10: 0.0043
- recall_at_k.50: 0.0208
- recall_at_k.100: 0.0359
- recall_at_k.200: 0.0636
- recall_at_k.500: 0.1454

value_date_quarter=2024Q3 - 1) Model Effectiveness - Explanations

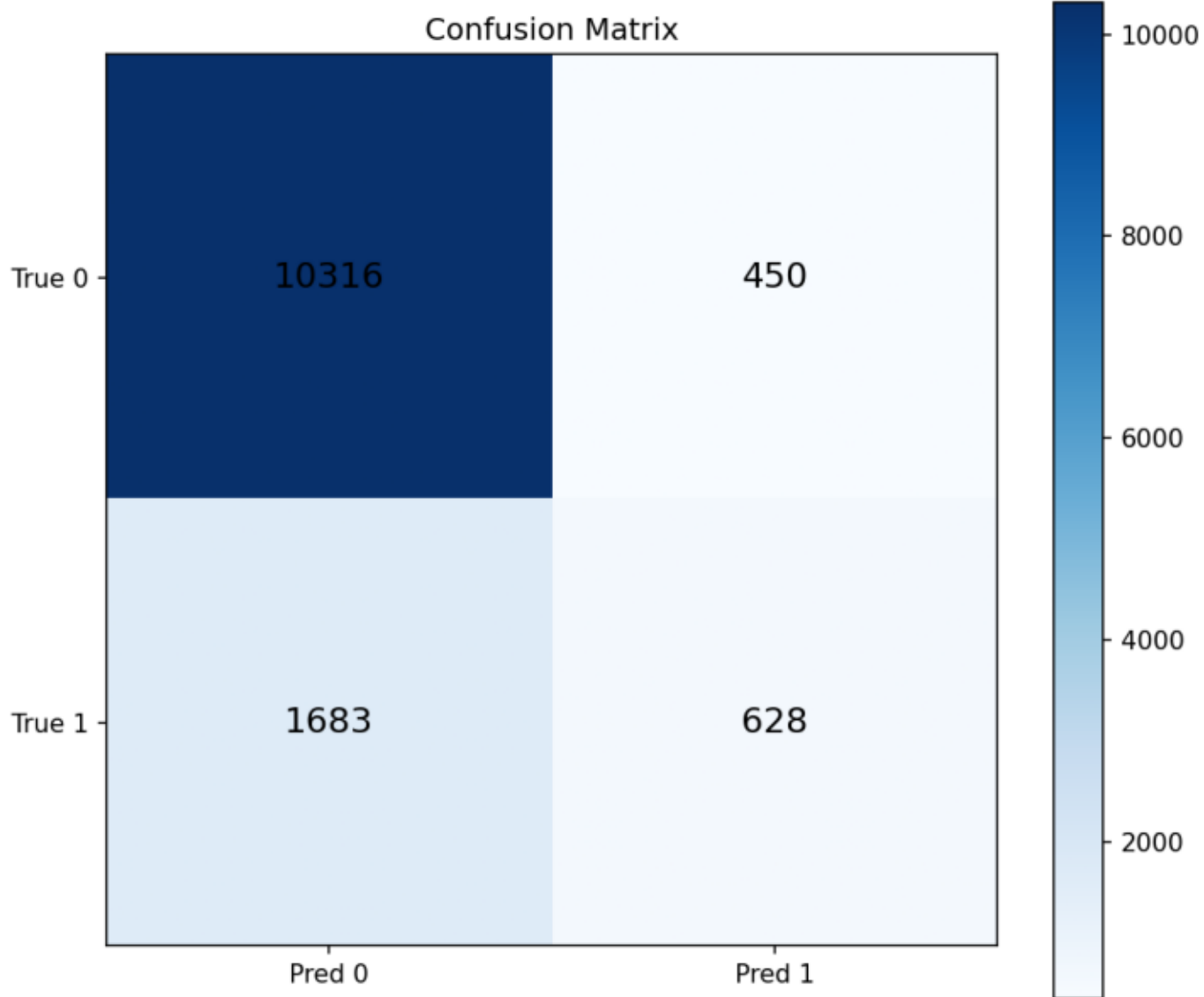
- Positive rate is 17.67%; AUC-ROC is 0.8128 (strong ranking).
- AUC-PR is 0.4926, above the base rate 17.67%, indicating meaningful lift.
- At threshold 0.50, precision=0.5826, recall=0.2717, F1=0.3706 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=628, FP=450, FN=1683, TN=10316 (FPR=4.18%, FNR=72.83%).
- KS is 0.4777 at threshold 0.14, indicating strong separation.
- Top-10: precision=1.0000, recall=0.0043 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.6720, recall=0.1454 shows how much coverage you get with a larger queue.



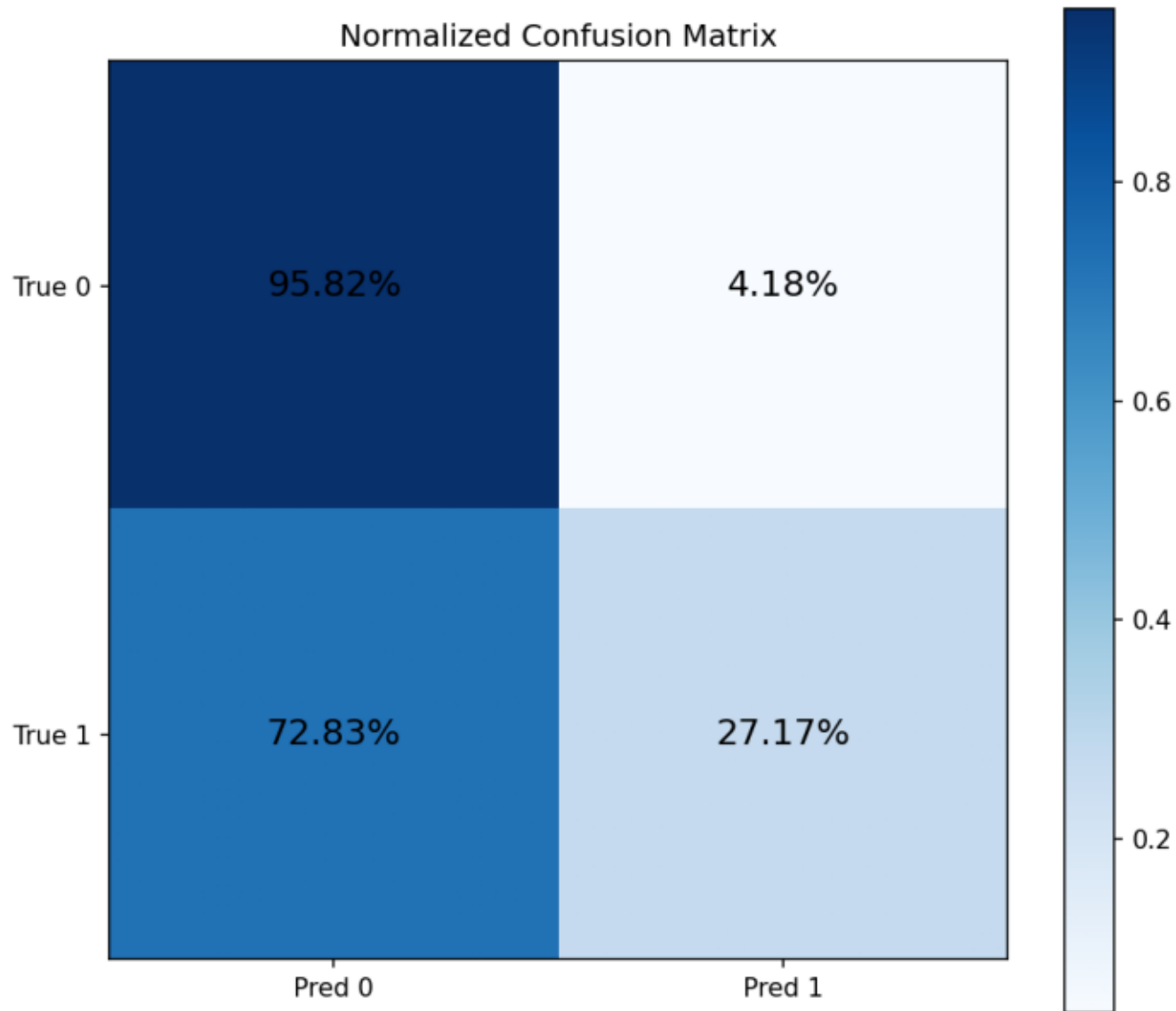
roc_curve: AUC-ROC=0.8128; curve above diagonal indicates strong separation.



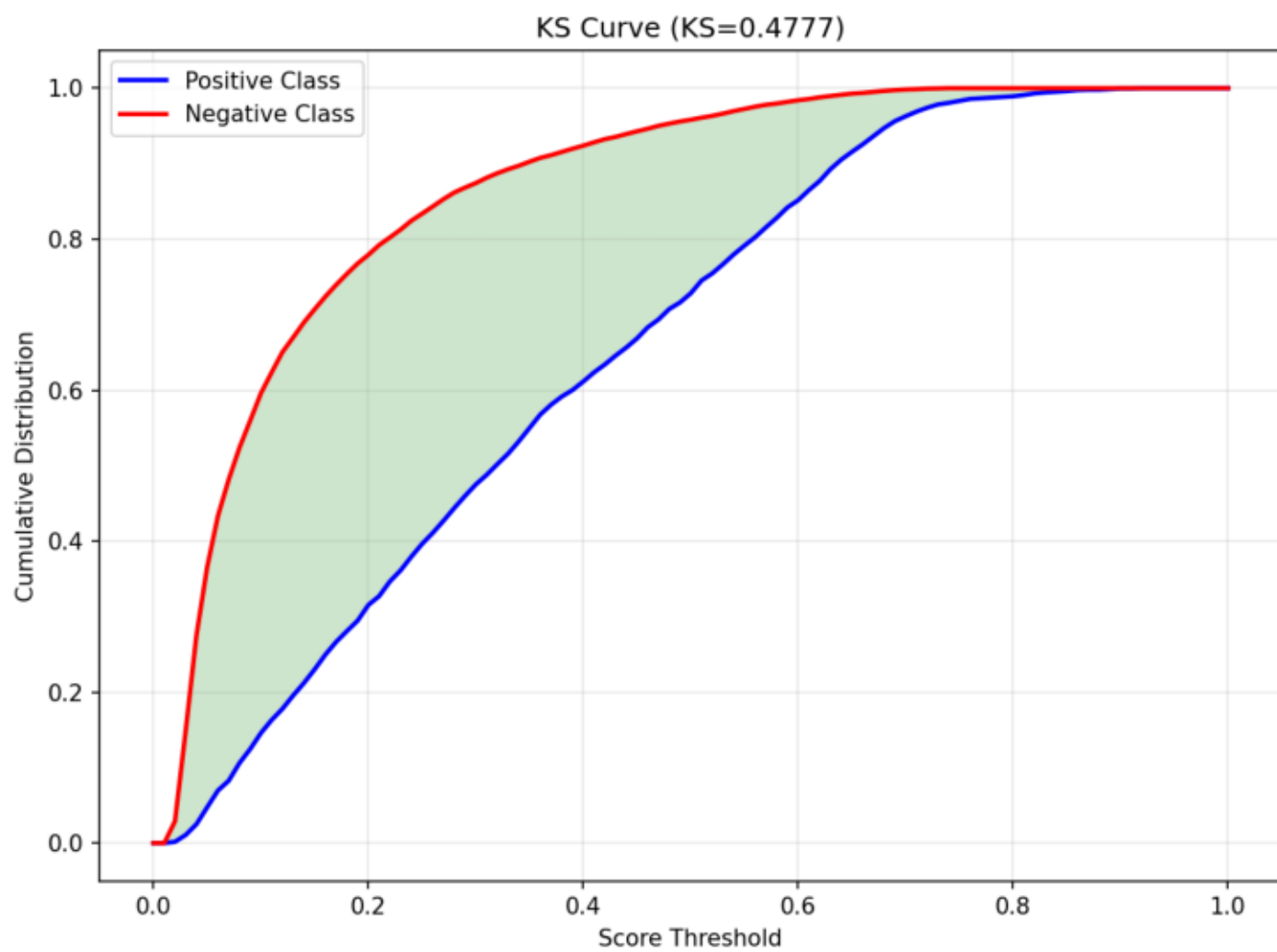
pr_curve: AUC-PR=0.4926 vs baseline 17.67%; higher curve implies better precision at recall.



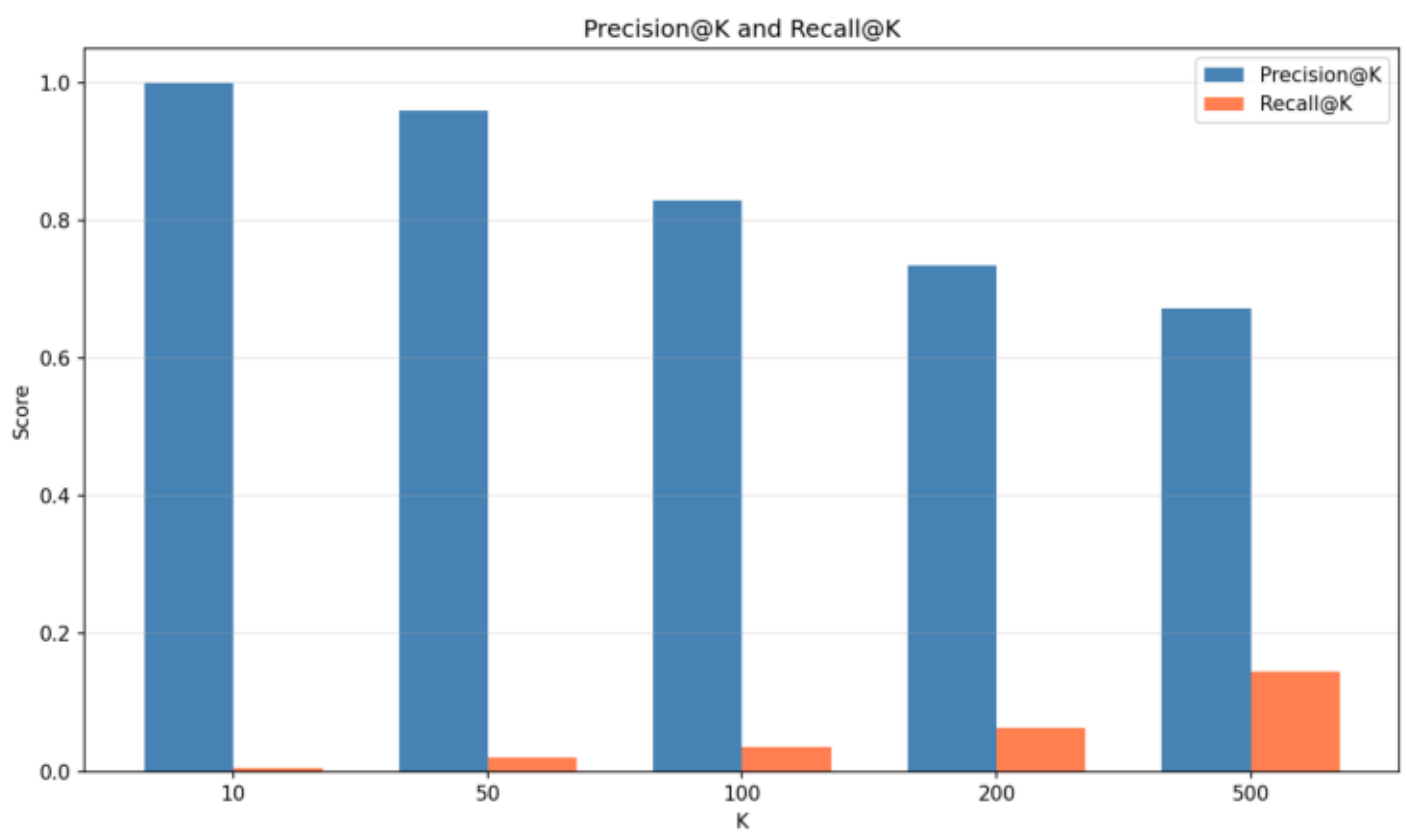
confusion_matrix: At threshold 0.50: TP=628, FP=450, FN=1683, TN=10316.



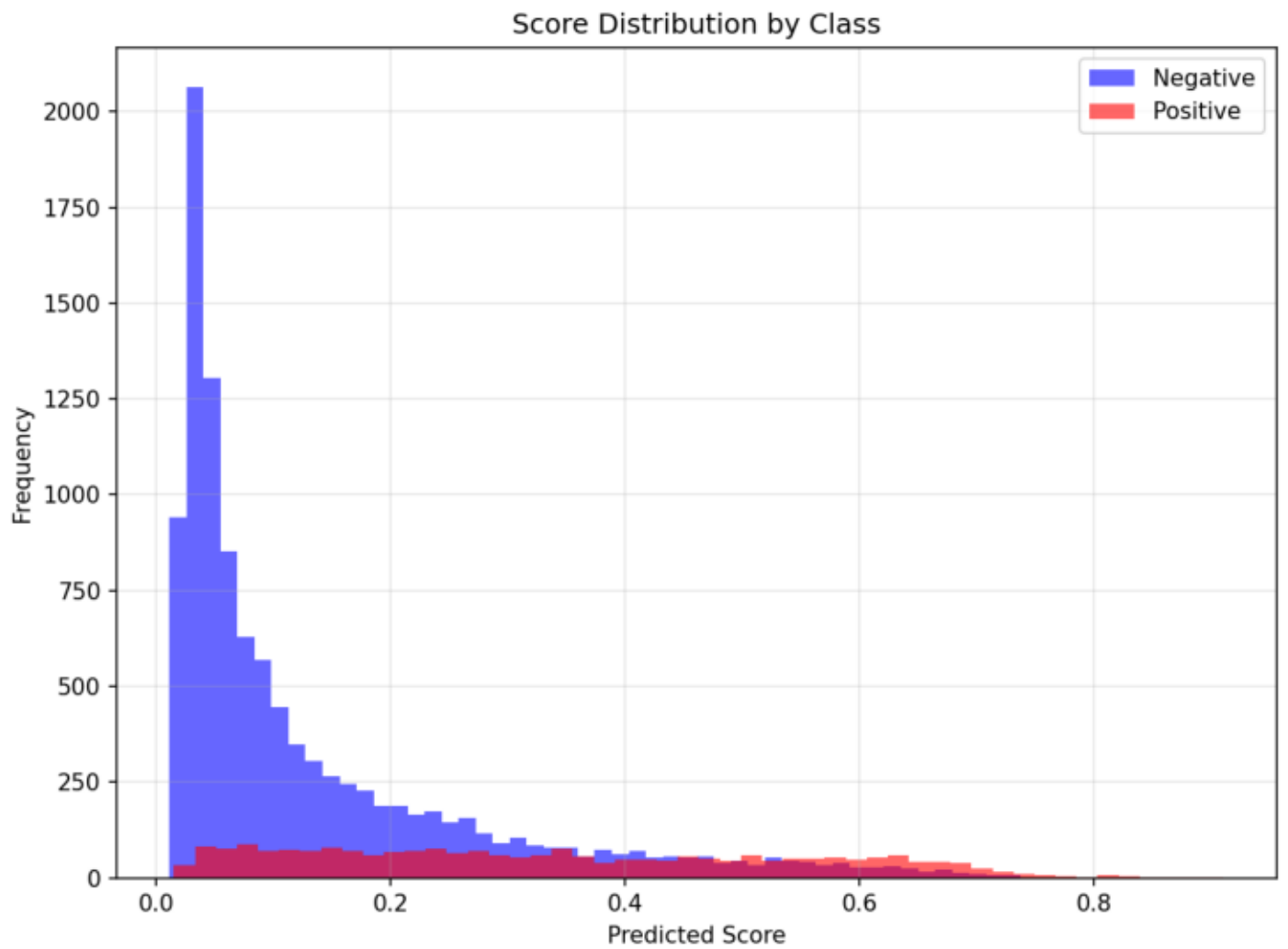
confusion_matrix_norm: Normalized rates: TPR=27.17%, TNR=95.82%.



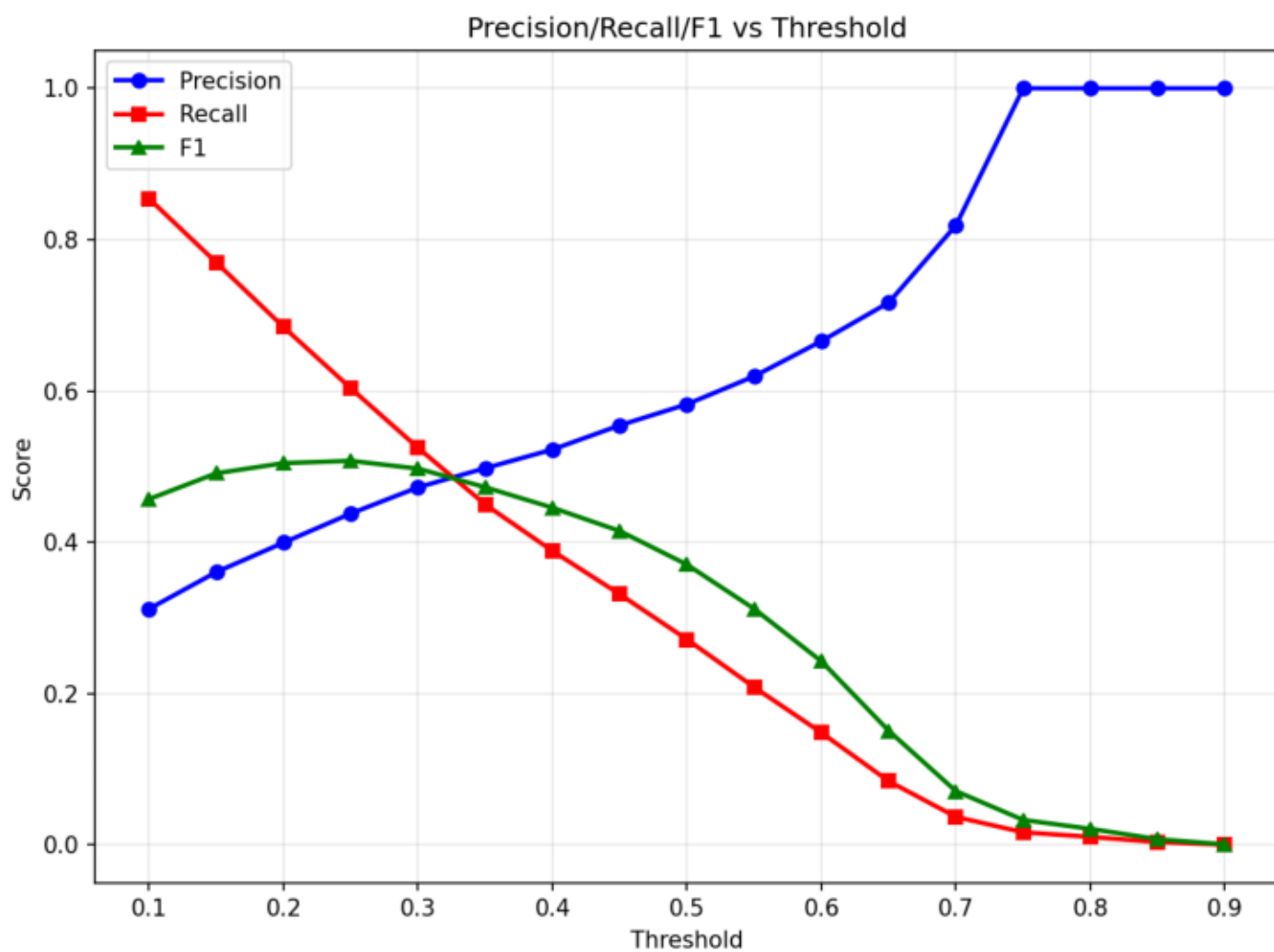
ks_curve: Maximum separation KS=0.4777 at threshold 0.14.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.342, negative=0.135 (larger gap is better).



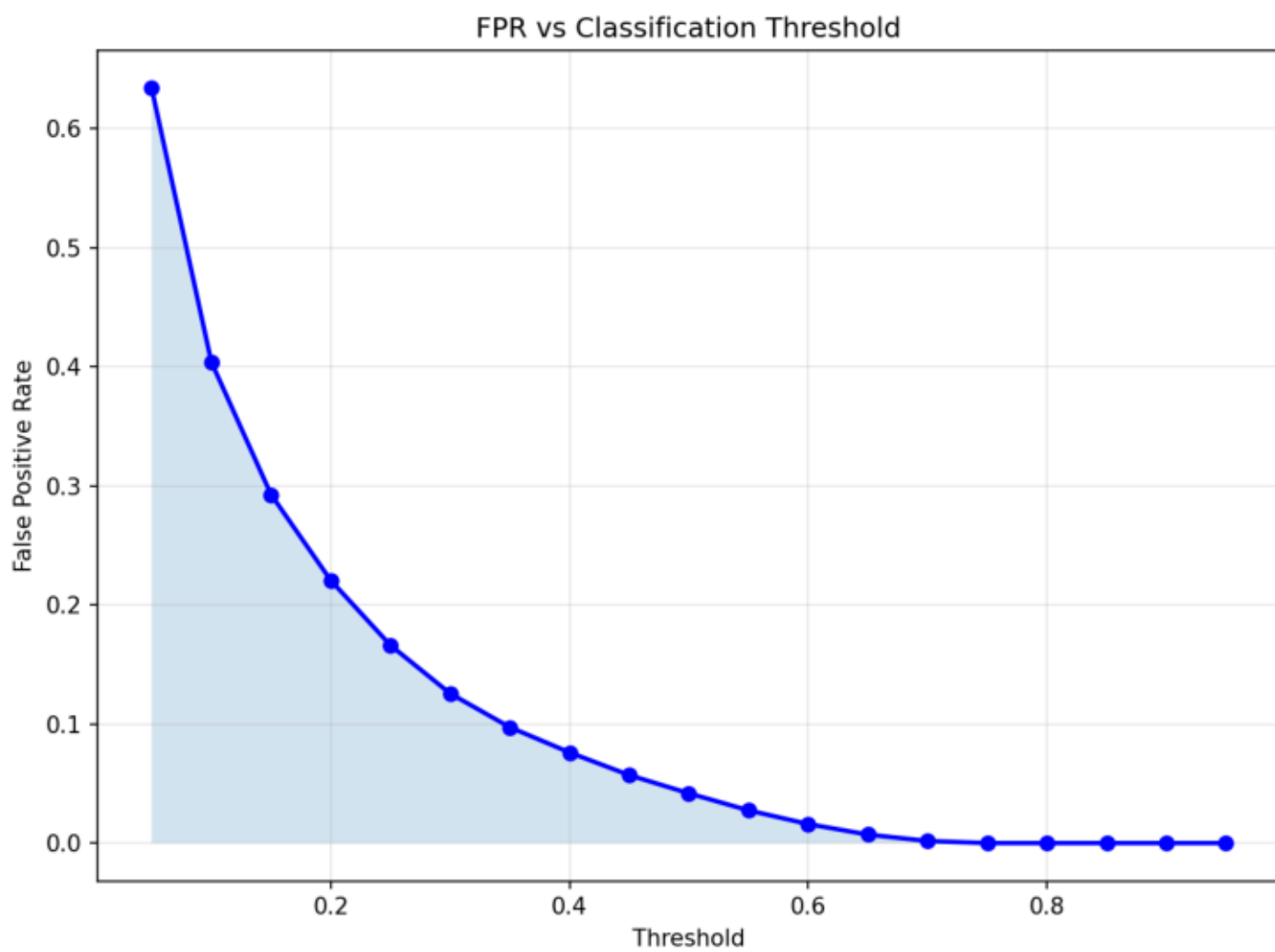
threshold_analysis: Best F1 in tested grid is 0.5077 at threshold 0.25.

value_date_quarter=2024Q3 - 2) Model Efficiency

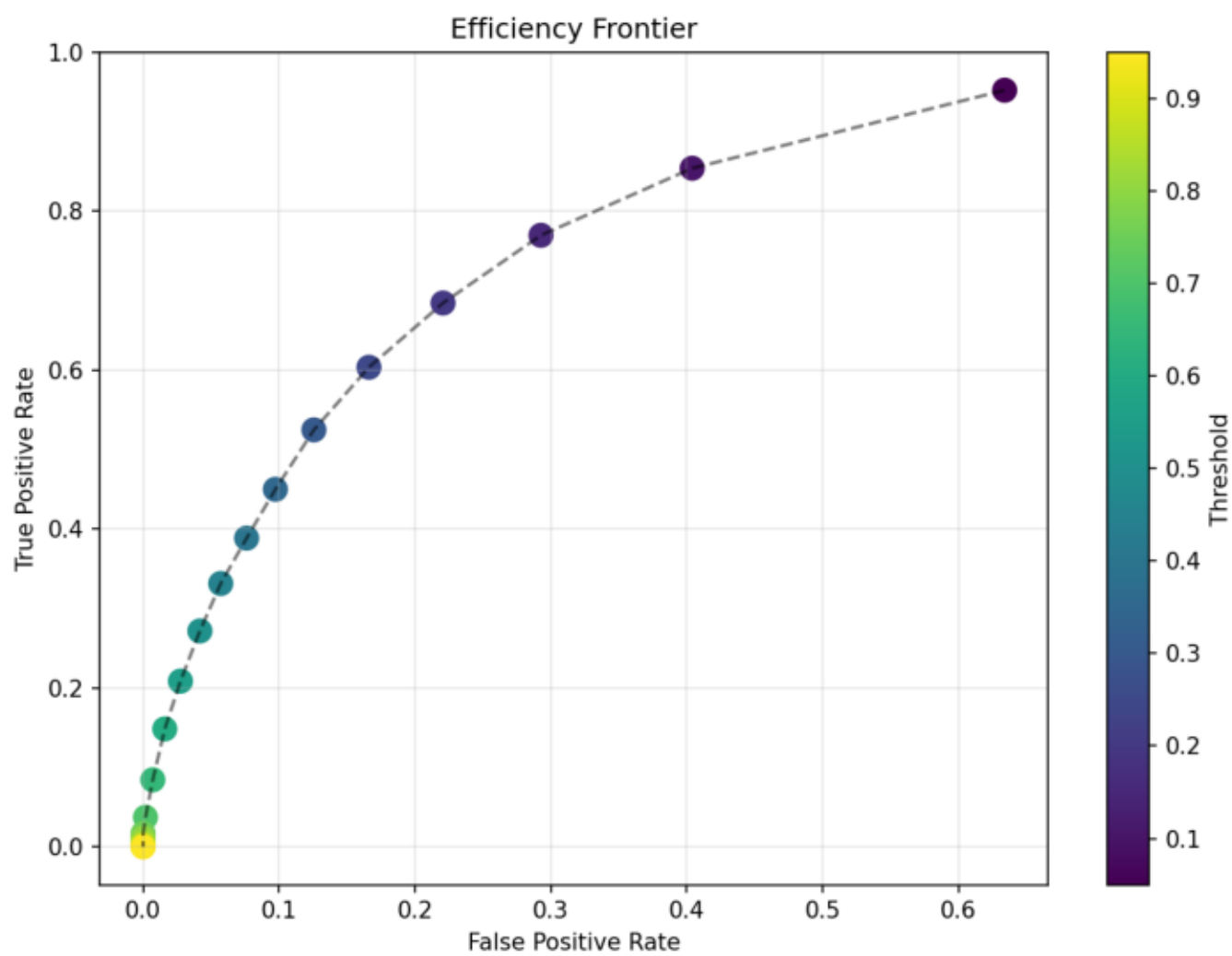
- fpr: 0.0418
- tn: 10316
- fp: 450
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.6339
- fpr_at_thresholds.t_0.10: 0.4040
- fpr_at_thresholds.t_0.15: 0.2928
- fpr_at_thresholds.t_0.20: 0.2207
- fpr_at_thresholds.t_0.25: 0.1662
- fpr_at_thresholds.t_0.30: 0.1257
- fpr_at_thresholds.t_0.35: 0.0973
- fpr_at_thresholds.t_0.40: 0.0762
- fpr_at_thresholds.t_0.45: 0.0571
- fpr_at_thresholds.t_0.50: 0.0418
- fpr_at_thresholds.t_0.55: 0.0275
- fpr_at_thresholds.t_0.60: 0.0160
- fpr_at_thresholds.t_0.65: 0.0072
- fpr_at_thresholds.t_0.70: 0.0018
- fpr_at_thresholds.t_0.75: 0.0000
- fpr_at_thresholds.t_0.80: 0.0000
- fpr_at_thresholds.t_0.85: 0.0000
- fpr_at_thresholds.t_0.90: 0.0000
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q3 - 2) Model Efficiency - Explanations

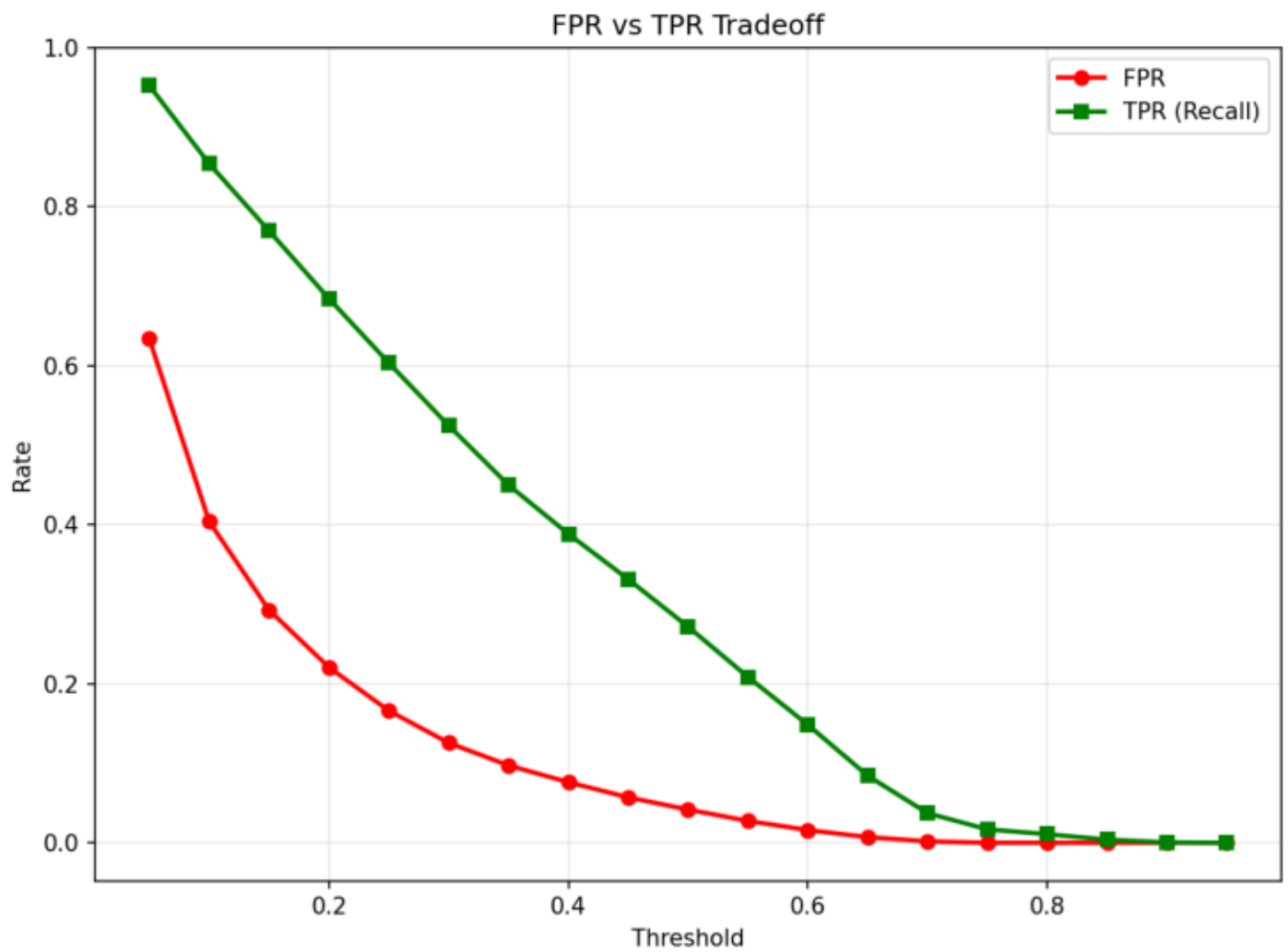
- At threshold 0.50, FPR=0.0418 (low); FP=450 out of 10766 negatives.
- At the same threshold, TPR=0.2717 with TP=628 and FN=1683, showing the capture rate of positives.
- A threshold near 0.45 yields FPR $\approx$ 0.0571 with TPR $\approx$ 0.3315 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

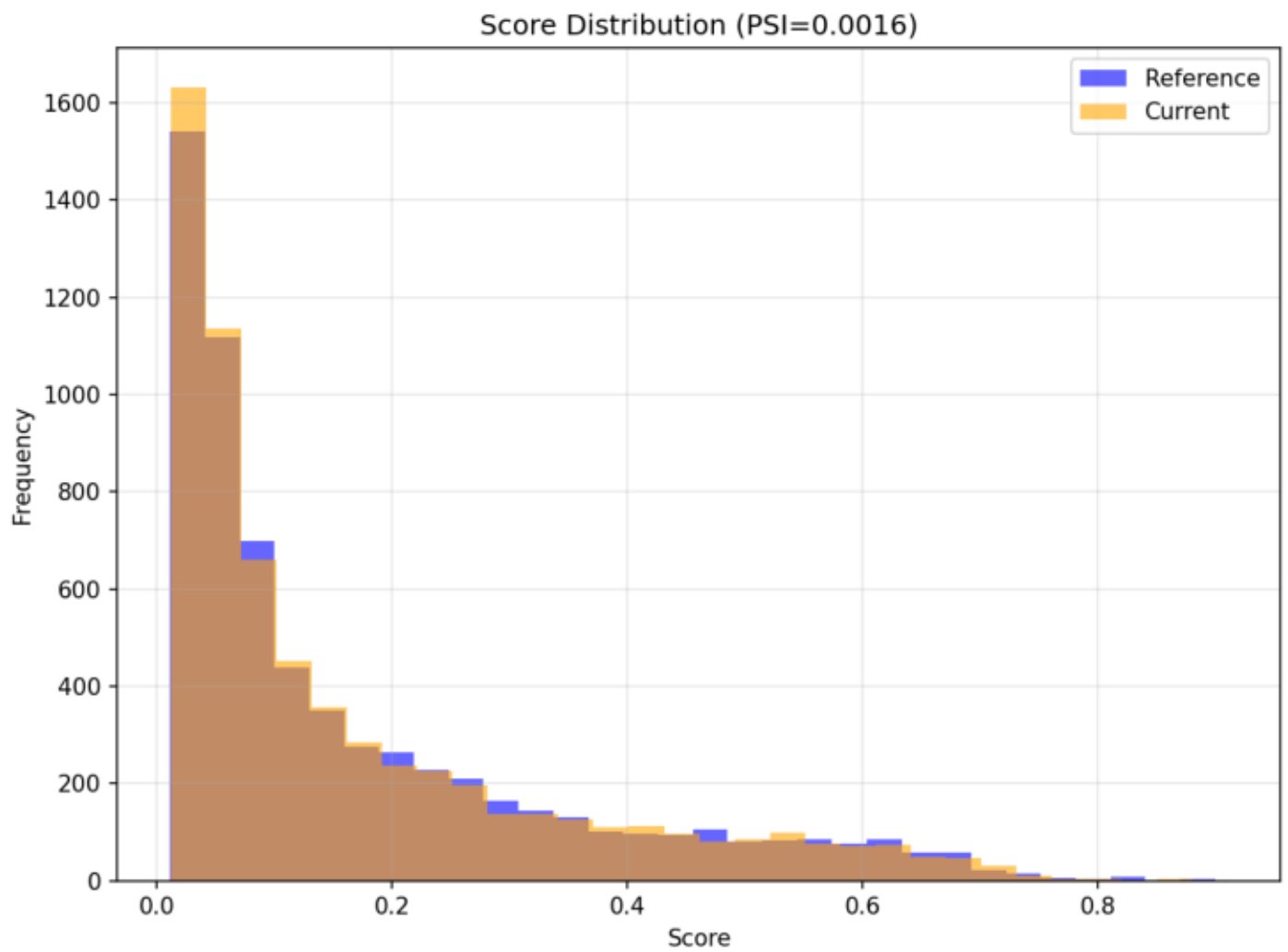


value_date_quarter=2024Q3 - 3) Model Stability

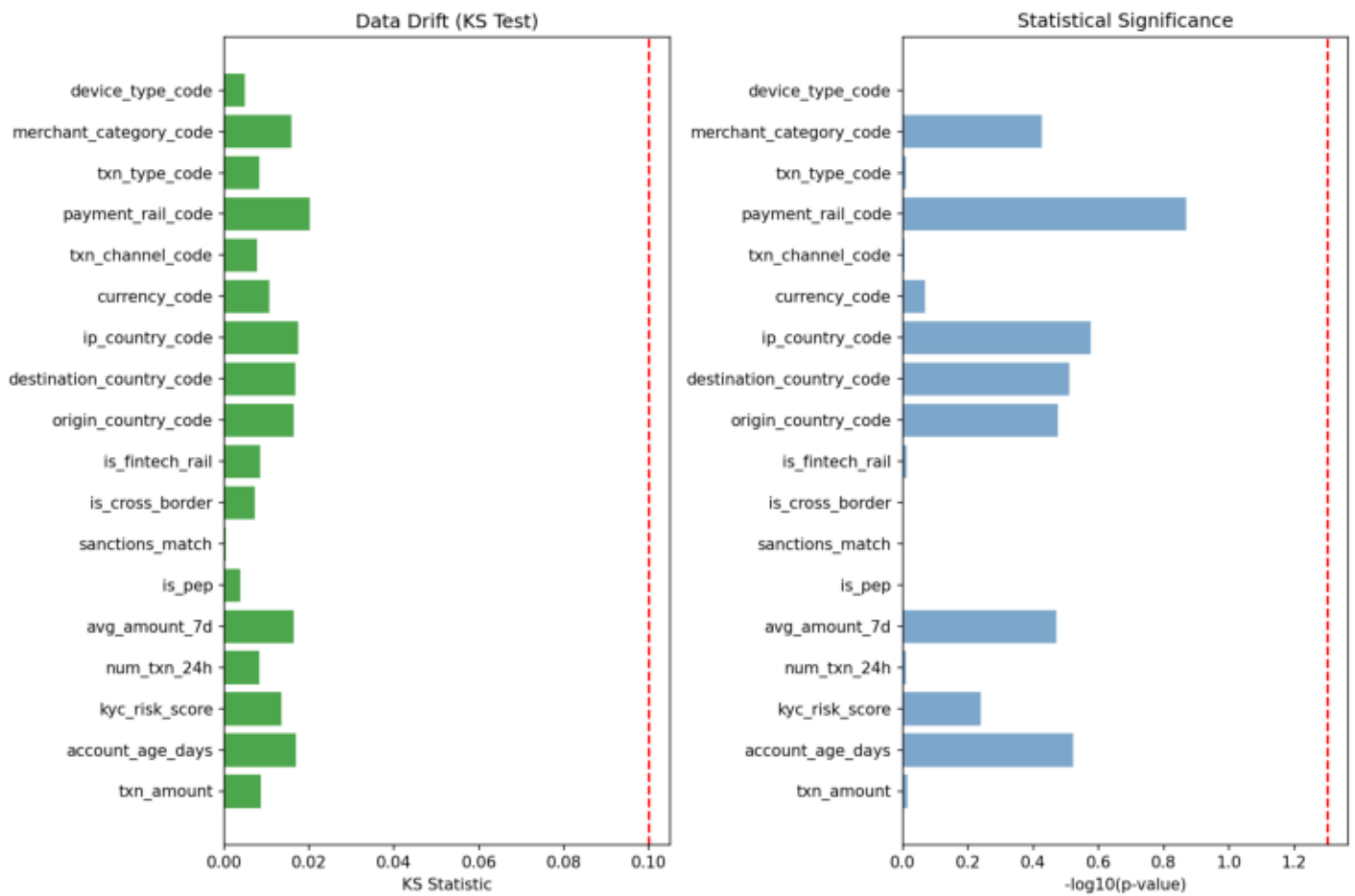
- psi: 0.0016
- cv_auc_roc_mean: 0.8126
- cv_auc_roc_std: 0.0139
- cv_auc_pr_mean: 0.4935
- cv_auc_pr_std: 0.0318
- bootstrap_auc_roc_mean: 0.8120
- bootstrap_auc_roc_ci_lower: 0.7996
- bootstrap_auc_roc_ci_upper: 0.8228
- concept_drift_detected: False
- concept_drift_score: 0.0309

value_date_quarter=2024Q3 - 3) Model Stability - Explanations

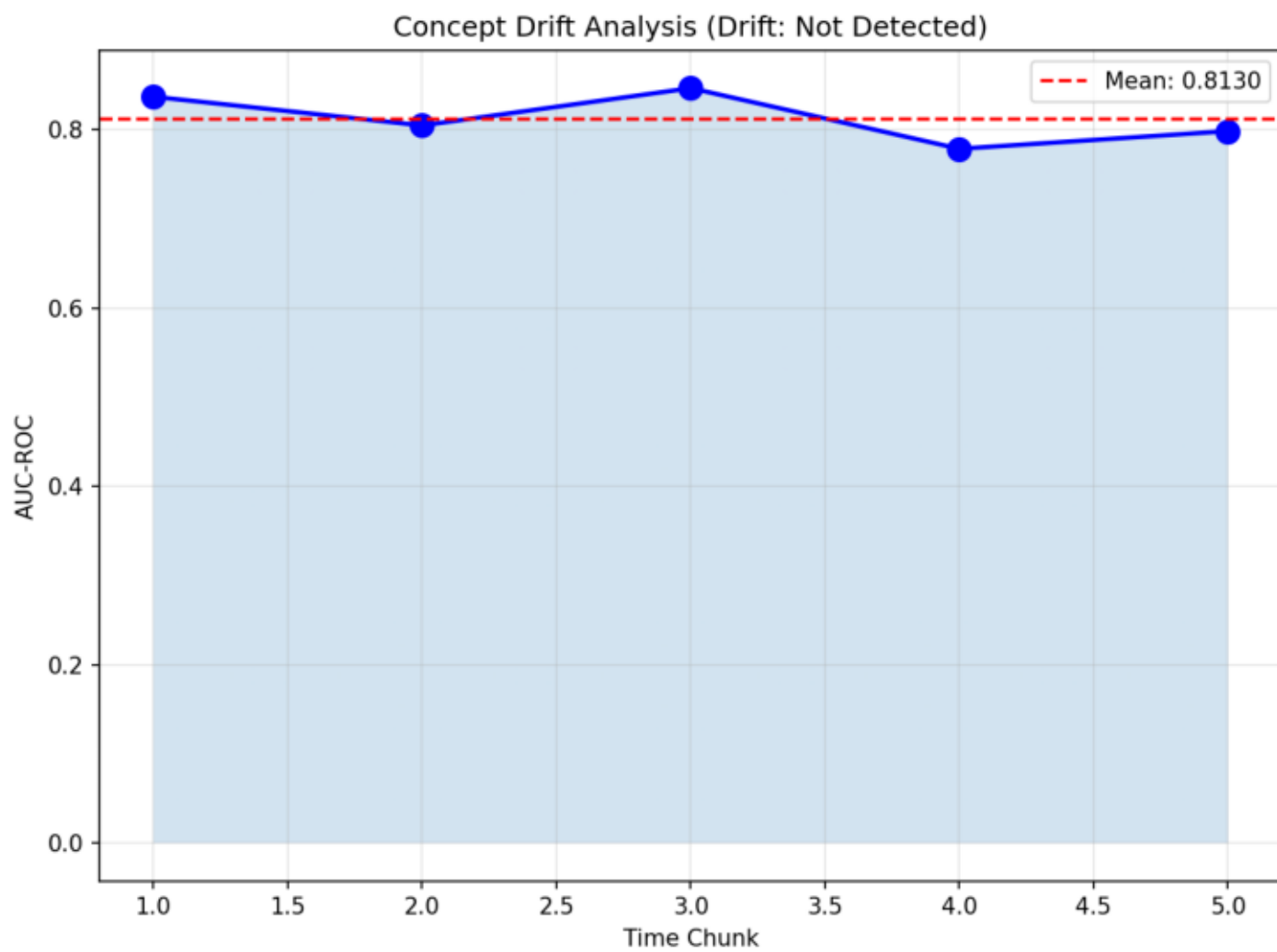
- PSI is 0.0016, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8126 ± 0.0139 (stable across folds).
- CV AUC-PR is 0.4935 ± 0.0318 (variable across folds).
- Bootstrap AUC-ROC mean is 0.8120 with 95% CI [0.7996, 0.8228] (width 0.0232).
- No concept drift detected (score 0.0309), suggesting stable performance over time.
- Data drift flagged 0/18 numeric features; top drift signals: ['payment_rail_code', 'ip_country_code', 'account_age_days'].



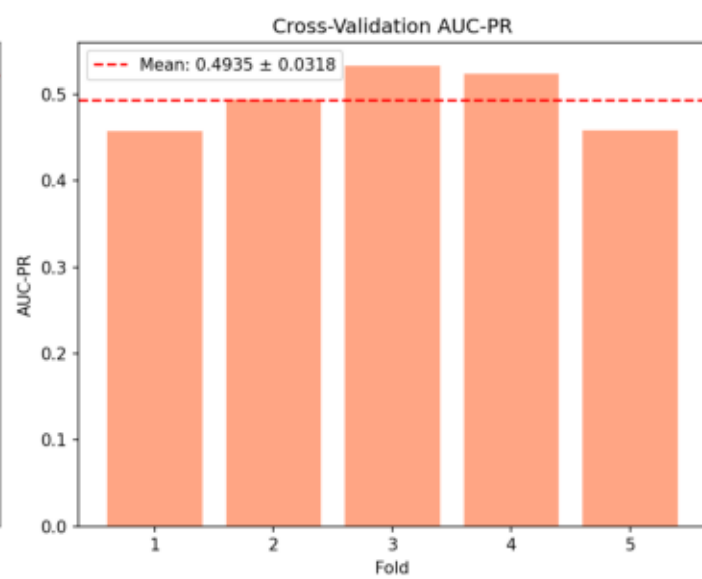
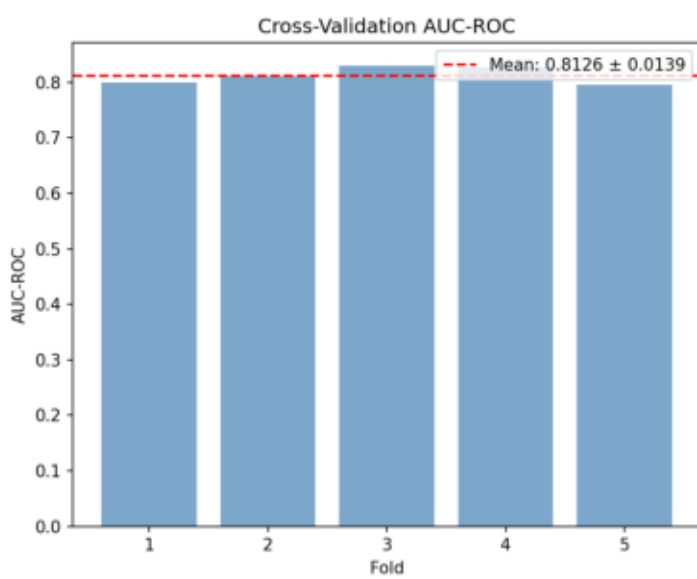
psi_distribution: Score distributions overlap consistent with PSI=0.0016.



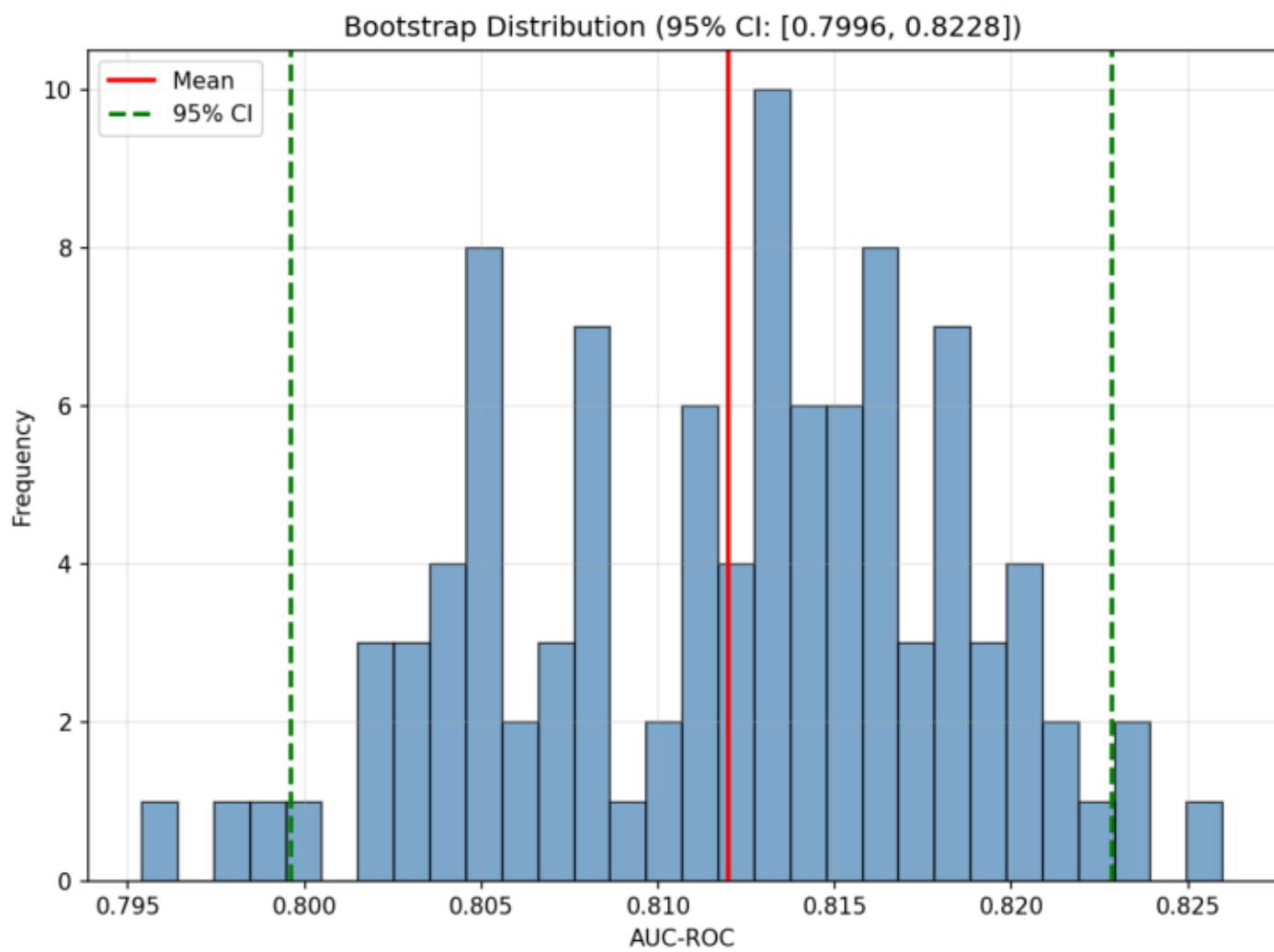
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.



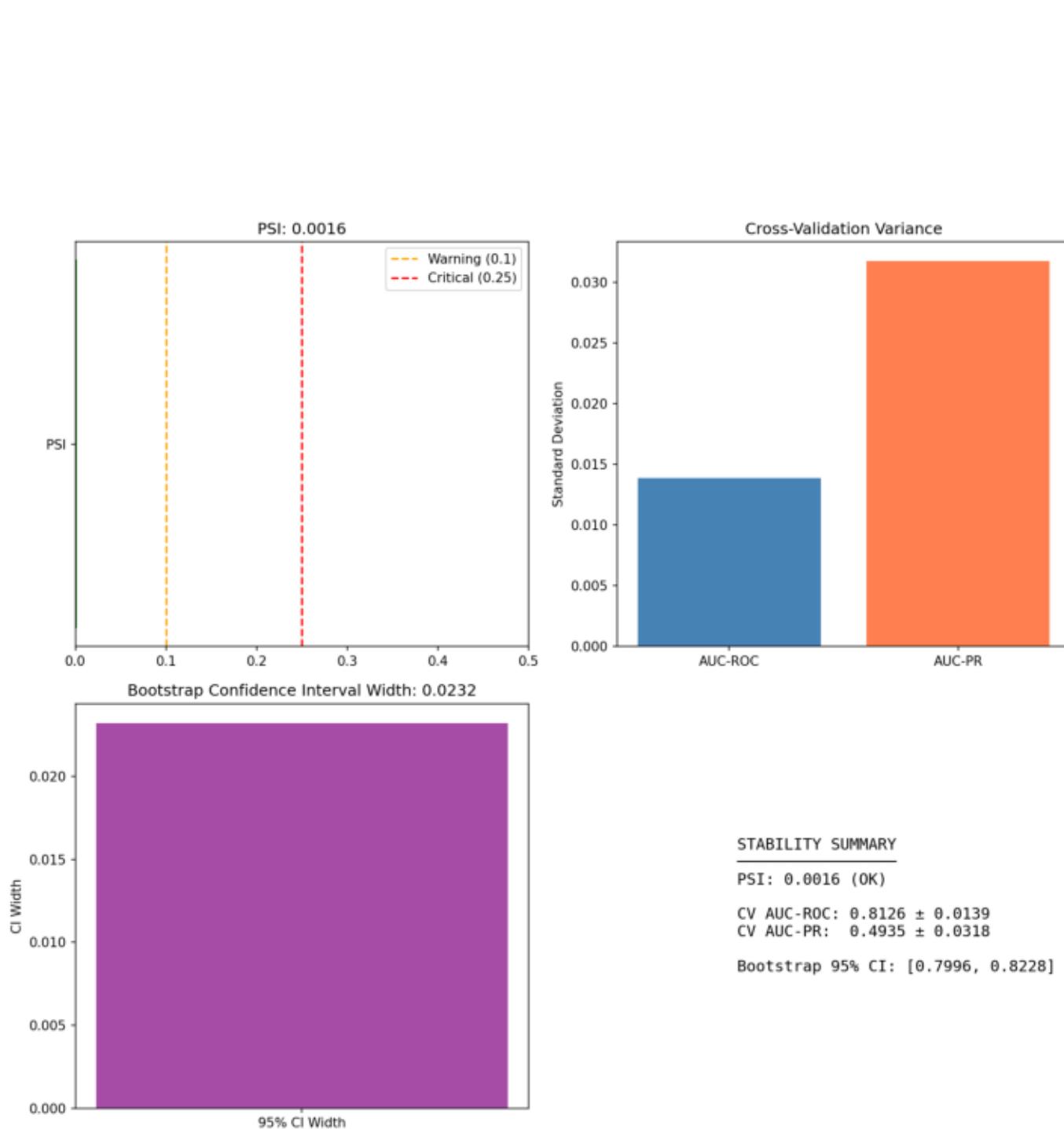
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8126 and PR 0.4935.



bootstrap_distribution: Bootstrap CI [0.7996, 0.8228] reflects performance stability.

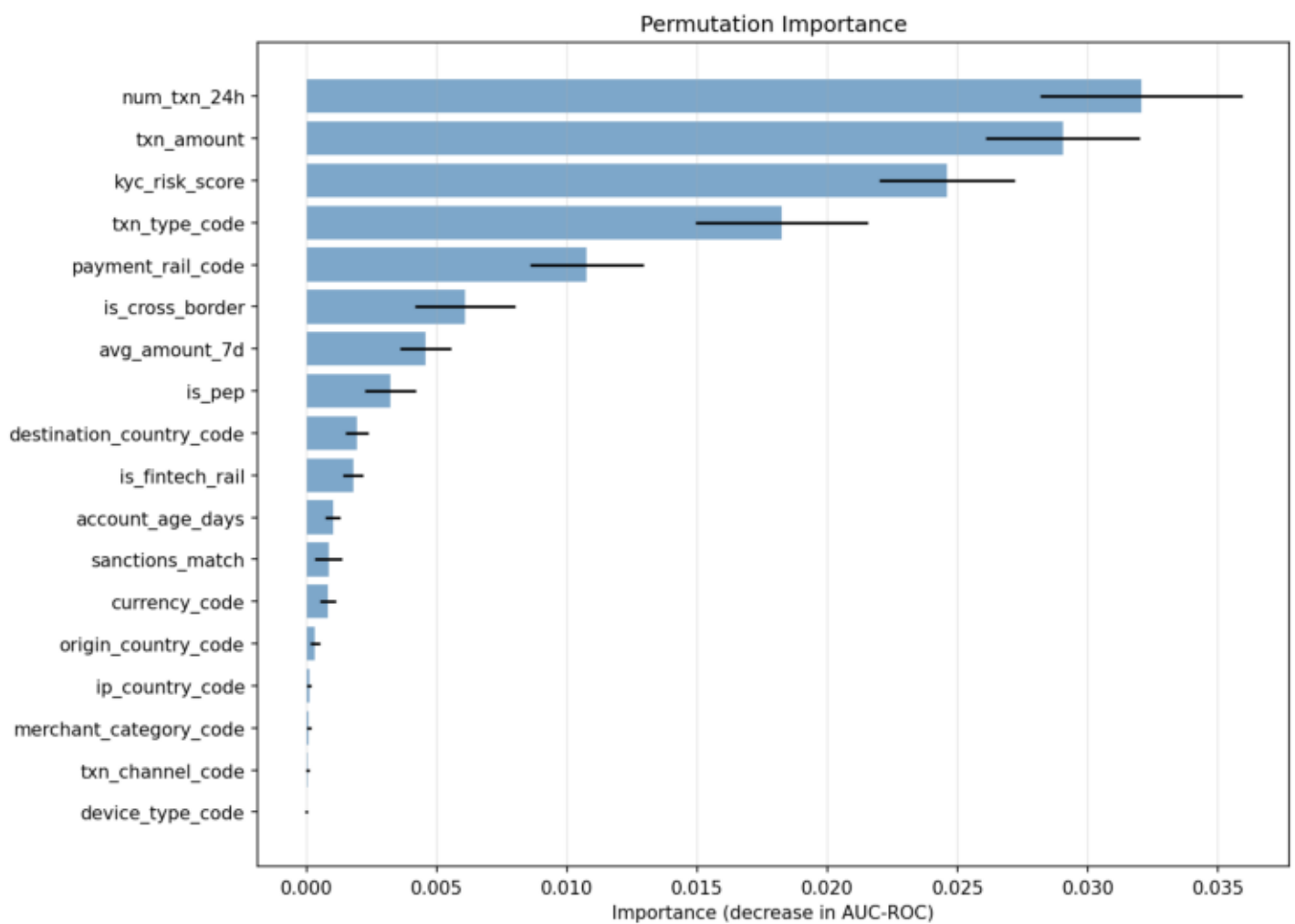


value_date_quarter=2024Q3 - 4) Model Interpretability

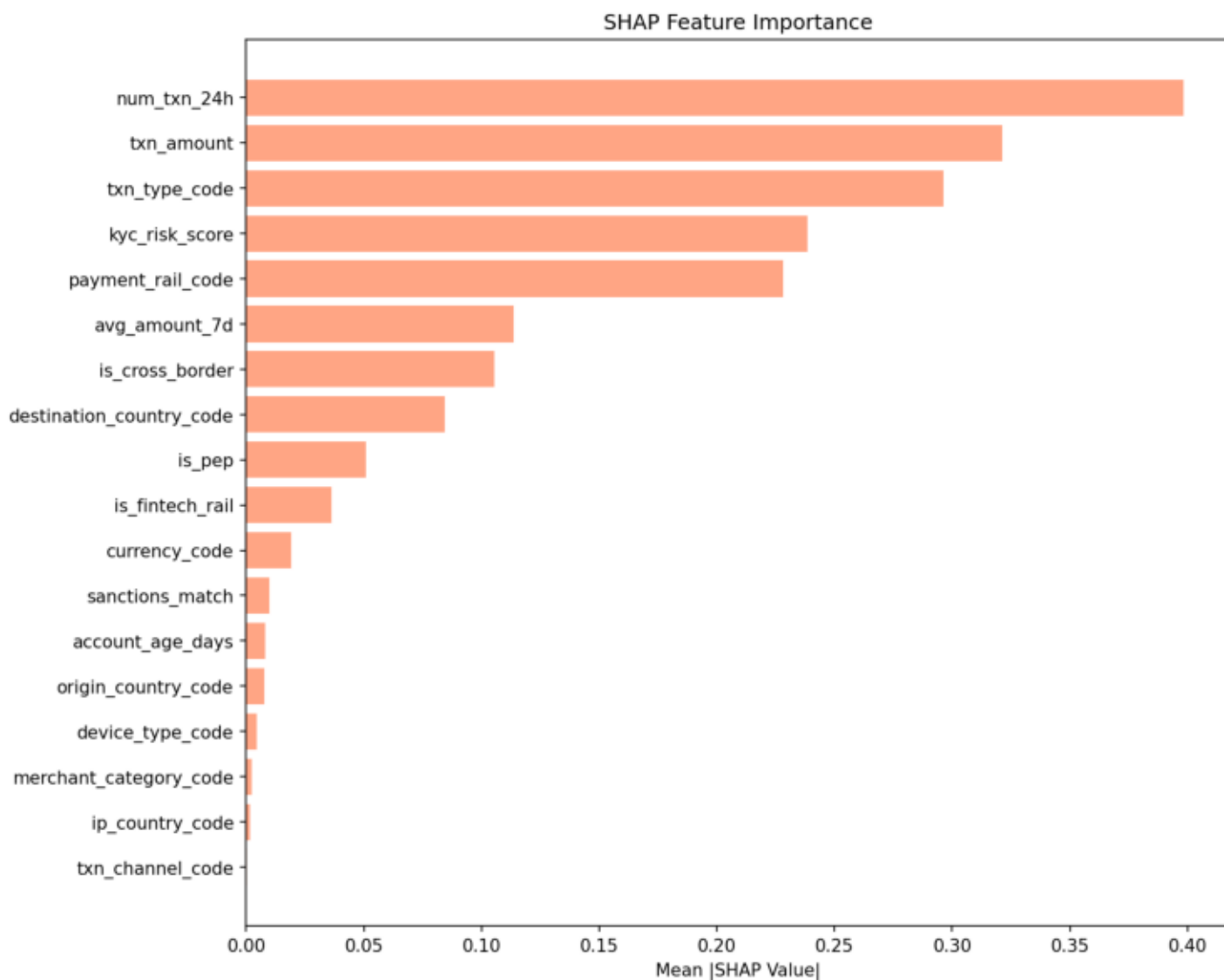
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'shap', 'lime']
- perm_top_features: ['num_txn_24h', 'txn_amount', 'kyc_risk_score', 'txn_type_code', 'payment_rail_code']...
- shap_top_features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code']...
- lime_instances: 3
- pdp_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q3 - 4) Model Interpretability - Explanations

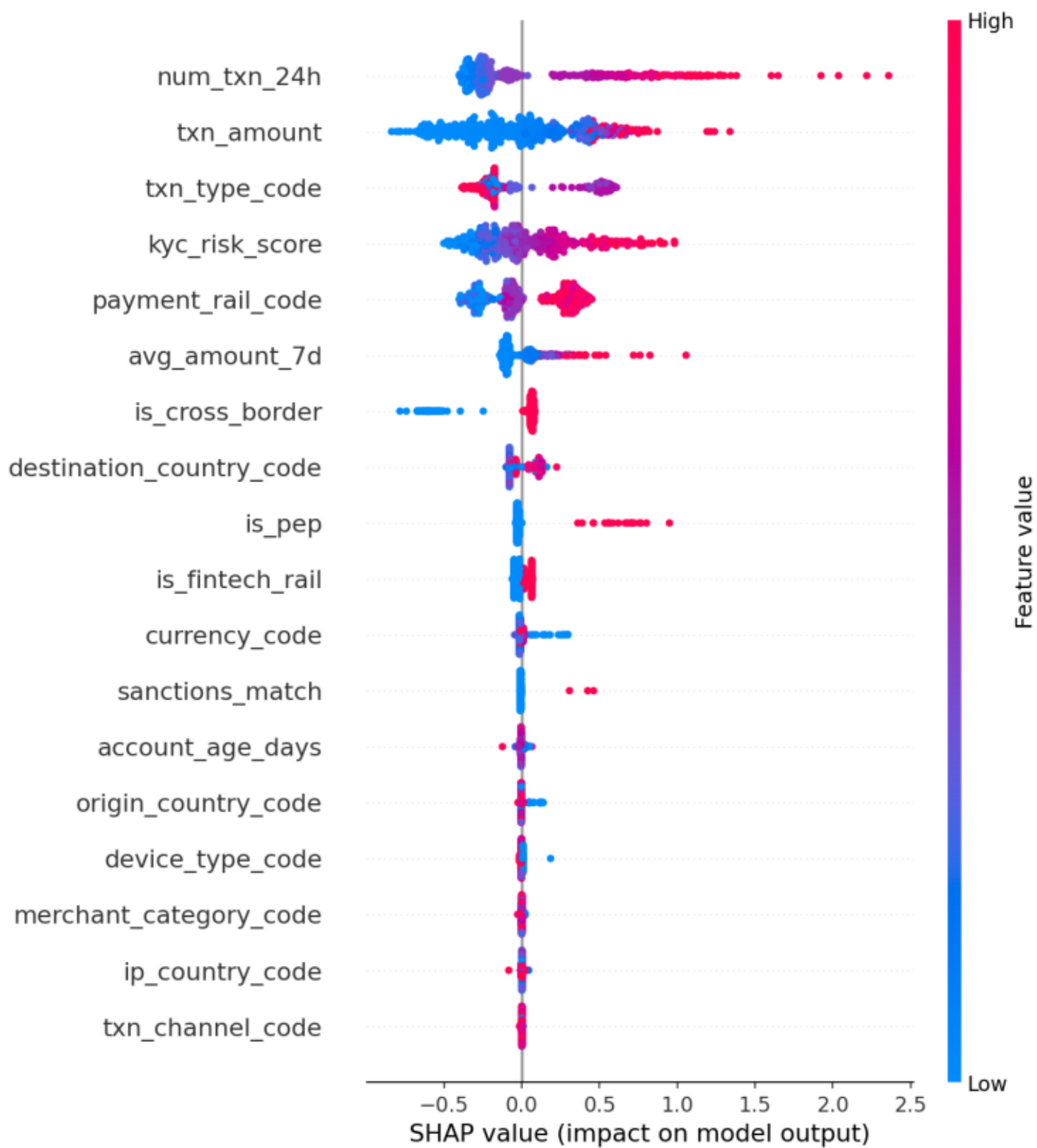
- Permutation importance highlights: ['num_txn_24h', 'txn_amount', 'kyc_risk_score', 'txn_type_code', 'payment_rail_code'].
- SHAP top features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code'].
- LIME generated local explanations for 3 instances; top contributors for instance_3: ['num_txn_24h > 5.00', 'sanctions_match <= 0.00', 'is_pep <= 0.00', 'kyc_risk_score > 51.40', '4.00 < txn_type_code <= 7.00'].
- PDP plots show average effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'origin_country_code'].
- ICE plots show individual-level effects for: ['avg_amount_7d', 'txn_amount', 'account_age_days', 'kyc_risk_score'].

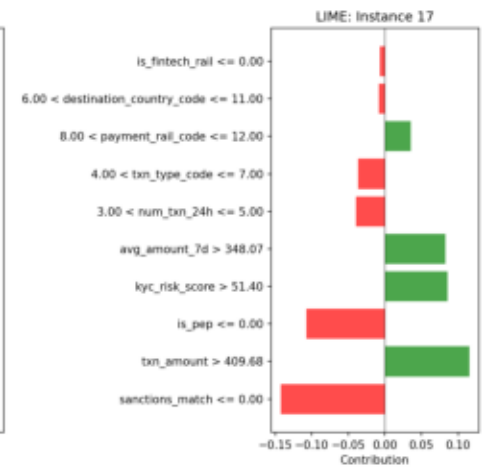
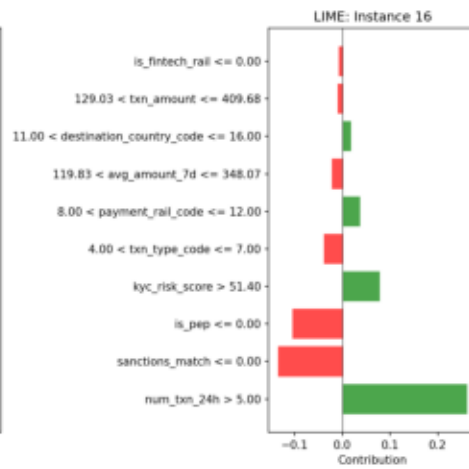
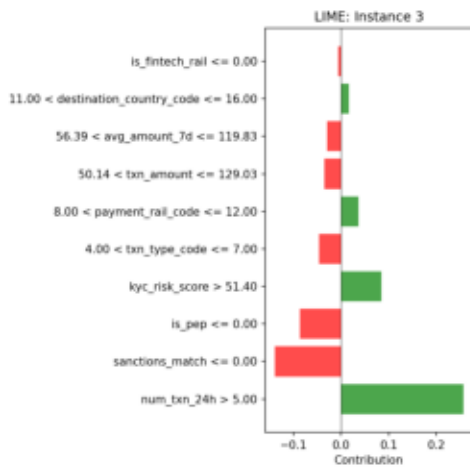


permutation_importance: Bars show how much AUC drops when each feature is permuted.



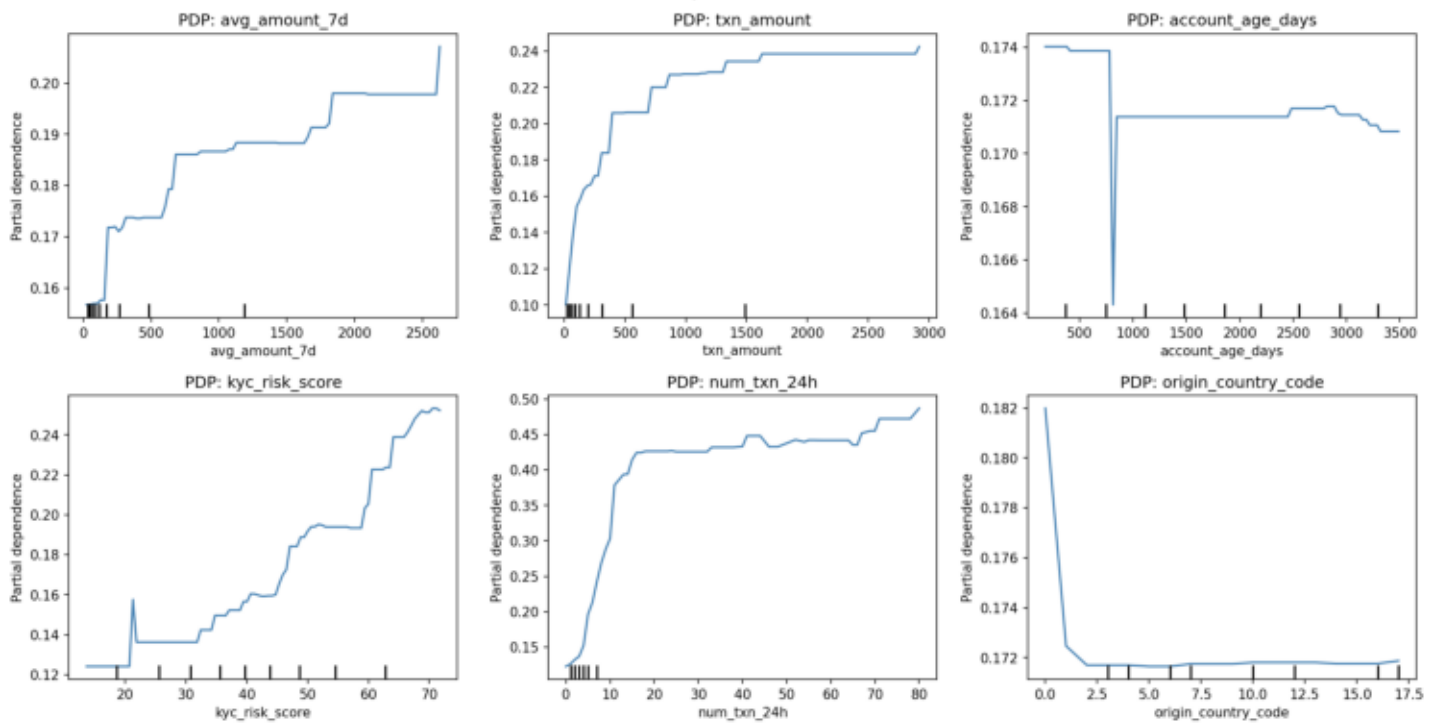
shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.





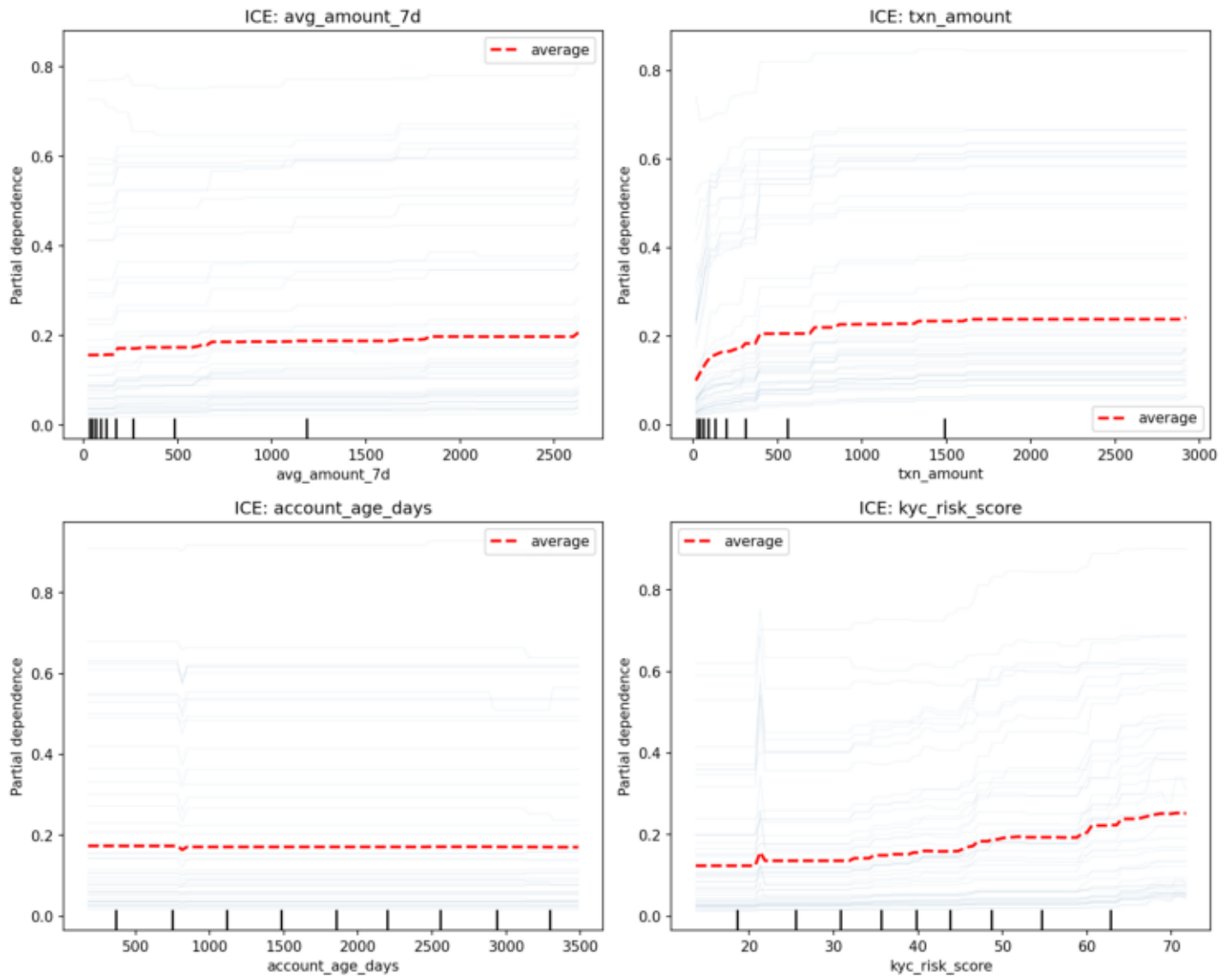
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



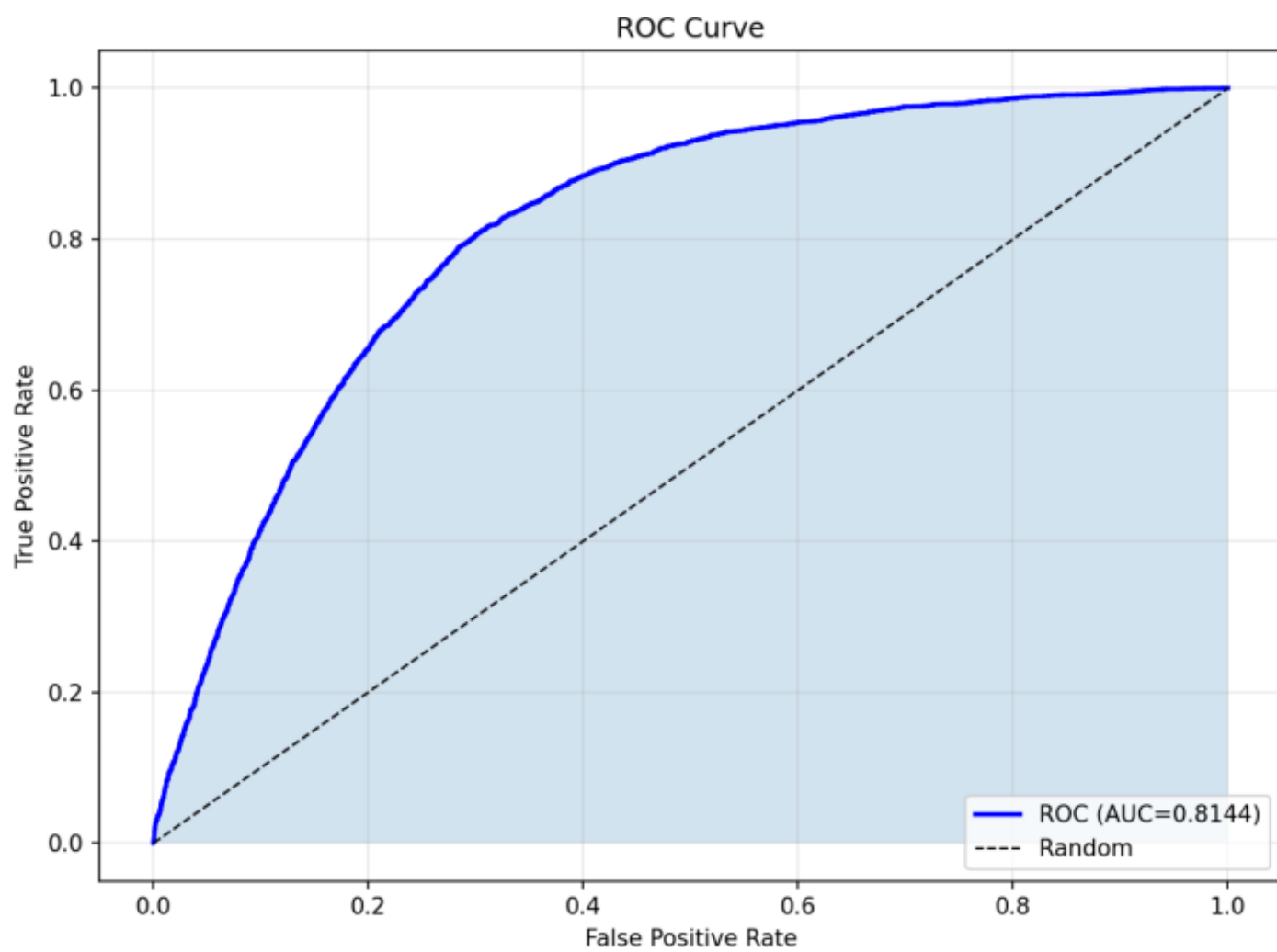
ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q4 - 1) Model Effectiveness

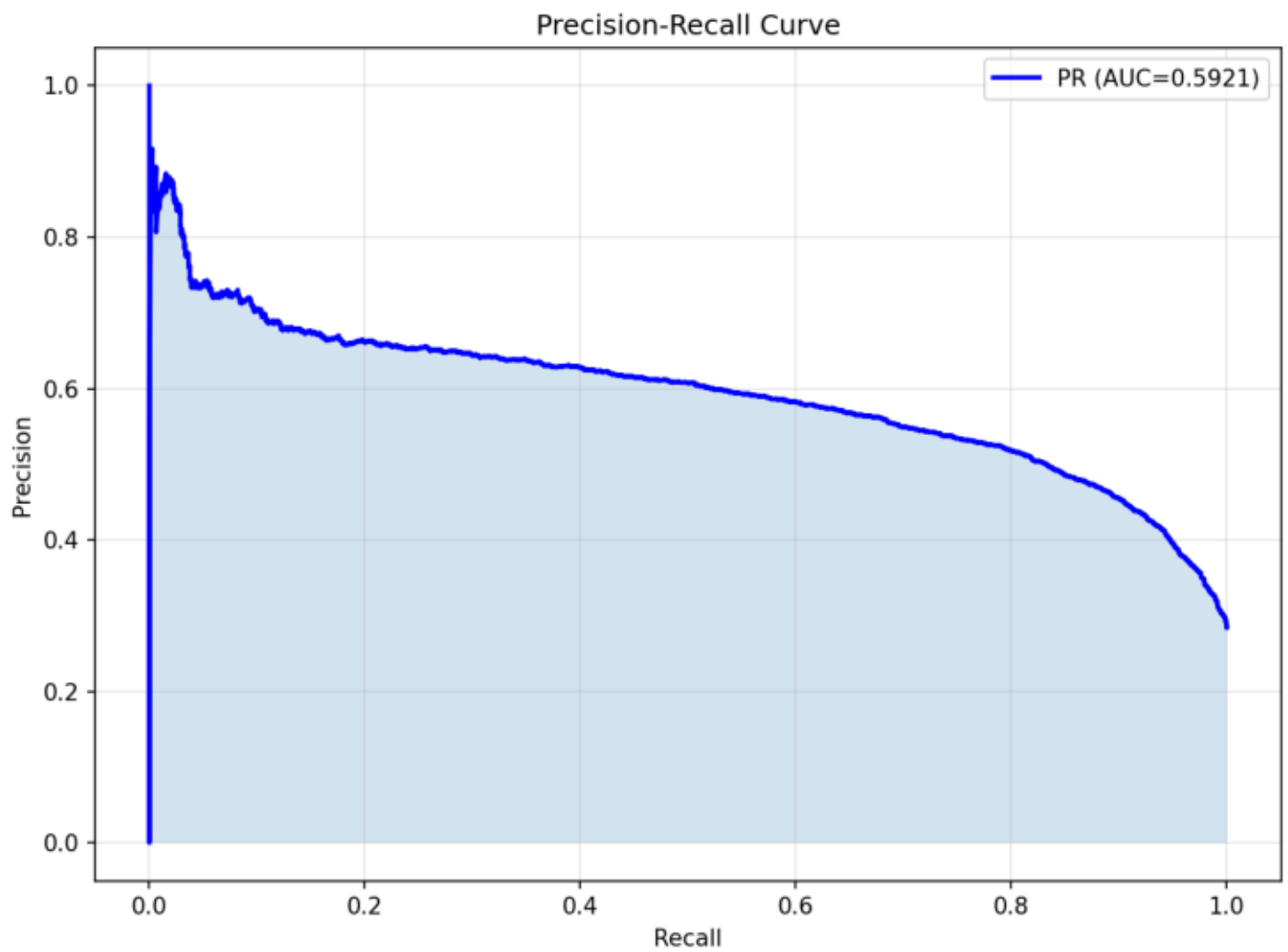
- auc_roc: 0.8144
- auc_pr: 0.5921
- precision: 0.6086
- recall: 0.4920
- f1: 0.5442
- ks_statistic: 0.5053
- ks_threshold: 0.2300
- confusion_matrix.TN: 8707
- confusion_matrix.FP: 1252
- confusion_matrix.FN: 2010
- confusion_matrix.TP: 1947
- precision_at_k.10: 0.9000
- precision_at_k.50: 0.8600
- precision_at_k.100: 0.8700
- precision_at_k.200: 0.7450
- precision_at_k.500: 0.7180
- recall_at_k.10: 0.0023
- recall_at_k.50: 0.0109
- recall_at_k.100: 0.0220
- recall_at_k.200: 0.0377
- recall_at_k.500: 0.0907

value_date_quarter=2024Q4 - 1) Model Effectiveness - Explanations

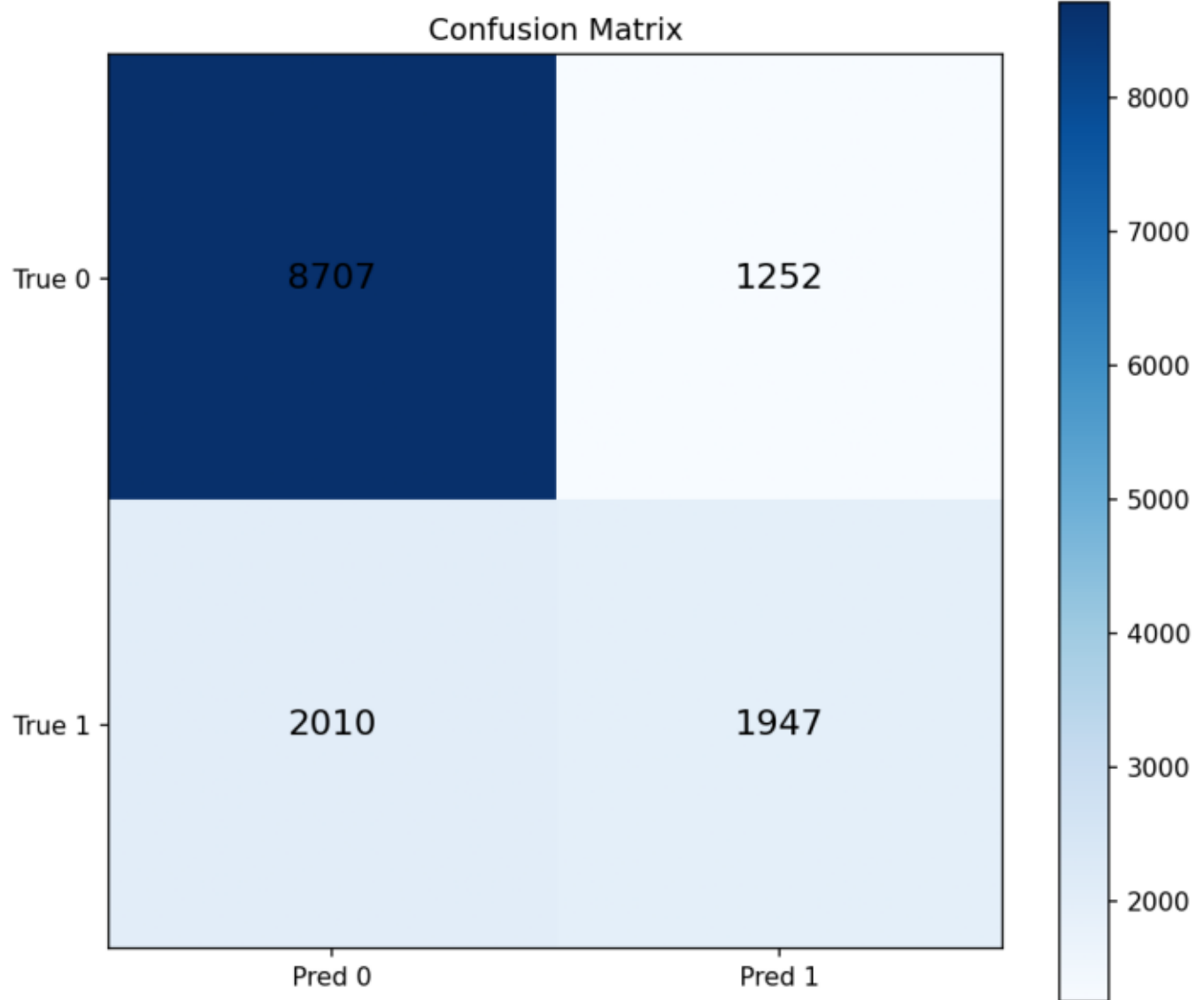
- Positive rate is 28.43%; AUC-ROC is 0.8144 (strong ranking).
- AUC-PR is 0.5921, above the base rate 28.43%, indicating meaningful lift.
- At threshold 0.50, precision=0.6086, recall=0.4920, F1=0.5442 (precision-heavy (fewer positives caught, fewer false alarms)).
- Confusion matrix: TP=1947, FP=1252, FN=2010, TN=8707 (FPR=12.57%, FNR=50.80%).
- KS is 0.5053 at threshold 0.23, indicating very strong separation.
- Top-10: precision=0.9000, recall=0.0023 shows the quality of the highest-risk shortlist.
- Top-500: precision=0.7180, recall=0.0907 shows how much coverage you get with a larger queue.



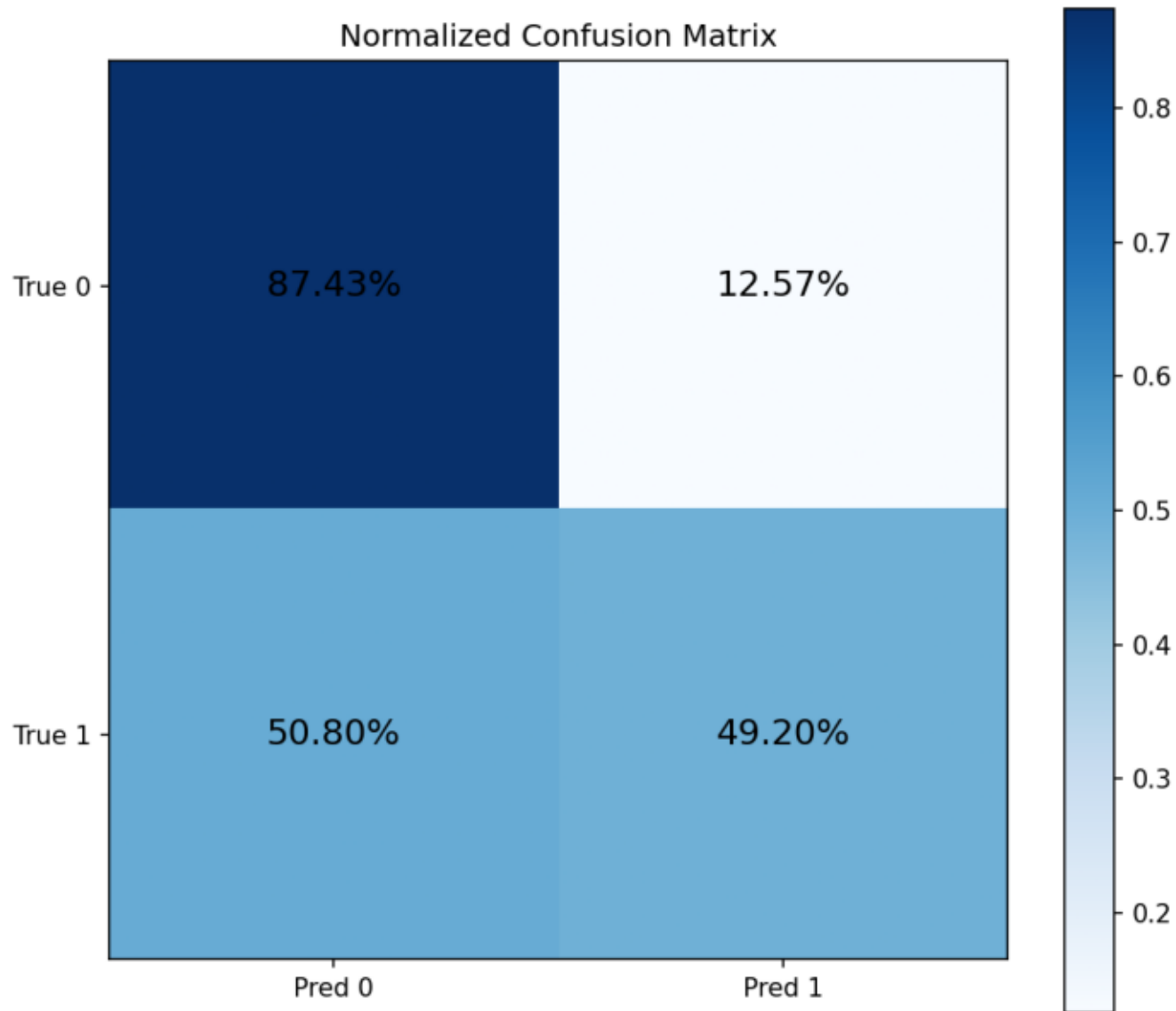
roc_curve: AUC-ROC=0.8144; curve above diagonal indicates strong separation.



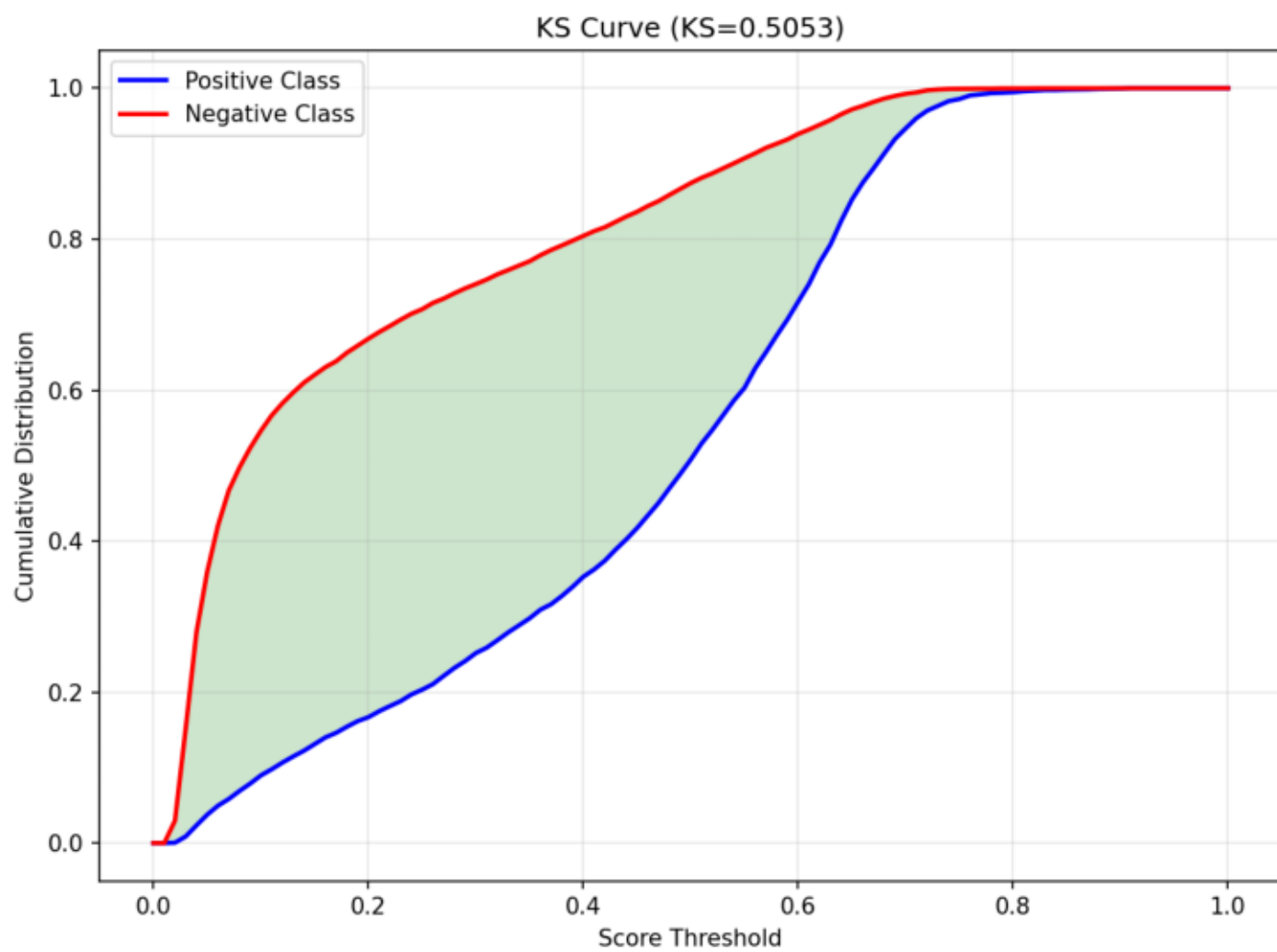
pr_curve: AUC-PR=0.5921 vs baseline 28.43%; higher curve implies better precision at recall.



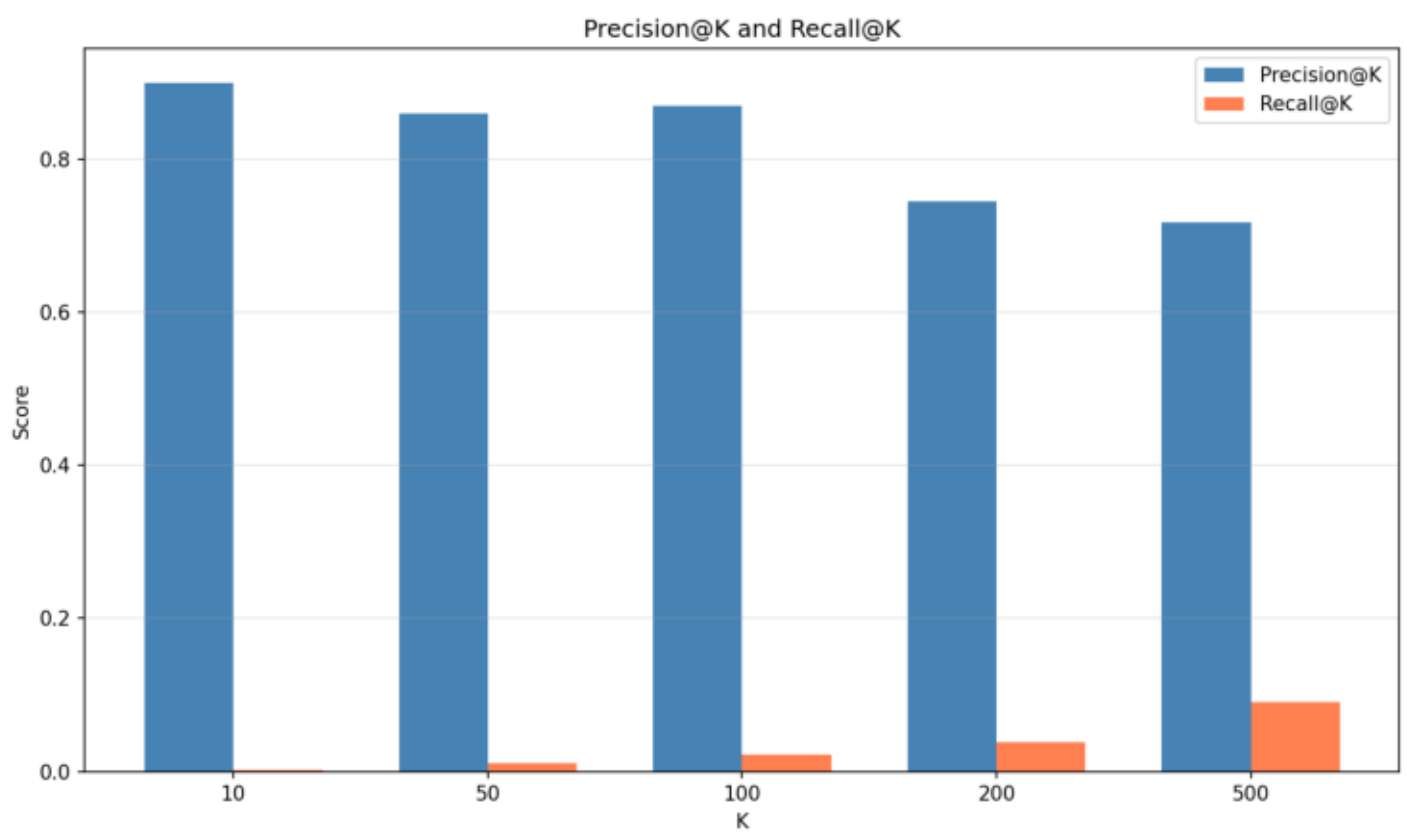
confusion_matrix: At threshold 0.50: TP=1947, FP=1252, FN=2010, TN=8707.



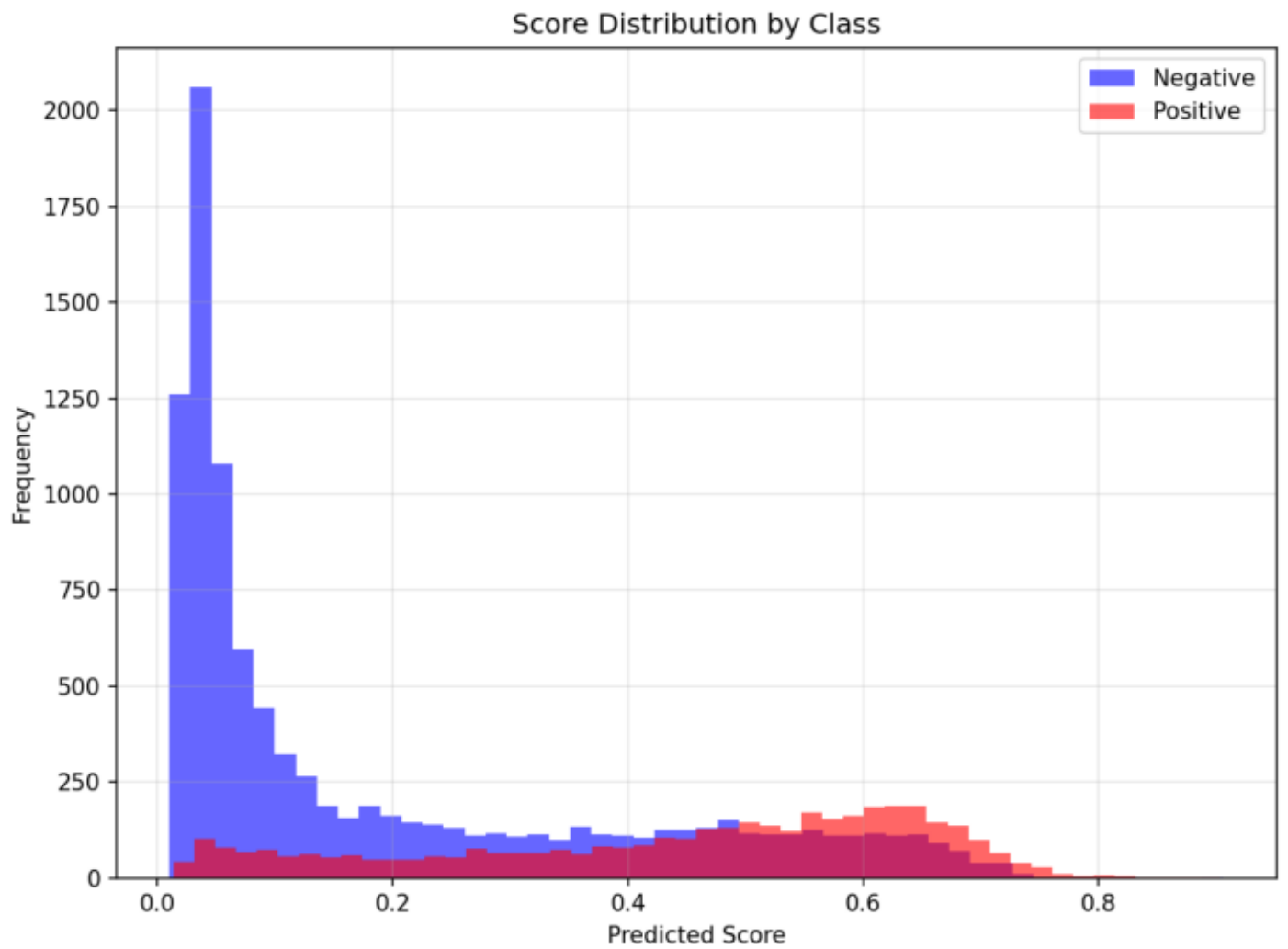
confusion_matrix_norm: Normalized rates: TPR=49.20%, TNR=87.43%.



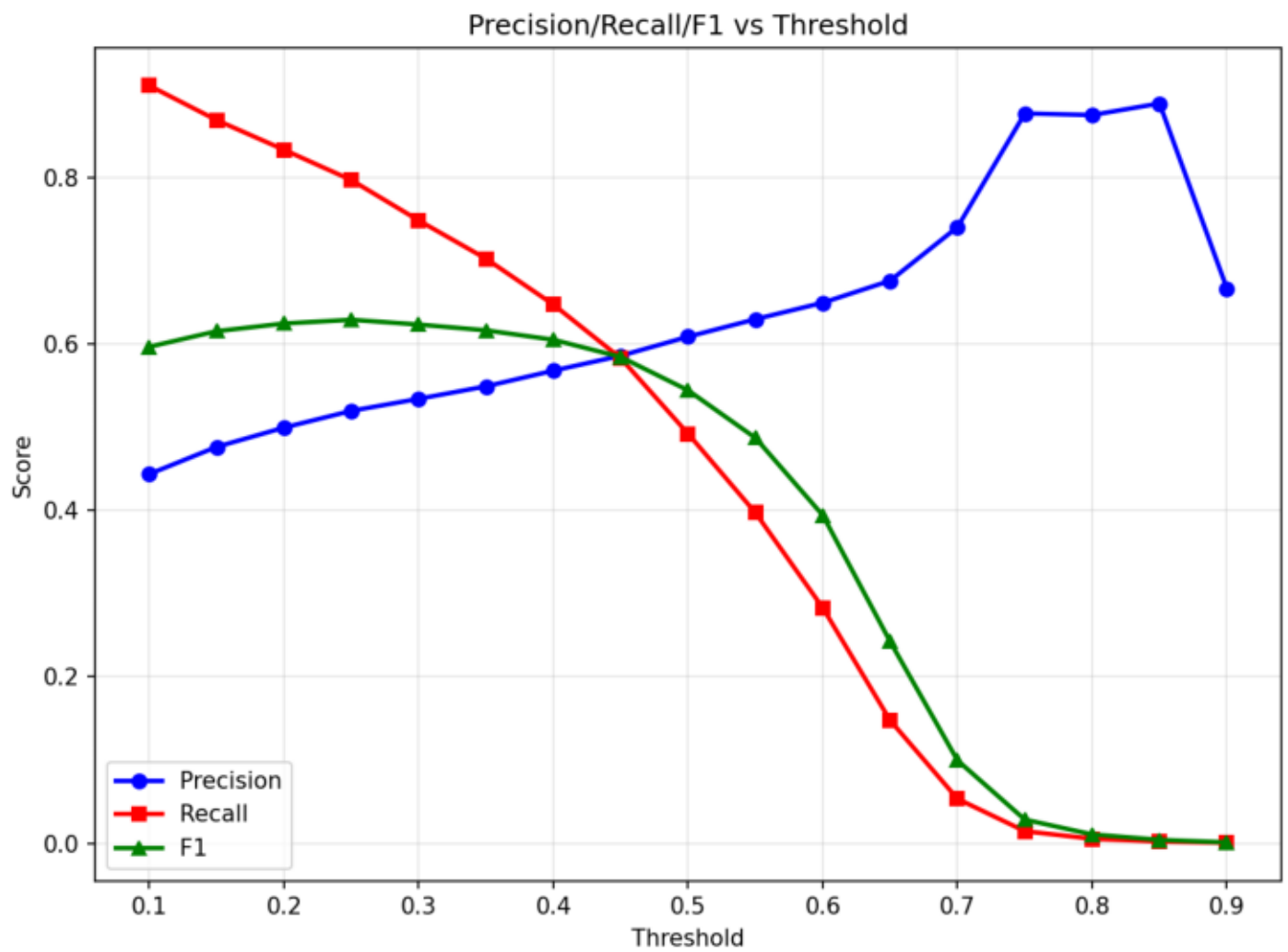
ks_curve: Maximum separation KS=0.5053 at threshold 0.23.



precision_recall_at_k: Bars show precision/recall as you expand the review queue.



score_distribution: Mean score: positive=0.448, negative=0.189 (larger gap is better).



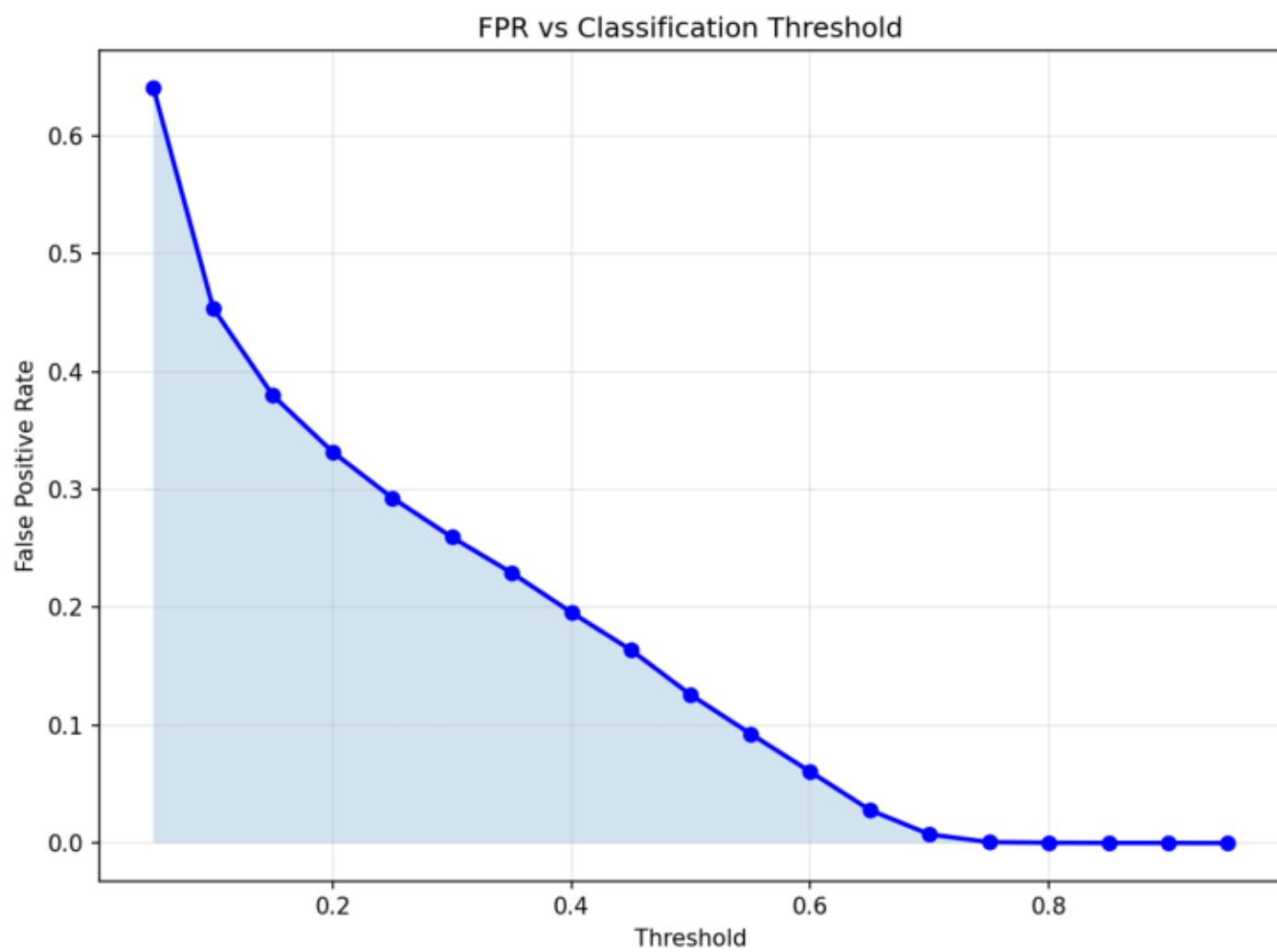
threshold_analysis: Best F1 in tested grid is 0.6290 at threshold 0.25.

value_date_quarter=2024Q4 - 2) Model Efficiency

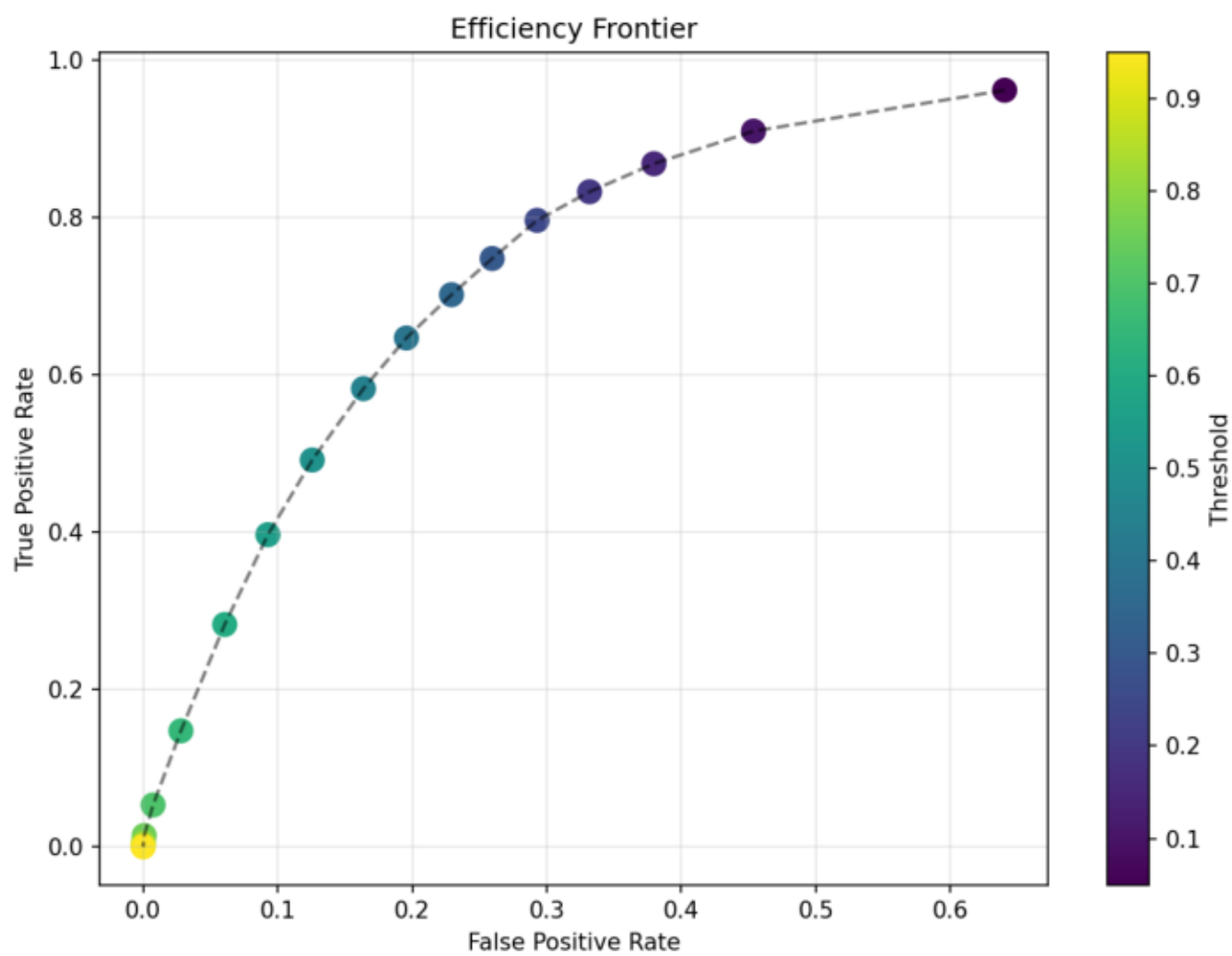
- fpr: 0.1257
- tn: 8707
- fp: 1252
- threshold: 0.5000
- fpr_at_thresholds.t_0.05: 0.6403
- fpr_at_thresholds.t_0.10: 0.4537
- fpr_at_thresholds.t_0.15: 0.3797
- fpr_at_thresholds.t_0.20: 0.3319
- fpr_at_thresholds.t_0.25: 0.2928
- fpr_at_thresholds.t_0.30: 0.2595
- fpr_at_thresholds.t_0.35: 0.2292
- fpr_at_thresholds.t_0.40: 0.1957
- fpr_at_thresholds.t_0.45: 0.1638
- fpr_at_thresholds.t_0.50: 0.1257
- fpr_at_thresholds.t_0.55: 0.0928
- fpr_at_thresholds.t_0.60: 0.0606
- fpr_at_thresholds.t_0.65: 0.0281
- fpr_at_thresholds.t_0.70: 0.0074
- fpr_at_thresholds.t_0.75: 0.0008
- fpr_at_thresholds.t_0.80: 0.0003
- fpr_at_thresholds.t_0.85: 0.0001
- fpr_at_thresholds.t_0.90: 0.0001
- fpr_at_thresholds.t_0.95: 0.0000

value_date_quarter=2024Q4 - 2) Model Efficiency - Explanations

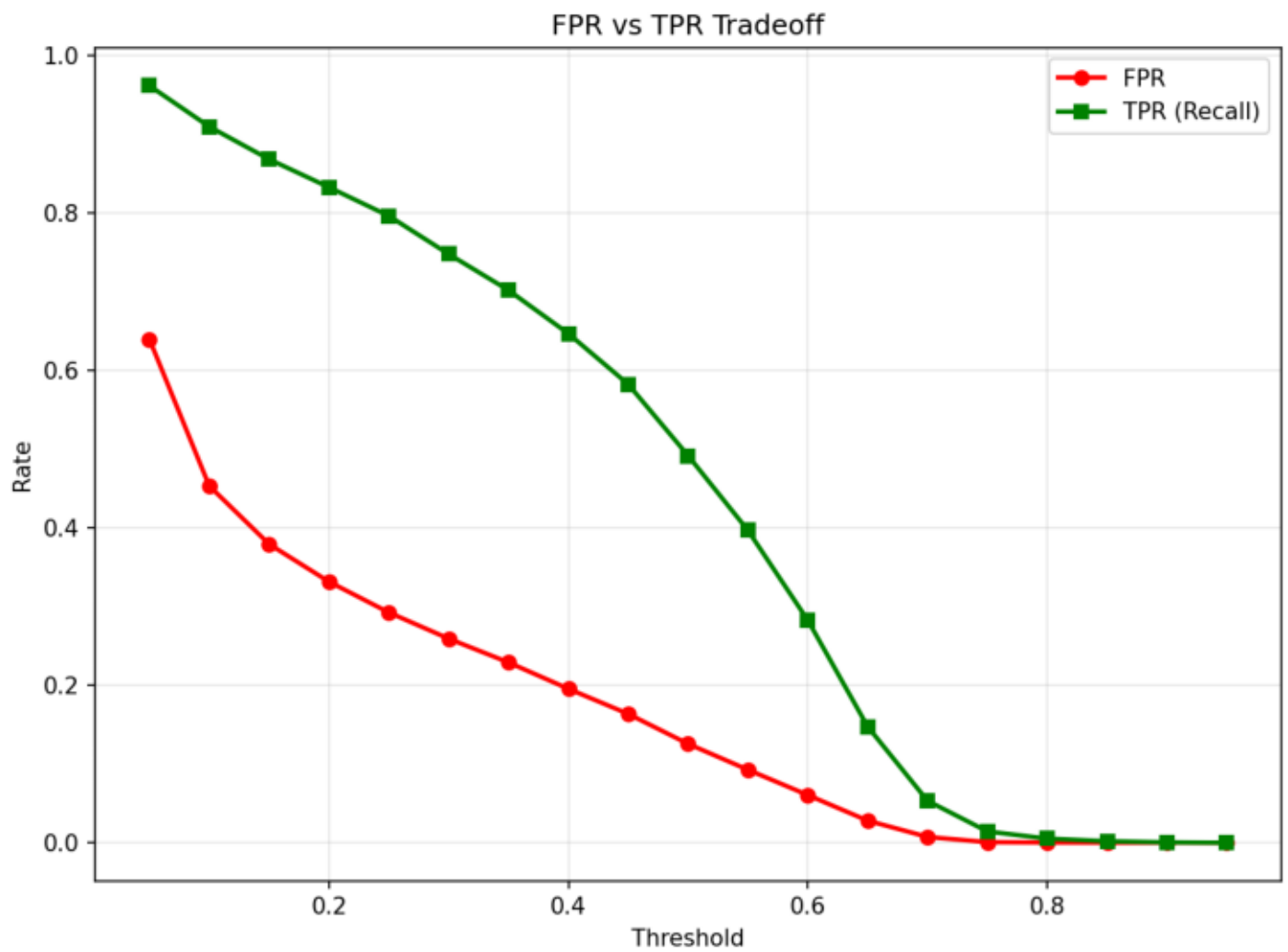
- At threshold 0.50, FPR=0.1257 (high); FP=1252 out of 9959 negatives.
- At the same threshold, TPR=0.4920 with TP=1947 and FN=2010, showing the capture rate of positives.
- A threshold near 0.60 yields FPR $\approx$ 0.0606 with TPR $\approx$ 0.2828 if you want to target $\sim$ 5% false positives.



fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

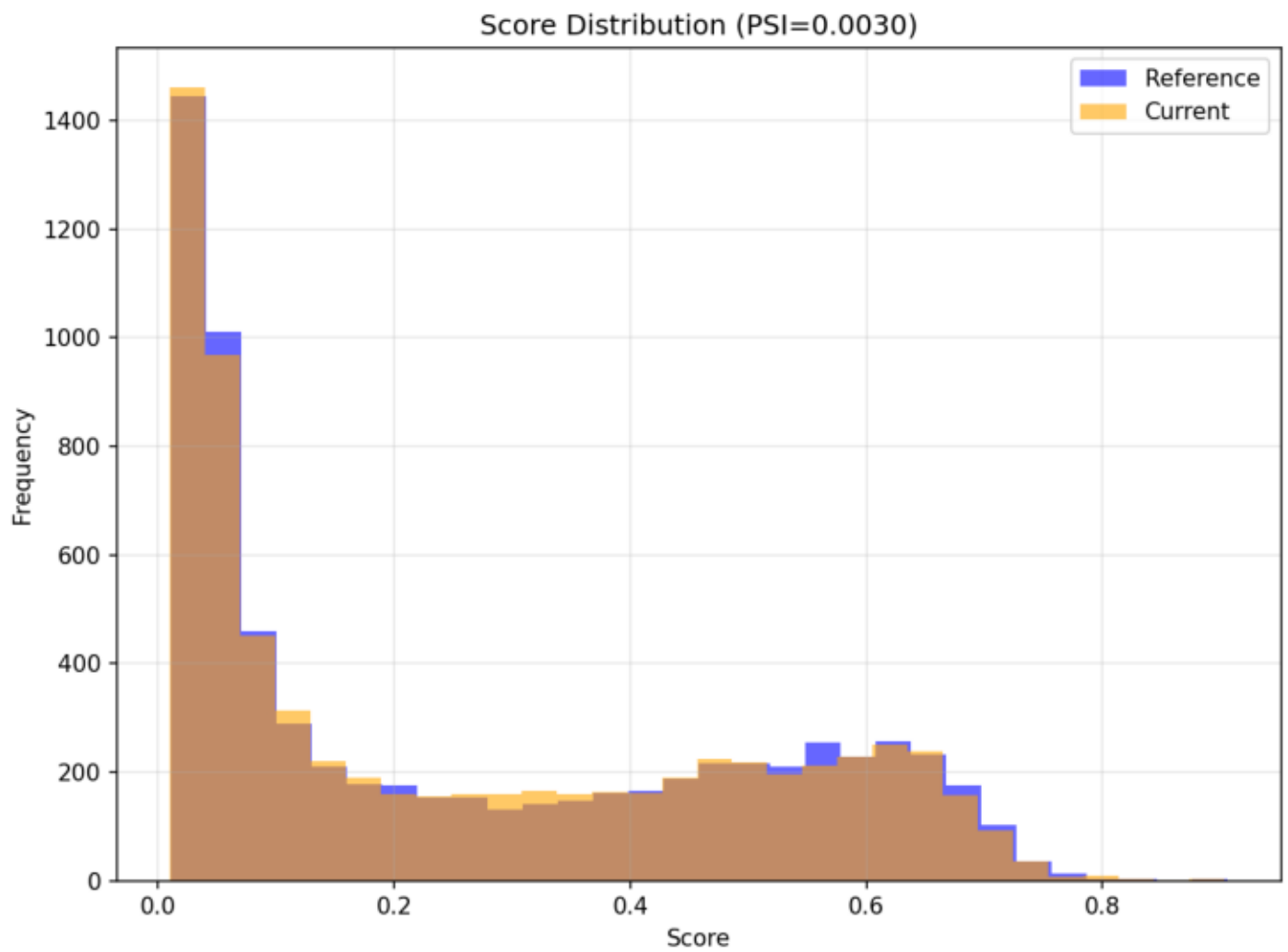


value_date_quarter=2024Q4 - 3) Model Stability

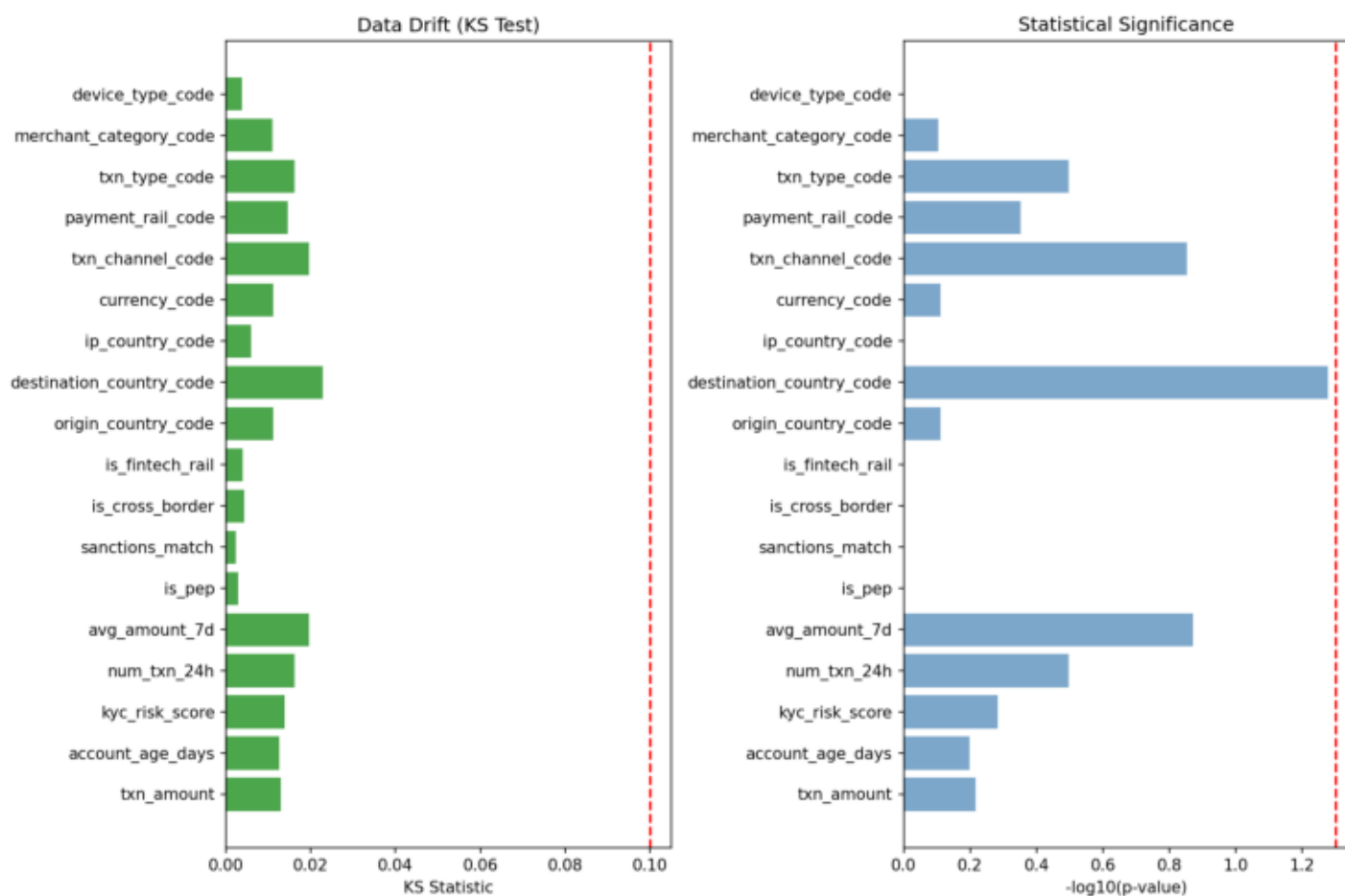
- psi: 0.0030
- cv_auc_roc_mean: 0.8144
- cv_auc_roc_std: 0.0107
- cv_auc_pr_mean: 0.5940
- cv_auc_pr_std: 0.0270
- bootstrap_auc_roc_mean: 0.8109
- bootstrap_auc_roc_ci_lower: 0.7997
- bootstrap_auc_roc_ci_upper: 0.8216
- concept_drift_detected: False
- concept_drift_score: 0.0076

value_date_quarter=2024Q4 - 3) Model Stability - Explanations

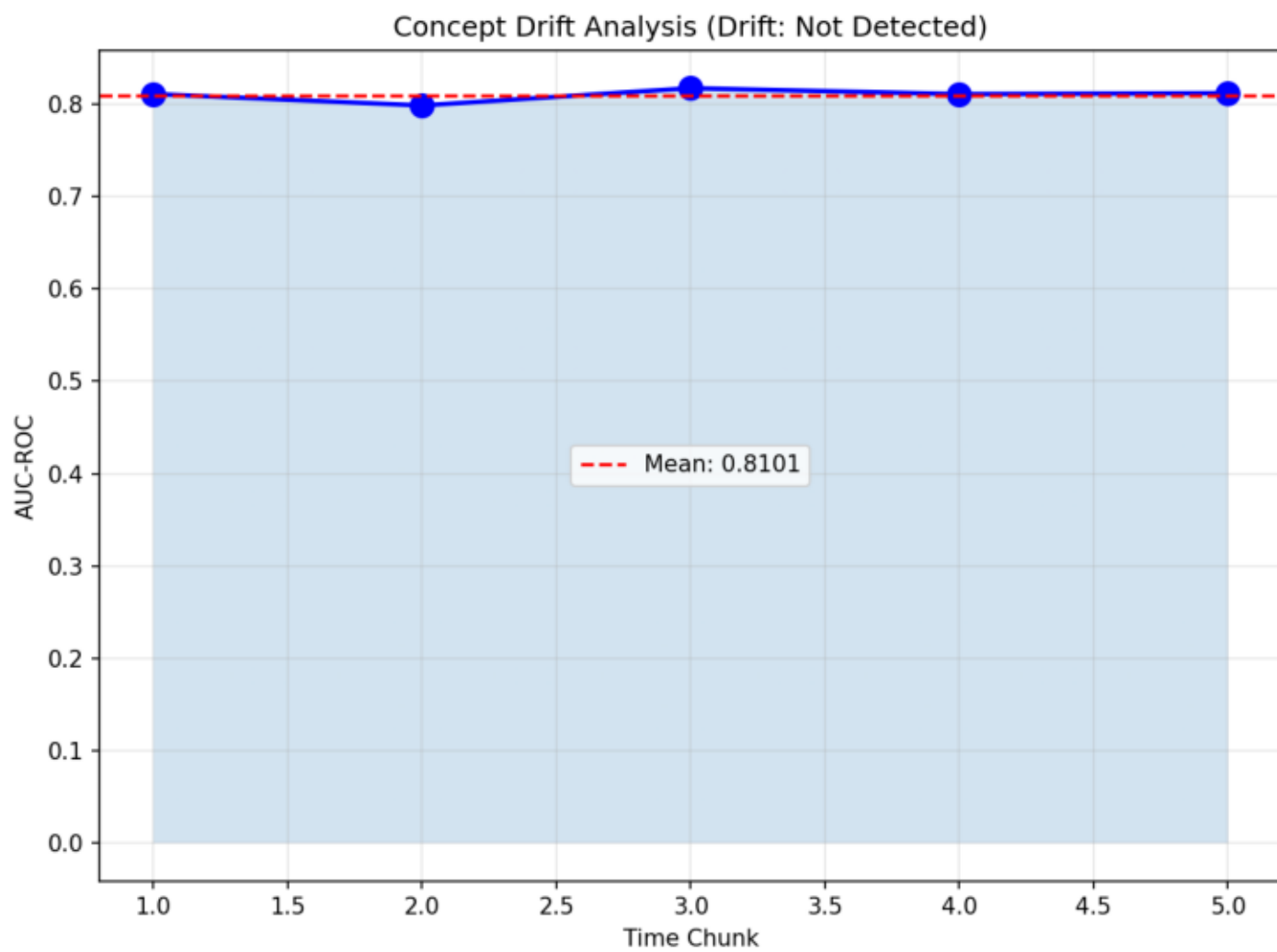
- PSI is 0.0030, indicating negligible score distribution shift between reference and current data.
- CV AUC-ROC is 0.8144 ± 0.0107 (stable across folds).
- CV AUC-PR is 0.5940 ± 0.0270 (moderate across folds).
- Bootstrap AUC-ROC mean is 0.8109 with 95% CI [0.7997, 0.8216] (width 0.0219).
- No concept drift detected (score 0.0076), suggesting stable performance over time.
- Data drift flagged 0/18 numeric features; top drift signals: ['destination_country_code', 'avg_amount_7d', 'txn_channel_code'].



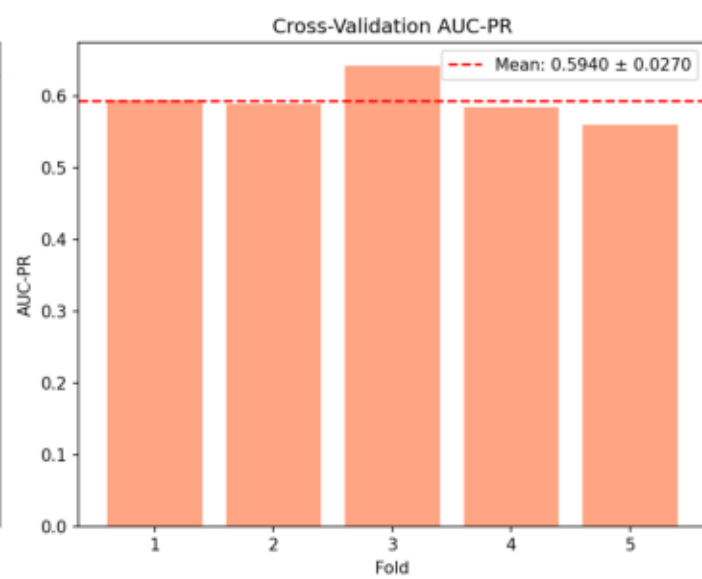
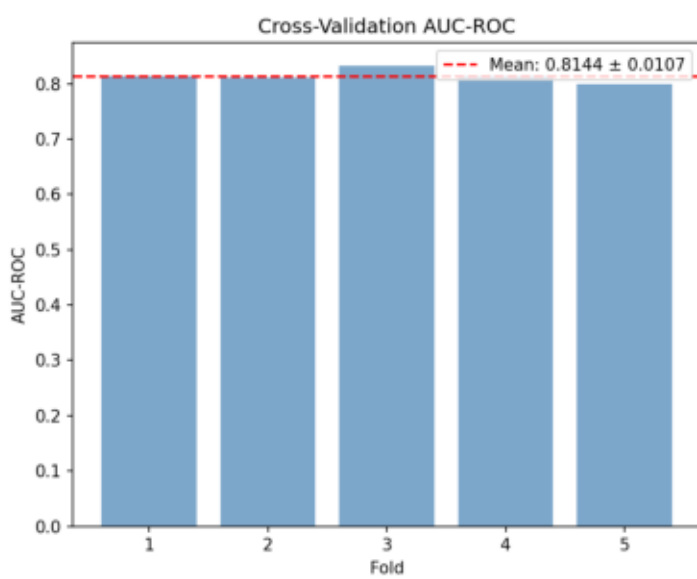
psi_distribution: Score distributions overlap consistent with PSI=0.0030.



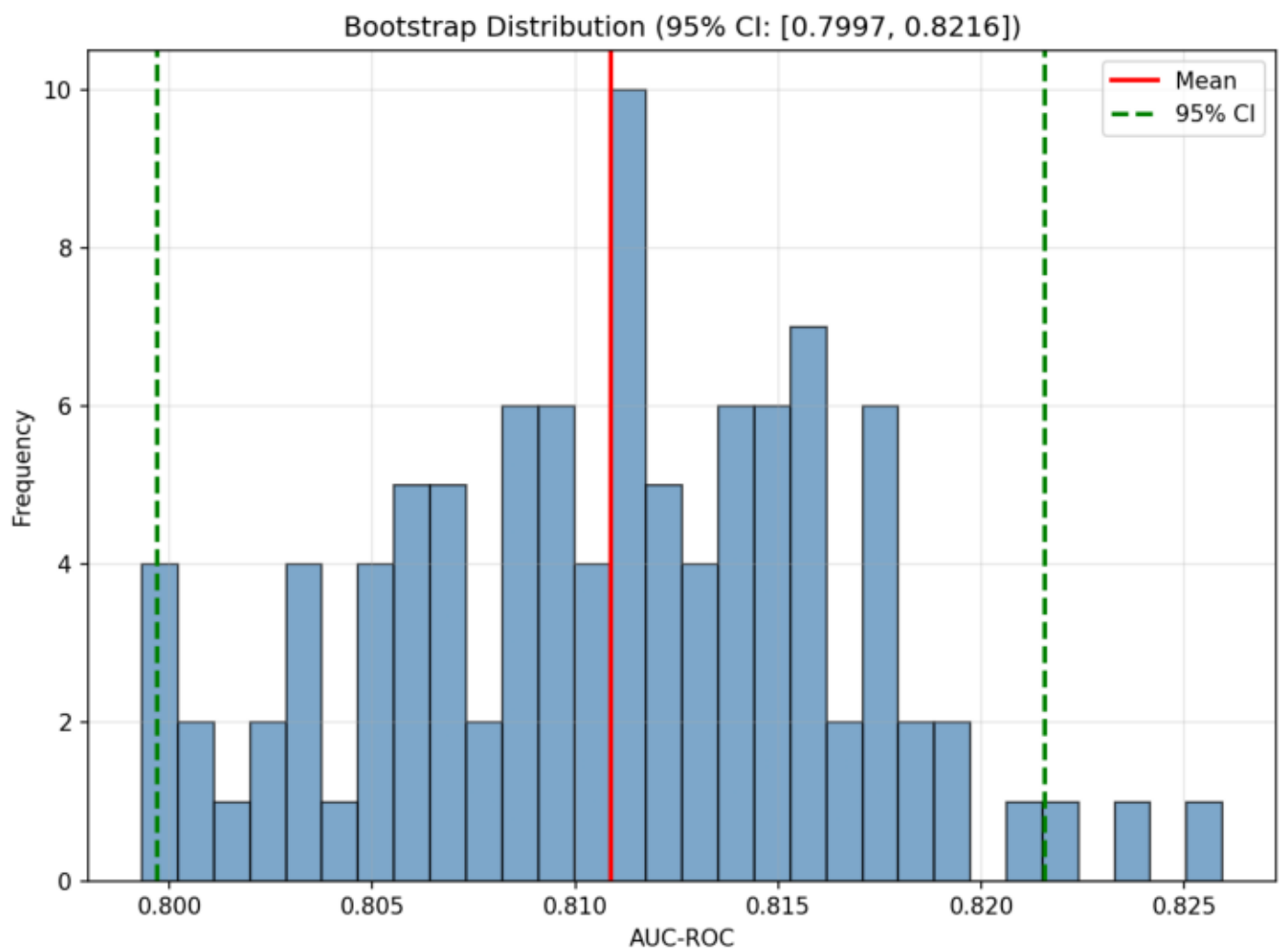
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.



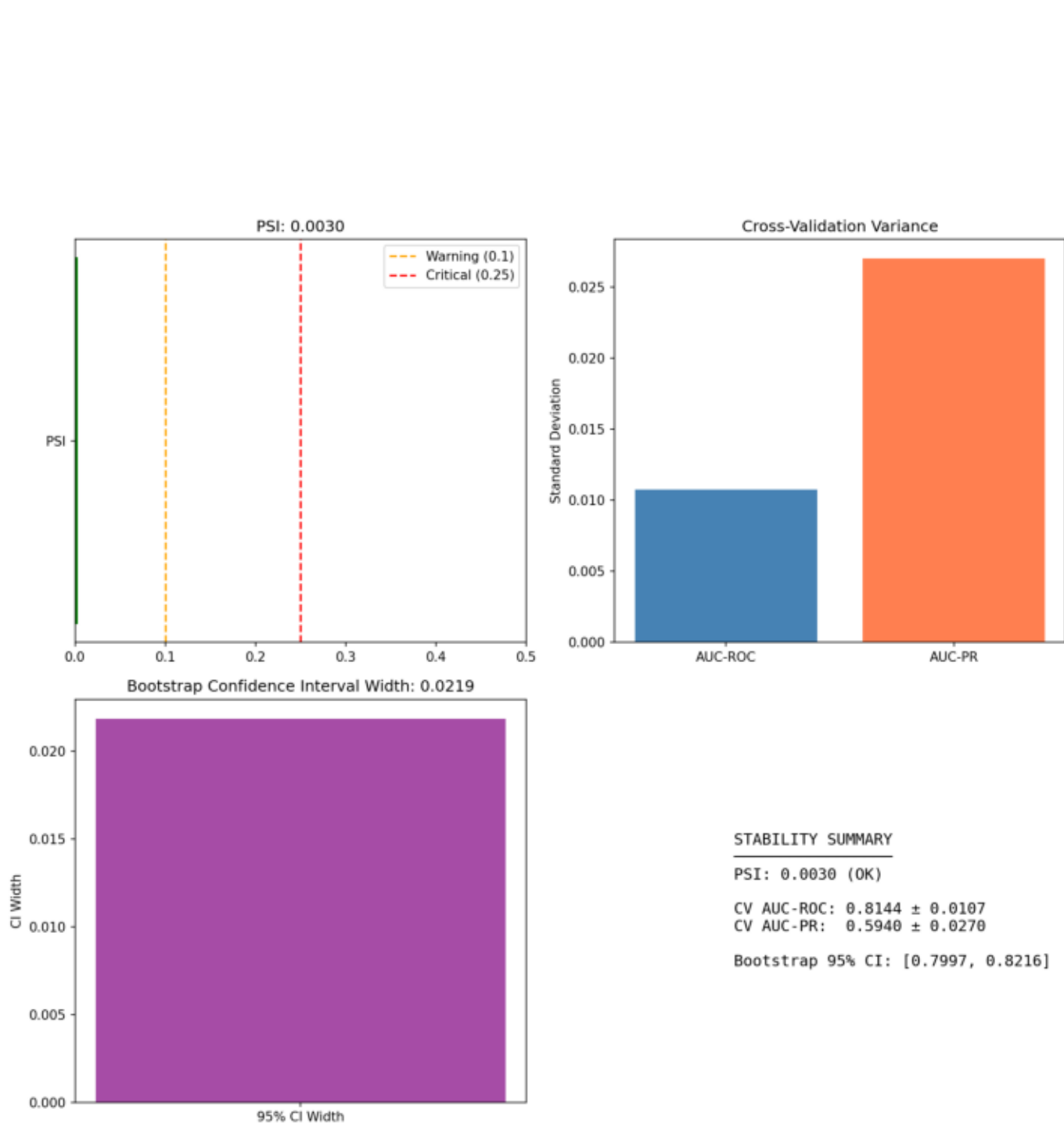
concept_drift: AUC across time chunks shows whether performance is stable over time.



cv_results: Fold scores cluster around ROC 0.8144 and PR 0.5940.



bootstrap_distribution: Bootstrap CI [0.7997, 0.8216] reflects performance stability.



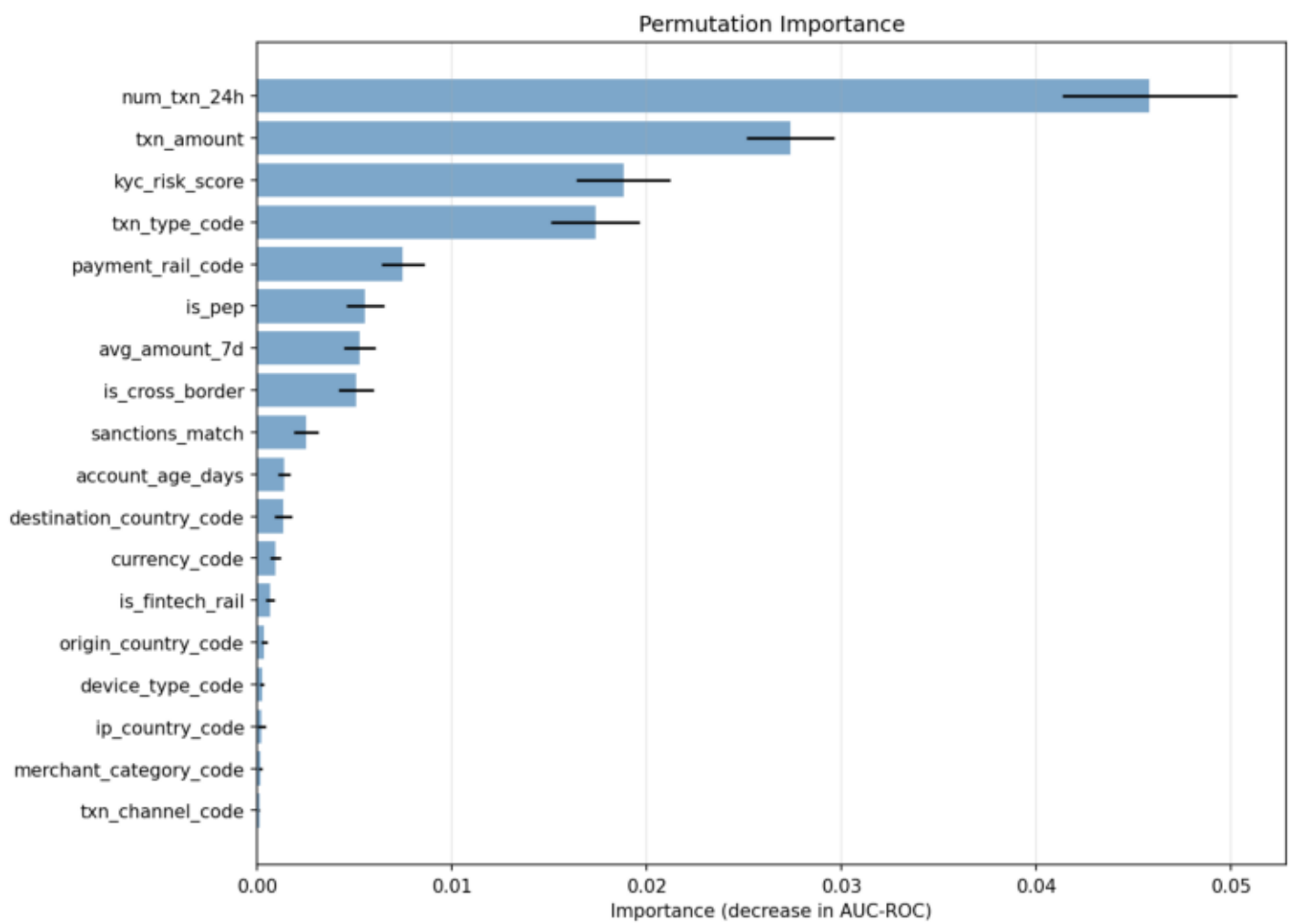
stability_summary: Dashboard summarizes PSI, CV variance, and bootstrap CI width.

value_date_quarter=2024Q4 - 4) Model Interpretability

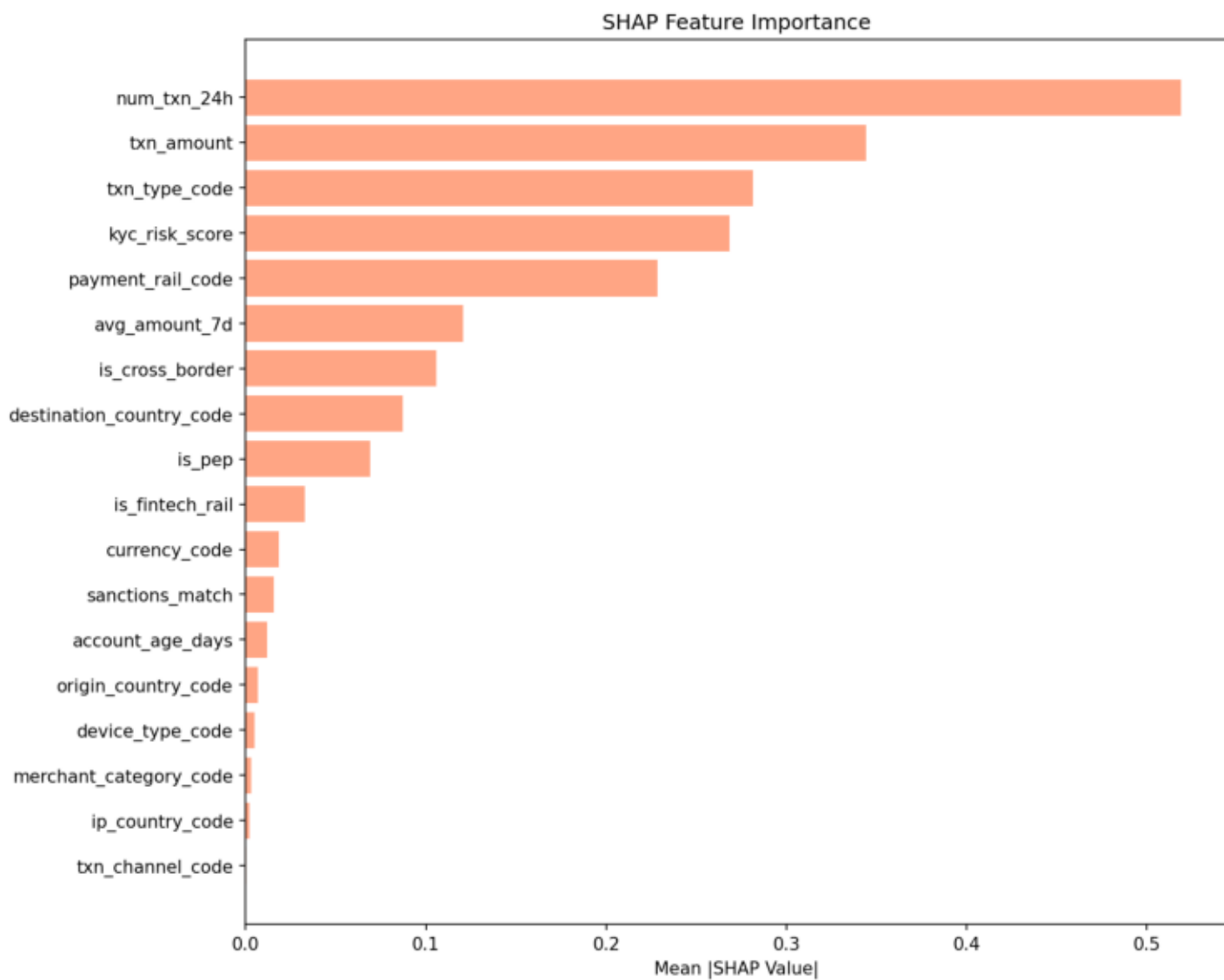
- model_type: tree
- methods_used: ['permutation', 'pdp', 'ice', 'shap', 'lime']
- perm_top_features: ['num_txn_24h', 'txn_amount', 'kyc_risk_score', 'txn_type_code', 'payment_rail_code']...
- shap_top_features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code']...
- lime_instances: 3
- pdp_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h']...
- ice_features: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score']

value_date_quarter=2024Q4 - 4) Model Interpretability - Explanations

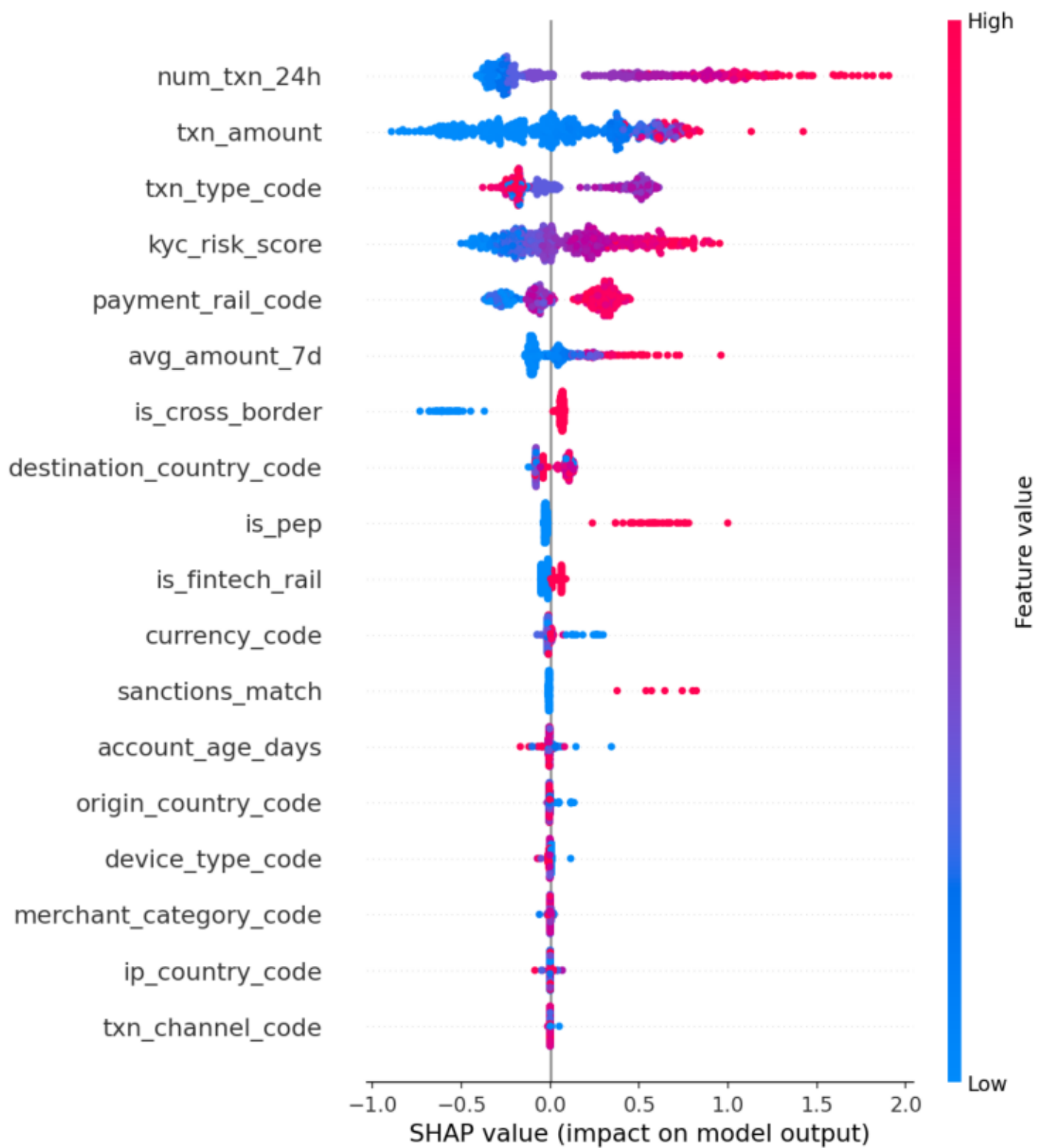
- Permutation importance highlights: ['num_txn_24h', 'txn_amount', 'kyc_risk_score', 'txn_type_code', 'payment_rail_code'].
- SHAP top features: ['num_txn_24h', 'txn_amount', 'txn_type_code', 'kyc_risk_score', 'payment_rail_code'].
- LIME generated local explanations for 3 instances; top contributors for instance_6: ['txn_amount > 583.84', 'sanctions_match <= 0.00', 'kyc_risk_score > 57.71', '2.00 < txn_type_code <= 4.00', 'is_pep <= 0.00'].
- PDP plots show average effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score', 'num_txn_24h', 'origin_country_code'].
- ICE plots show individual-level effects for: ['txn_amount', 'avg_amount_7d', 'account_age_days', 'kyc_risk_score'].



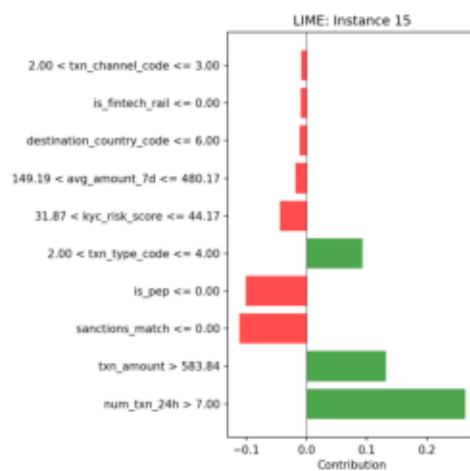
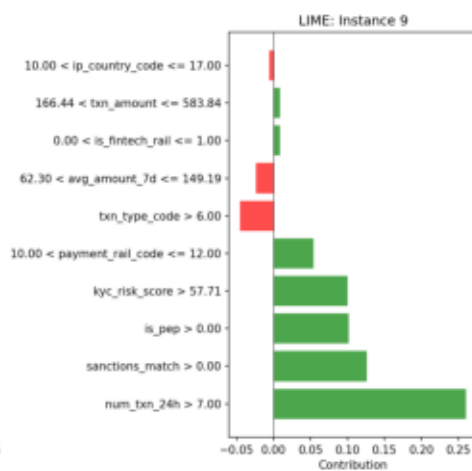
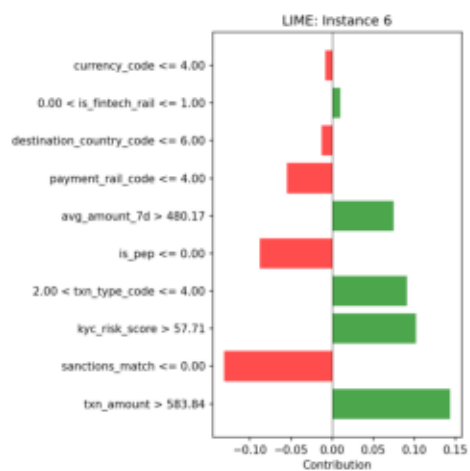
permutation_importance: Bars show how much AUC drops when each feature is permuted.



shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

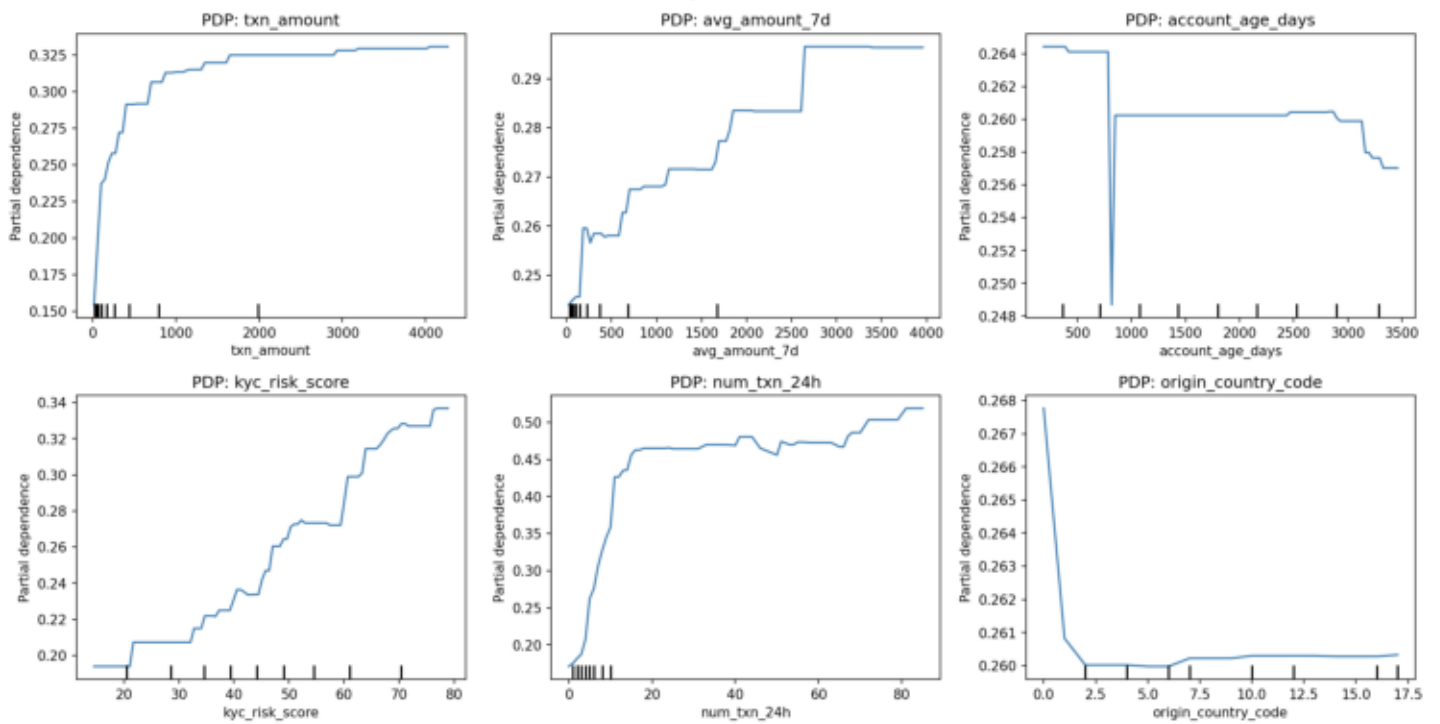


shap_beeswarm: SHAP beeswarm shows both impact size and direction per feature.



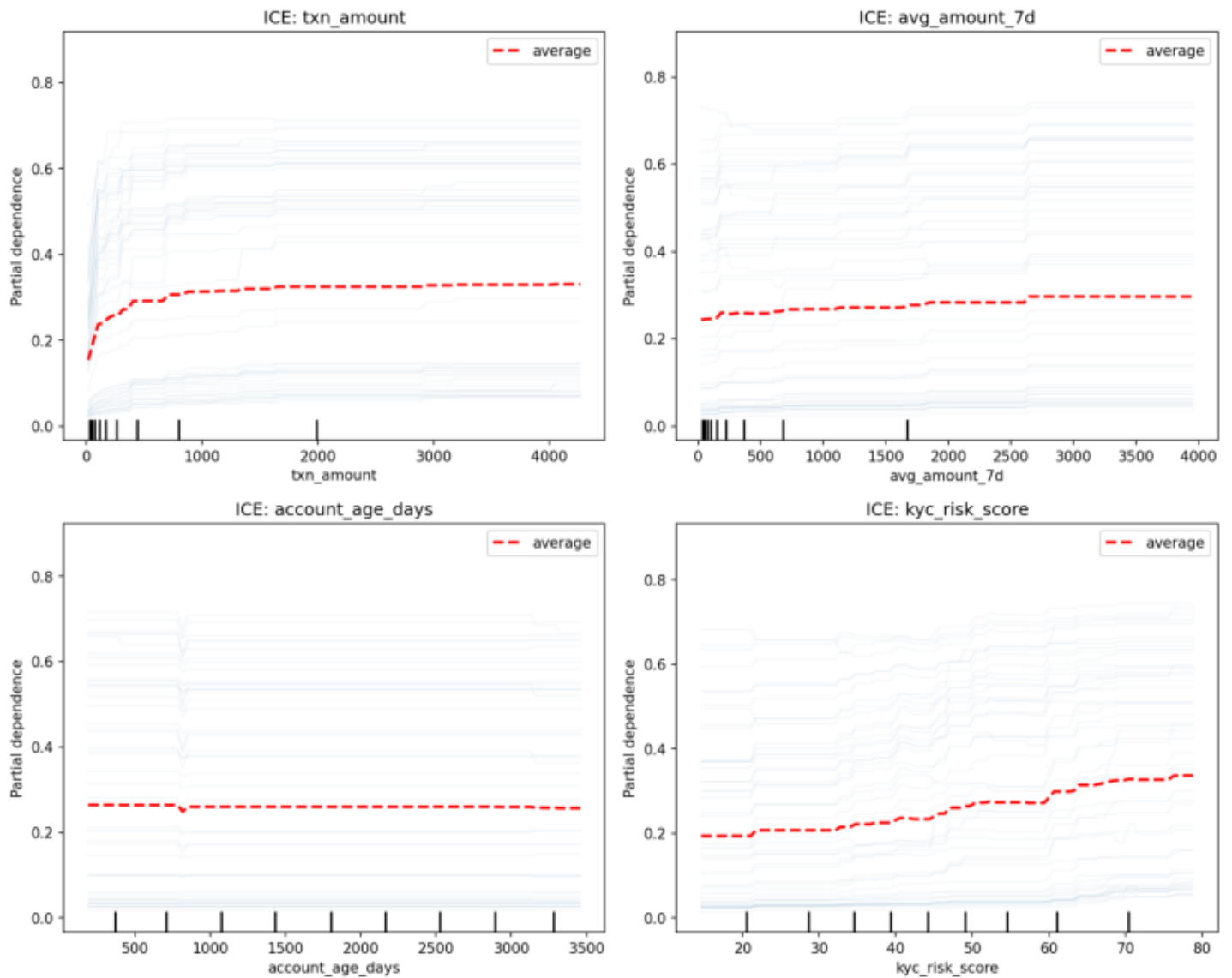
lime_explanation: LIME bars show local feature contributions for selected instances.

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.